



CITY UNIVERSITY LONDON

MSc Computer Games Technology with VR (Full Time)

Project Report

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Dynamic NPC Touch Interaction in Virtual Reality

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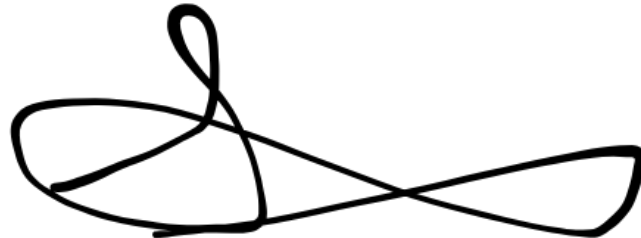
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Abstract

Virtual Reality (VR) is a rapidly advancing field with potential to dominate the digital market. It offers unparalleled immersion by allowing the players to physically interact with a virtual world. However, a significant gap remains in the sophistication of interactions with Non-Playable Characters (NPCs), which are often static or unresponsive to physical touch, thereby breaking the player's immersion. This project addresses this limitation by designing and implementing a dynamic NPC interaction system for VR that enables characters to react realistically to player's physical touch and thrown objects.

The interaction system, developed in Unreal Engine 5 utilizes a custom collision detection system to interpret the touch area and intensity of the player's hand. These inputs execute a multi layered response system combining a Finite State Machine with different behaviors with a procedural, physics-based ragdoll movement for dynamic interactions. Furthermore, a bone-grab system allows players to physically move NPC's limbs. The prototype was evaluated with user testing, using an Immersive Experience Questionnaire (IEQ) and qualitative questions. The results indicate this system enhances immersion and believability and should be incorporated in future games.

Keywords

Virtual Reality, Non-Playable Character (NPC), Immersion, Physics-Based Interactions, Touch Interaction

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1 Chapter 1 – Introduction and Objectives:

1.1 Project Scope:

Virtual Reality (VR) has gained a lot of popularity among gamers and companies around the world because of its ability to immerse the player in a fictional world or 3D replicas of real worlds. This immersion has been improved by adding further additions to each part of the VR world. And what started out with viewing a VR world through your phone attached to a small headset has now reached a point where VR headsets have their own system hardware and motion controllers with which you can interact with objects in the VR world. Now immersion is driven by the player's ability to interact with the virtual world in ways that are physically engaging and realistic such as through hand tracking or motion controllers. However, when it comes to non-playable characters (NPC) in VR games, there is still a significant gap between the player and NPC interactions. There is a lack of sophistication of NPC interactions in many big VR games. They are often static or non-responsive breaking the immersion or the illusion when the player engages with them physically.

This particularly affects games when other objects and mechanics are made to be realistic and immersive. While non-VR games like Red Dead Redemption 2(Rockstar games, 2019), or Grand Theft Auto games and some old assassin's creed games feature NPCs that react dynamically to player presence and behavior or touch. This feature is a priority for immersion in VR, but it hasn't yet been leveraged to its extent to create believable social or physical encounters with virtual characters. Some VR games such as Superhot VR(Superhot Team, 2017), Half-life: Alyx (Valve, 2020), and BONEWORKS (Stress Level Zero, 2019) have explored this space with ragdoll physics and interaction with objects, this led these games to become very popular because of the immersion and realism fans often highlight how they felt while playing these games making them successful, but these interactions are often limited to specific situations like combat. This limitation is clear with NPCs perception, and interrupting responses realistically to all kinds of player interactions, from a friendly tap to shove.

This project addresses this gap by developing NPCs with reactive, realistic and immersive movements to player's physical touch interactions. The player can interact with the NPC in many ways e.g.: grab, push or shove NPC body parts such as arms, head, shoulder etc. They are perceptive NPCs; they can sense the pressure of player's touch intensity and location with the ability to sense objects thrown toward them and react to their hit. This prototype used Unreal Engine which is an industry standard game engine taking advantage of all the different tools and features. Leveraging modern game development techniques, including physical animation, Finite state machines, behavior trees etc. These were used to create NPCs that enhance interactions.

1.2 Aim of the Project

The primary aim of the project is to enhance immersion in VR worlds by designing and implementing a system for NPCs that react realistically to a player's physical touch interactions. And also react to objects thrown at the NPC. They should be able to interpret the location and intensity of touch to trigger appropriate behavior, animated responses and move their body realistically beyond simple ragdoll physics to create a believable character.

This leads to these research questions:

1. How non-playable character interaction in virtual reality can be more realistic and immersive using a custom changeable system that reacts to player's actions?
2. Is it possible to design and implement a non-playable character which reacts to player's actions based on force and area that triggers a more realistic and believable reaction in virtual reality?
3. To what extent can we increase the realism of virtual reality environment by including more reactive non-playable characters?

This project's purpose is to enhance the player's gameplay experience by creating a small VR demo by creating a new system to handle interactions.

1.3 Reasons for the choice of project

This project was chosen by the author for several key reasons. Firstly, it provides an opportunity to gain deep, advanced practical experience with Unreal Engine 5, an industry standard tool while implementing a complex system which is a combination of AI, animations, Game Physics, and VR interaction. Furthermore, it addresses the identified gap in current VR games, offering a tangible contribution to the field. Finally, creating a reusable template for NPCs has the potential to benefit other developers and advance the state of the art in VR immersion.

1.4 Beneficiaries

The main beneficiary of this project is the author who is benefited by gaining experience in Unreal Engine 5 which is an industry standard game engine. Also understand the complexities of procedural animations, finite state machines and VR prototyping.

Other benefactors include VR developers that can use this template as a starting point for games, also contributing to future VR studies in understanding the effects of NPC interaction to immersion which in turn benefits future game titles, creating a better experience for the VR community.

1.5 Project Objectives

List of objectives, the results of the objectives are part of the report and the prototype that was implemented.

- Literature review and game design
- Implementing VR environment and Character controls, animations
- Implementing Non-Playable Character AI
- Implementing new custom collision system
- Implementing Finite State Machines (FSMs), Behavior Trees and animations
- Implementing FSM with new collision system
- Implementing Physical animation using the new collision system
- Implementing Physical animation when object is thrown
- Implementing Demo level for testing

- Player testing for immersion, realism and experience

1.6 Methods Used

This project was implemented using Unreal Engine 5.5 and Visual Studio Community 2022. Agile development was used. Each part of the project was implemented separately with many iterations and incrementing or modifying with each step then followed with a sufficient amount of testing and refactoring. Then each part of the project was joined by the main body. The reason for this was to avoid errors and overlap with other parts of the project. Each system functions on its own so it is easily modifiable for a designer or a developer in the future. This also mitigates many risks, errors and problems that might arise when testing or modifying project further. This approach is well-suited for game development as it accommodates constantly evolving requirements.

Standard or inbuilt Unreal Engine functions were explored through documentation and game community articles studying the implementation so modifications or new functions can be created to accommodate the new system, by exploring similar implementation in videos and seminars. Some of these are familiar with the author and others a new topic.

Analysis was based on both quantitative and qualitative questions by using immersion analysis questions called Immersion Experience Questionnaire (IEQ) with some qualitative questions for more insight and some additional questions that will help with future use.

1.7 Project Timeline

The project followed a structured work plan as outlined in the proposal, the implementation was very close to the work plan, only varying a slightly for the main component of the project. Some early tasks were completed quicker than anticipated but the time was used to improve the implementations. The Agile development process allowed for flexibility within this plan, with regular milestones for steady progress towards the submission. One change was made to the planned schedule: The testing was postponed accommodating for the lab schedule and also to create a better test environment in VR to show the versatility of the NPC system.

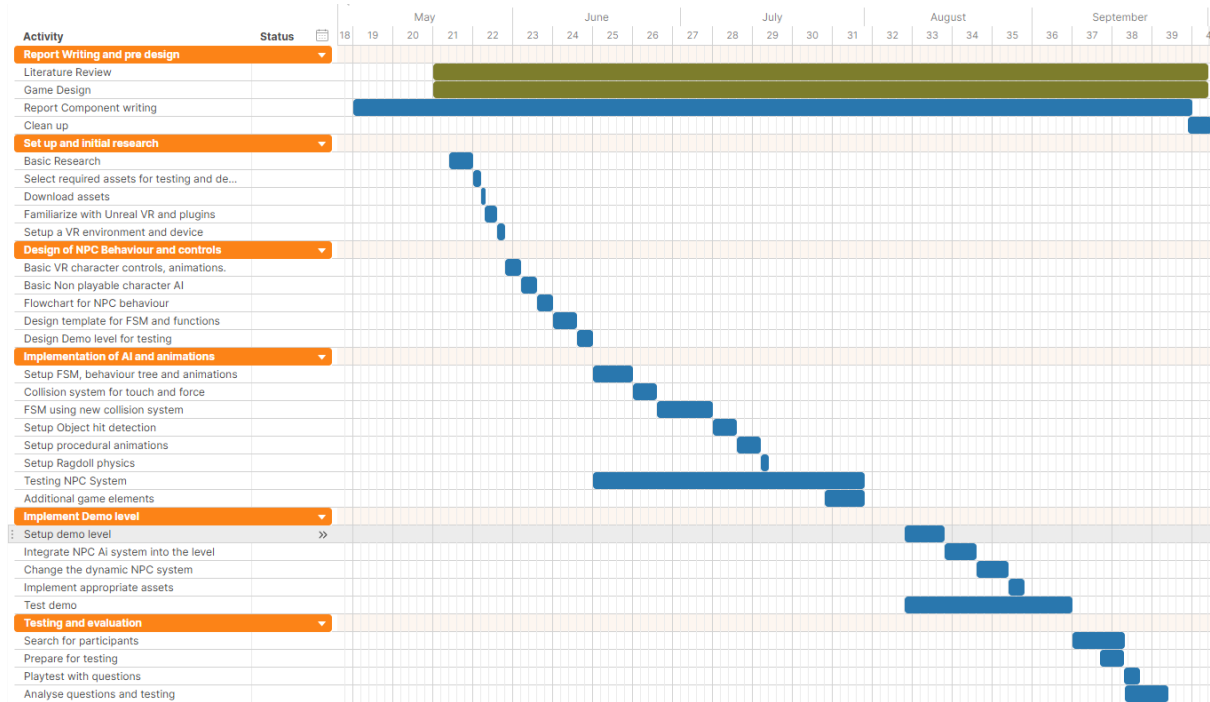


Figure 1. 1 Project Work Plan

1.8 Report Overview

Following this introduction, Chapter 2 covers the critical context for the project through a review of relevant literature. Chapter 3 then details the implementation methodology for the prototype; the results are presented in Chapter 4. Ending the report with a discussion of the findings in Chapter 5 and a final evaluation and reflection on the process in Chapter 6.

2 Chapter 2 – Context:

2.1 Introduction

This chapter establishes the literature and the real-world applications of immersion in Virtual Reality, with a focus on interactions with non-playable characters and how it affects the player's sense of immersion. It reviews the relevant literature surrounding this topic, analyzing the current VR games or experiences. The chapter identifies the gap in the field that is being addressed by this project.

2.2 Virtual reality and immersion

“Virtual reality (VR) is a computer-generated environment accessed through a head-mounted display (HMD), such as a helmet or glasses, which allows users to become immersed in a digital simulation. VR can be used for educational, entertainment, and even work purposes.” (Coursera, n.d.). This VR world is experienced through special equipment explained in 2.3 Virtual Reality Hardware.

The goal of every VR developer is immersion. Every feature, technique and modification made to the technology or the world created is to improve immersion and bring it closer to reality. Researchers Witmer and Singer (1998) describe it as a psychological state where you feel enveloped by and involved in the virtual environment. A key part of this is the "**plausibility illusion**" (Slater, 2009)—the feeling that what is happening in the VR is real. This illusion is powerful but fragile. If something in the VR world behaves in an unrealistic way—like a character not reacting when you touch them—the illusion can be broken, and the sense of immersion is lost.

2.3 Virtual Reality Hardware

There are many components in Virtual Reality that are used to navigate the virtual world and to increase immersion and playability. The most common and easily available VR hardware components are Headsets and Motion Controllers.



Figure 2. 1 Meta Quest 3 Headset and Motion Controllers

2.4 Non-Playable Characters in Games

Non-Playable Character (NPC) are characters in games that cannot be controlled by the player. These are characters in games that act programmatically or through Artificial Intelligence. NPCs are used in almost every game in many different forms to convey the story or missions or for interactions.

In non-VR games the NPCs have made significant progress from enemies in Space Invaders (Atari, Inc , 1978) which are targets for the players to games now where NPCs act, react, make decisions and store information etc. The evolution of NPCs has been significant throughout the years from just static models that were not reactive and to NPCs now where they move or fall depending on their environment for example: characters in Red Dead Redemption 2 (Rockstar Games, 2019). NPCs in this game even have complex routines and react to players in a human like way: flee or fight when threatened, move aside when you bump into them. Researchers note a shift from NPCs as mere functional obstacles to “believable agents” that enhance the player’s sense of immersion (Lankoski, Helio, 2002)

2.5 Non-Playable Characters in Virtual Reality

With VR players have an advantage that is not available when it comes to non-VR games. Players have the chance to embody the character almost completely, not separated by a screen or a keyboard. With the ability to control, move and see their environment. The players have an advantage with immersion.

- **Static Objects:** Since this is still a field that's in its early stages of development, the NPCs lack depth, reactions or movement. This is the case with many VR games and worlds, breaking the illusion and making gamers feel like they have been transported to late 2000s gaming where NPCs were nothing more than 3D models.
- **NPCs are limited to combat:** Some games do feature NPCs that have physics-based combat like *Blade and Sorcery* (WarpFrog, 2024) and *BONEWORKS* (Stress Level Zero, 2019). In these games NPCs react to being hit but it features a simple physics ragdoll reaction sometimes they play just an animation. The NPCs don't have a complex reaction system reacting the same way to a gentle tap or a shove.
- **Programmed Reaction:** Very few games like *Half-Life: Alyx* (Valve, 2020), have more advanced NPCs where they flinch if an object is thrown or some specific NPCs react more dynamically (in combat scenarios).

This lack of dynamic interaction is a known problem. Research by Carroll (2022) found that unresponsive NPCs directly disrupt immersion. Moreover, studies like Aburumman et al. (2022), state that non-verbal communication such as eye contact, touch is critical for making virtual characters feel real. Detailed analysis of each notable game later in this document.

2.6 Realistic Non-Playable character design concerns

A common concern among many developers and game studios is the computational expense. VR is computationally heavy when compared to screen games because of its complex computations of both real-world surroundings and the need to render the world twice (once for each eye). This is being overcome by inventing faster computational chips or by playing the game tethered to a computer which usually has higher computational power.

Another concern is the development time and cost for developing dynamic NPC because of the engineering complexity behind constructing natural movement, reactions and variations.

2.7 Notable Non-Playable Characters in Games

Here are some of the notable NPC's that inspired this idea. These are games that pushed the barrier for NPCs and interactions with the player this includes both VR and non-VR games to show the effect it had on each game and its reception in the gaming community.

2.7.1 SUPERHOT VR



Figure 2. 2 SUPERHOT VR Gameplay from Steam Page

SUPERHOT VR (SUPERHOT team, 2017) is a VR strategic, puzzle-like action game that has one of the most advanced and notable physics movements in VR. One of the first games that introduced a complete physics-based combat system to VR.

The game's focus is on physical interaction between the player and NPC. In the game you are stationary and time moves when you move. The goal of the player is to strategize and eliminate enemies. The enemies have glass-like bodies that break at the place of contact eliminating the NPCs. While the game features very other world like visuals but it still has a strong player embodiment and realism because of its consistent movement and reactions to player's movement.

Although this game's main interaction is based on physics of the enemies, it is limited to the player's touch or attacks. It does not feature any advanced NPC interactions like grabbing the

NPC or touch and force-based reactions. SUPERHOT VR has pushed the VR game standards and inspired many games that featured physics-based gameplay.

2.7.2 Half Life: Alyx



Figure 2. 3 Half Life: Alyx Gameplay from Steam Page

Half Life: Alyx (Valve, 2020) is another VR game which takes place in the same world as non-VR games Half Life (Valve, 1998) and Half Life 2 (Valve, 2004). This is a narrative driven game with core mechanics focusing on physics-based interactions such as grabbing, throwing and advanced AI.

There are two types of NPCs in these soldiers and aliens both exhibiting advance AI and physics reacting realistically to player's presence. The game is known for being the most advanced and ahead of its time immersive interactions with the world, no other game reaching its level of immersion. Enemies have a great perception and reaction to the world and objects around them. Stumbling or tilting when hit with throwables or bullets. It maintains natural movement without crossing the threshold and reaching uncanny valley. The interactions are consistent with high-fidelity maintaining the illusion. The player has the ability to throw objects towards the NPC in different angles with the NPC taking the hit and reacting with a semi procedural system.

The game still lacked a more advanced interactive combat system with no melee attacks or close interactions with the NPC. The player interactions with enemy are limited to shooting

and throwing objects to stun or kill enemies. Another limitation is the interactions are limited to combat scenarios or some rare situations where player has the ability to talk to the NPC, instructions are mostly carried out through voice over which reportedly broke immersion for many players reminding them they were playing a character. According to the in-game developer commentary videos provided to the owners of the game stated due to limitations of both VR technology and breaking immersion the developers chose to limit player interaction.

This game served as one of the inspirations for some methods developed in this study's project. The enemy interactions were studied and analyzed for this study. They were implemented in this project with some optimizations, combining additional features that push the movement and realism.

2.7.3 Blade and Sorcery



Figure 2. 4 Blade and Sorcery Gameplay from Steam Page

Blade and Sorcery (WarpFrog, 2024), another VR game that has physics driven melee, ranged and magic combat. It has a progression-based system focused on fighting enemies and collecting items. The gameplay is driven by player created scenarios providing a challenging combat driven by physics.

The player fights with a variety of human enemies. Player also has the ability to spawn different types of NPCs to train and test the physics of NPC with combat dummies, archers and mages. The physics-based interaction is complex, and the player has to account for many

game mechanics while fighting such as weapon weight and swing speed. The enemies react to many different combat styles. Swords, axes and shields can be used against the enemies with one of the most realistic physics-based combat. Enemies have full body physics allowing you to manipulate bodies in creative ways, the ability to grab and wrestle with enemies to gain an advantage in the fight. It also features a similar interaction to Half Life: Alyx (Valve, 2020) where you can fight ranged battles or throw objects to deal damage. The punches in the game are advanced and take account of both location of the punch and intensity of player's hand movement.

Although this game has one of the most advanced enemy physics, there are still a few limitations. The friendly NPCs are completely non-reactive to player's touch or other interactions. The enemy interactions work well with weapons but when it comes to punches it is less immersive, the game enemies just fly off with ragdoll physics and makes the reactions more comical than realistic.

Even though the physics of this game is impressive and groundbreaking, some interactions are lacking, and outside solo developers have created many additions to the game. The developers of the game gave players the freedom to modify and add custom features or improvements. This study's project has many of the same features with some advanced modifications to the NPC for more immersive and realistic responses.

2.7.4 BONEWORKS



Figure 2. 5 BONEWORKS Gameplay from Steam Page

BONEWORKS (Stress Level Zero, 2019), is an experimental physics VR adventure game that uses advanced physics-based tools and weapons. Gameplay is a single player campaign.

Game's priority and interaction is based on realistic and physics-driven VR mechanics. The game has a full body inverse kinematic system. Unlike many other VR games, the player is given control of a full virtual body that can be manipulated and used to interact with the game world. It features the most advanced virtual body movement with climbing, vaulting and taking damage from environment obstructions.

The NPCs in the game react to punches, kicks and weapon strikes with realistic and physics driven reactions similar to Blade and Sorcery (WrapFrog, 2024). The game has one big advantage that other games lack. The use of a virtual body for the player allows for a very interactive response for pushes and collision with the NPC body. Similar to Blade and Sorcery the player can grab and manipulate NPC joints and physical damage or harm them. You can also throw objects at the NPC and get reactions similar to other VR games

The main emphasis in this game is player's full body system. Though it lacks friendly NPC interactions. The interactions work only under combat scenarios.

2.7.5 Batman: Arkham Shadow VR



Figure 2. 6 Batman: Arkham Shadow VR Gameplay from Meta Store

Batman: Arkham Shadow VR (Rocksteady Studios, 2024), is Batman action-adventure VR game which released exclusively in Meta Quest 3 and 3S. The game features both narrative and combat approach. You embody Batman, interact with friendly NPCs, and fight different types of enemies.

A heavily narrative driven game that limits the player to follow a certain path and interactions. As player you can interact with many friendly and enemy NPCs. These interactions are mostly conversations or combat scenarios that are very standard and controlled. Combat interactions feel more click-based. The attacks are prompted by the developer for maximum damage with User Interface appearing in front of you in VR directing your attacks.

The game was made in Unreal Engine (The same game engine used in this study's project) and it has many flaws when it comes to player and world interactions. Combat is mostly melee with the responses given by the enemy NPC nothing more than a scripted animation. The interaction with friendly NPC is very unnatural. Their only role being a static model that delivers dialogue. No response to player's touch or interaction. The enemy NPC lacks any kind of interaction with objects and world around them.

The game's interactions are another reason that prompted this study. The potential in the game with more interactive responses can improve the immersion and hugely benefit the games story.

2.7.6 Red Dead Redemption 2



Figure 2. 7 Red Dead Redemption 2 Gameplay from Steam Page

Red Dead Redemption 2 (RDR 2) (Rockstar games, 2019) is an open world, story game. “NPCs in RDR2 are different. They behave more like people who operate according to certain storylines that give meaning to their actions” (Velardo. V| Medium.com, 2019). The NPCs are very complicated and believable; they have daily routines and interactions with environment around them. They react and remember player’s responses for example: if the player insults them continuously or tries to shoot or kill them, they remember the interaction and react accordingly when they see the player again.

The NPCs also have a complex ragdoll system where each part of their body has a realistic movement to environment around when in free motion or unconscious. They move and react realistically when the player bumps into them falling or interacting naturally. Each part of their body has its own health and damage reaction when shot, moving dynamically to that action.

This made the game deeply immersive almost like a living world. This project explores a particular part of this complex movement.

2.7.7 The Last of Us Part 2



Figure 2. 8 The Last of Us Part II Gameplay from Steam Page

The Last of Us Part 2 (Naughty Dog, 2020) is a sequel to the very popular game The Last of Us, which also has a famous TV adaptation. It is a semi-open world and story game that focuses on combat with humans and zombies.

The NPCs in this game are highly intelligent from the way they act and react with each other and the players to how they move and react when they are shot, stabbed etc. The body physics in this game is groundbreaking. Since the game is a combat-based game, the developers have spent a lot of time focusing on both player and NPC movement making this game more realistic. The NPCs react very naturally and human-like in combat scenarios. They hop, or can't control their limbs when broken etc.

This was one of the biggest advances in NPC realism in screen games, featuring both realistic and immersive visuals with great combat situations. These features are more valuable in Virtual Reality, giving the player the ultimate sense of immersion.

2.8 The Science of Believability in Virtual Characters

The importance of responsive NPC is also supported by academic research in human-computer interaction and virtual reality. Several studies provide evidence as to how this benefits the user.

Nonverbal cues between humans have always been an important aspect of communication. Reacting to a gentle touch or a shove through gestures or bodily reactions is realistic and human. This is backed by Nonverbal communication in virtual reality (Aburumman, 2022) where different head nods were used to get feedback from the participants in a controlled virtual environment, the feedback showed a positive impact for naturalistic nodding. It showed that virtual characters that were more responsive were more trusted and realistic, establishing a social presence in VR. Current VR games lack this feature, breaking the immersion and reminding the player they are in virtual reality.

Further supporting this, a study Artificial Intelligence and Interactive Digital Entertainment conference, Finite State Machines for Building Believable NPC in Game (K Fathoni et al, 2020) explores believable virtual characters highlighting believability happens when the characters are consistent and contextually appropriate. Also, that characters need to feel unscripted. This project uses Finite State Machines and Procedural animation to overcome this issue by reacting to the player's responses in a consistent manner while allowing the reactions to be dynamic.

Moreover, a critical concept in character design is Uncanny Valley (M Mori, 1970), Mori suggested that as a robot or virtual character becomes more human-like our emotional response becomes increasingly positive till a particular point. Human minds have an incredible sense of detection for anything unnatural so any subtle imperfections to the virtual character can make it eerie. This is something that has to be considered when creating a realistic virtual character. Making sure to not enter Uncanny Valley and lose player immersion.

Finally, the practical implementation of these systems is still an area that is being explored. A paper on The OPPORTUNITY Framework (Kunze et al., 2014) talks about the use of on-body sensing to recognize human activities and interactions. While this project uses VR headset and controller for input rather than complex body sensors, they still have the same principle at its core i.e., interpreting human activity through input and receiving appropriate

responses. This is done in this project using game engine's collision and physics for touch-based interactions.

2.9 Measuring Immersion: Immersion Experience Questionnaire

The concept of immersion has been studied extensively for gaming and virtual reality, but it remains complex requiring careful measurement. Immersion is a progression through stages of engagement, from investing time and effort to engrossment where emotions are directly affected (Brown and Cairns, 2004). To help measure immersion in digital games, Jennett et al. (2008) developed the Immersive Experience Questionnaire (IEQ) as a validated instrument.

IEQ measures immersion through various dimensions, dissociation of time, reducing focus on external environment. Emotional connection with the virtual world. The presence of the user in the virtual environment. This questionnaire is widely adopted in many studies related to gaming, showing its importance in VR environments.

2.10 An Innovative Interaction System

This project goal is to create a realistic and dynamic non-playable character system that includes the strong aspects of the other games stated above, taking them a step further by combining the interactions that made the games successful and immersive. Also, adding other subtle interactions to the system gives an artificial perception to the NPC. Though each game has its own way of interacting with NPC this project will strive to cover most of the foundations interactions and will serve as a modifiable template for a designer or a developer, making this system versatile. This project is a prototype combining the different interactions and will be a huge step forward in NPC interactions.

The author's idea is to create a NPC that can react and respond to any type of unpredictable behavior from the player while not breaking immersion. A system that detects and responds to both non-combat touch and combat touch, giving each NPC a behavior and a personality. Further, by using procedural and animated responses to any kind of touch response which supports any animation for example: the same system generates a realistic procedural animation when the NPC is in idle state or in a box stance and is punched giving the future developer full freedom, moving beyond simple animations or ragdoll reaction.

3 Chapter 3 – Methods:

This chapter details on how everything was designed, developed and evaluated. It explains the overall approach behind the prototype. Unreal engine is a complex software with a steep learning curve, so some concepts have been overlooked focusing only on the core parts of the project.

3.1 Development Environment

The development methods for this project were chosen based on the author's experience and the requirements of the dissertation. While many devices and frameworks are available for this prototype and testing, Unreal Engine 5 and Meta Quest 3 were used due to their flexibility and developer freedom to create and run applications, and the author's has worked with these for different projects. Meta Quest 3 (2023) is a Virtual Reality Headset with motion controllers developed by Reality Labs, a Meta Platforms company. Meta Quest 3 can be used as both a standalone device with its own hardware that can run VR apps or as a tethered device to stream PC games or apps that are VR supported.

Unreal Engine was chosen for the game development software explained later in this document. The compatibility with Meta Quest 3 is seamless and helps with rapid prototyping and testing.

The hardware development environment for this project was Windows 10, Unreal Engine 5, C++, Laptop (Intel 11th Gen processor, 32GB RAM, NVIDIA GeForce RTX 3060 – 6GB VRAM), Meta Quest 3.

3.2 Unreal Engine Framework

This project uses Unreal Engine because of the author's prior experience with the game engine and the freedom it provides when working with complex game mechanics. The game engine offers many built in features that heavily benefit the project such as VR rendering, base input handling, animation tools, collision and physics. Unreal Engine is developed by Epic Games as free for all game development software that includes all base tools necessary to make the development process easy.

Unreal Engine is an industry standard software written in C++, Assembly used by many game studios as part of their development process. It is flexible and is used by all fields of game development from artists, designers and even architects to game developers. The software covers many different fields Augmented Reality (AR), Virtual Reality (VR), 3D and 2D games, and computer simulation. Unreal Engine 1 was developed in 1995 by Epic Games founder Tim Sweeney. Unreal went through many upgrades releasing new versions of the engine on a regular basis (Unreal Engine 2,3, and 4). Unreal Engine 4 (2014) released and made it free to download with royalties on game revenue. It also introduced Blueprints, a visual scripting node-based language that hugely helped developers with prototyping, this is an upgraded version of a previous visual scripting language called Kismet released in Unreal Engine 3.

3.3 Unreal Engine 5

Unreal Engine 5 (UE5) is the current version of Unreal Engine. After Unreal Engine 4 the company rewrote many parts of the engine to adapt for their future plans and technological optimizations, introducing many groundbreaking concepts and further venturing into VR, AR development.

Blueprints in UE5 which was introduced in UE4 were optimized and included many options making it easier for artists, designers or even developers to prototype or test game mechanics. Blueprints nodes are C++ snippet codes that can be joined with each other and create logic without having to work with C++.

3.3.1 Unreal Engine Editors

Unreal Engine has different editors that make the development process quick and easy.

3.3.1.1 Level Editor

This is main editor where you build gameplay levels, defining the levels for the game with 3D models, blueprints, and game actors.

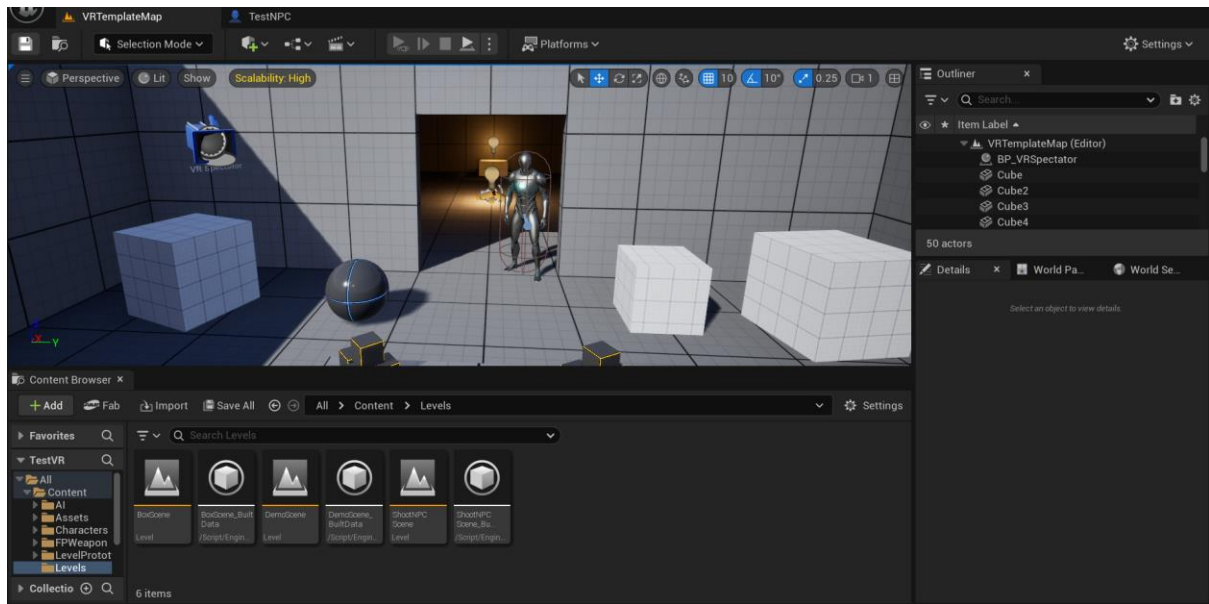


Figure 3. 1 Unreal Level Editor

3.3.1.2 Blueprint Editor

The visual script language Blueprints can be worked on and modified in this editor. You can create game mechanics or any game elements that function as a replacement for C++. It has 3 different window editors.

Viewport to incorporate or align static meshes, collision components, cameras, etc.

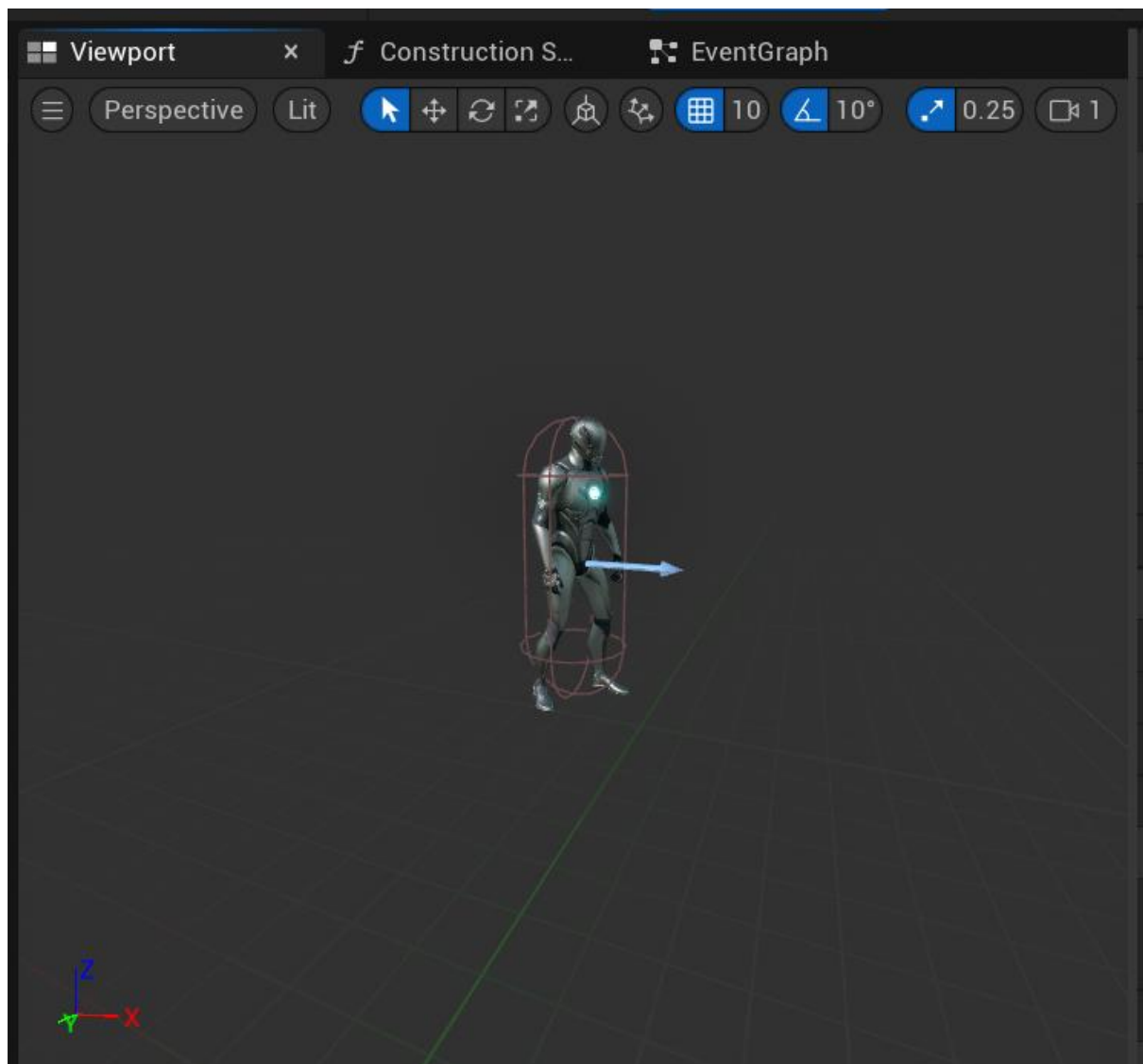


Figure 3. 2 Blueprint Viewport Editor

Event Graph is the main editor to program logic and connect blueprint nodes to construct behavior of the game actor. Blueprints have many in-built functions with underlying C++ code.

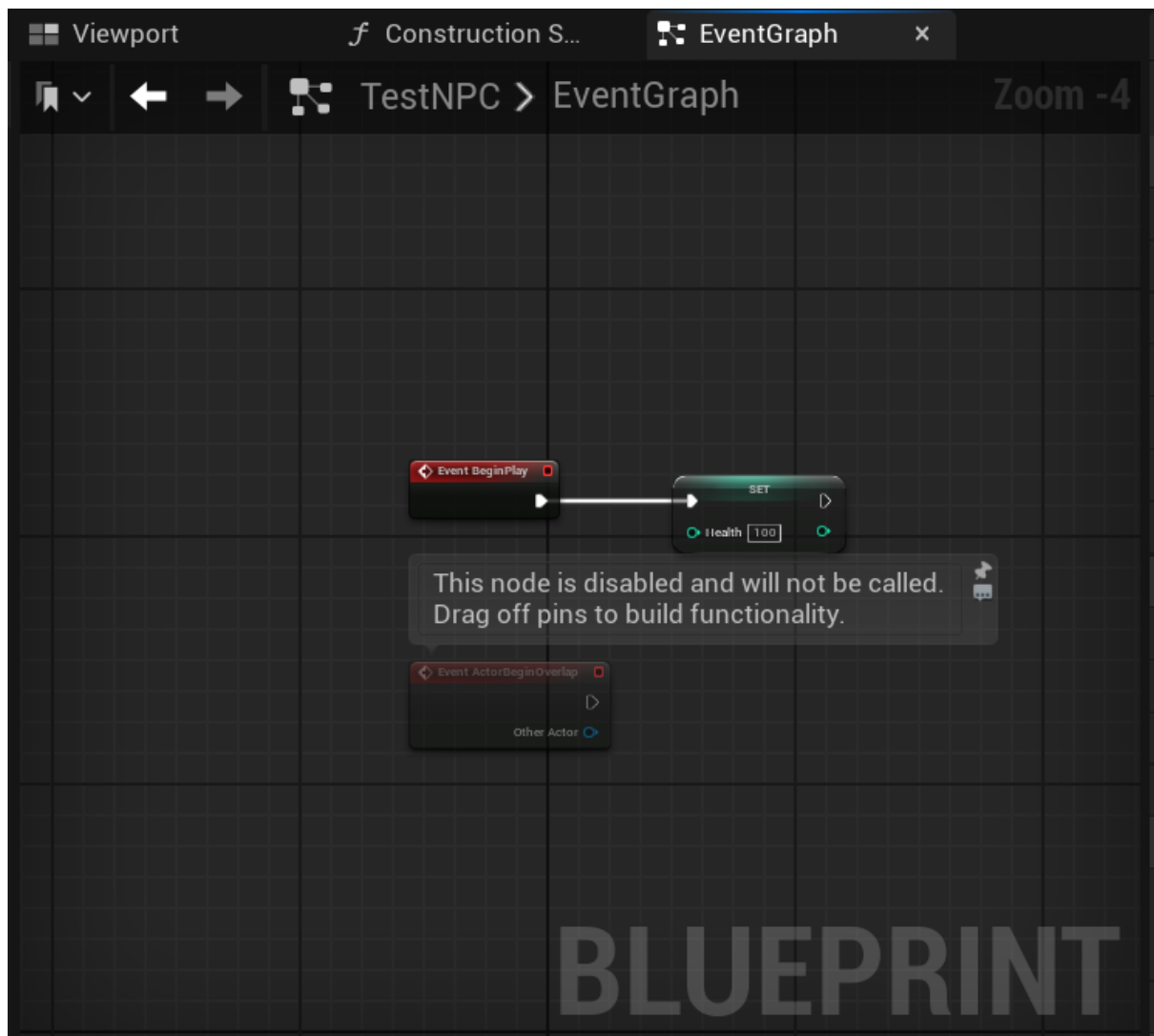


Figure 3. 3 Blueprint Event Graph Editor

Construction Script is used to modify appearance of the game actor and it can be used for other options like procedural design.

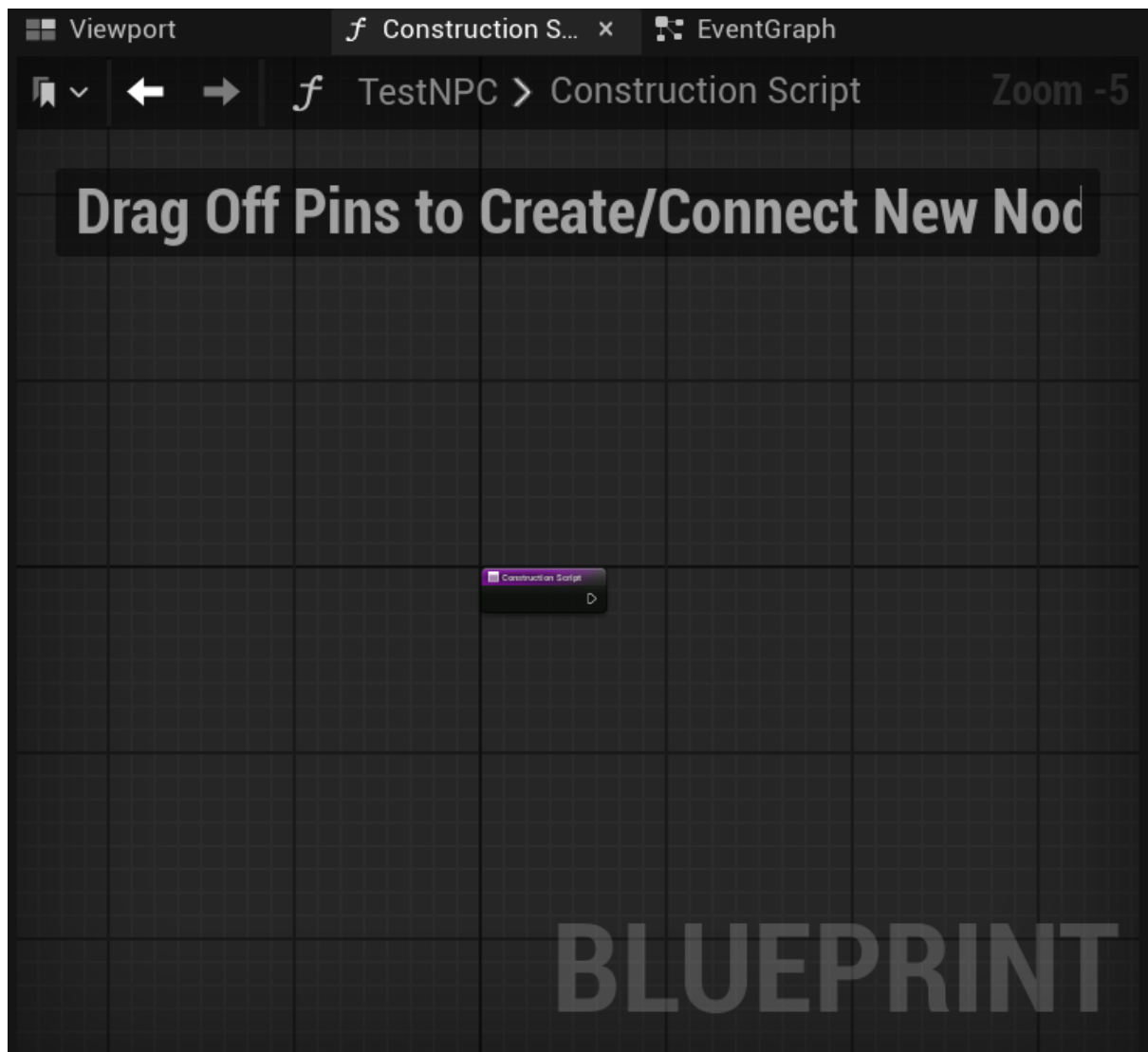


Figure 3. 4 Blueprint Construction Script Editor

3.3.1.3 Physics Asset Editor

This editor is used to create Physics assets for Skeletal Meshes. You can implement features like collisions, ragdoll for skeletons for human like bone structure and weights.



Figure 3. 5 Physics Asset Editor

3.3.1.4 AI Behavior Tree Editor

A Behavior Tree (BT) editor is similar to Blueprints node-based editor. This can be used for behavior of NPCs such as movement, attacking or defending or patrolling. The Behavior tree has a Blackboard and an AI Controller that are used as a pair with Behavior tree to control the NPC. Blackboard is a list of variables like Booleans, integers, floats, vectors or other program variables acting as a bridge between the AI Controller and BT. AI Controller is used to get values of the NPC or the world and assigned to the Blackboard. BT can be used to create services or tasks or assign conditions to states or behaviors. These sections are explained in detail later.

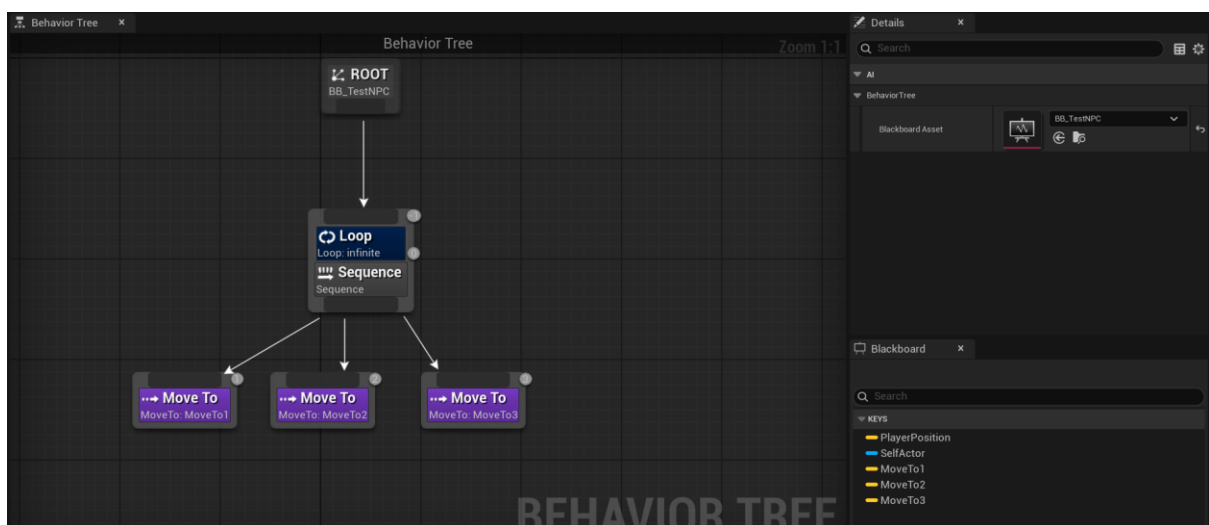


Figure 3. 6 AI Behavior Tree Editor

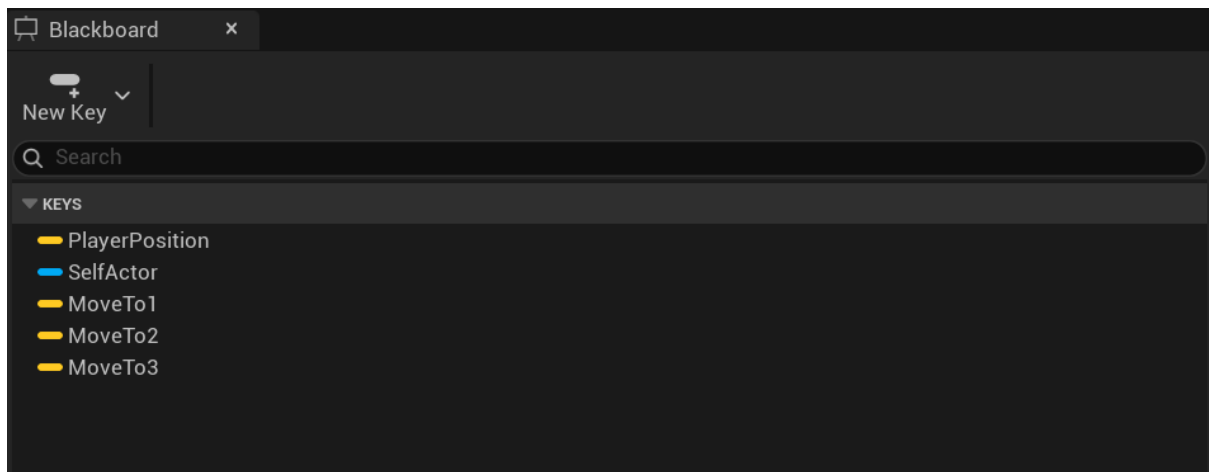


Figure 3. 7 AI BlackBoard Editor

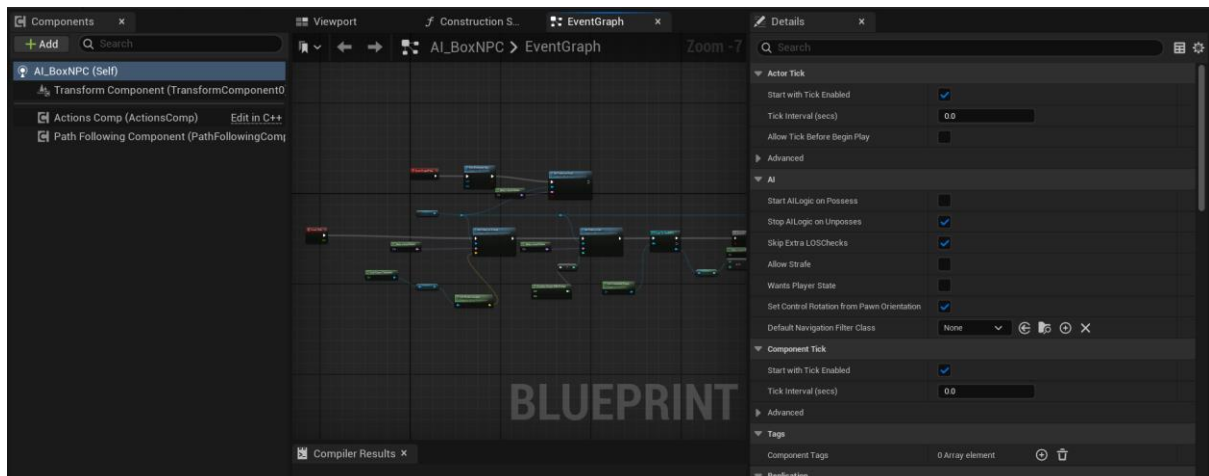


Figure 3. 8 AI Controller Blueprint Editor

3.3.1.5 Animation, Control Rig Editor

Animation editor is used for editing animation assets for skeletal meshes or animation blueprints.

Control Rig editor is an individual editor for a skeletal mesh can be utilized to animate characters or use animations imported externally.

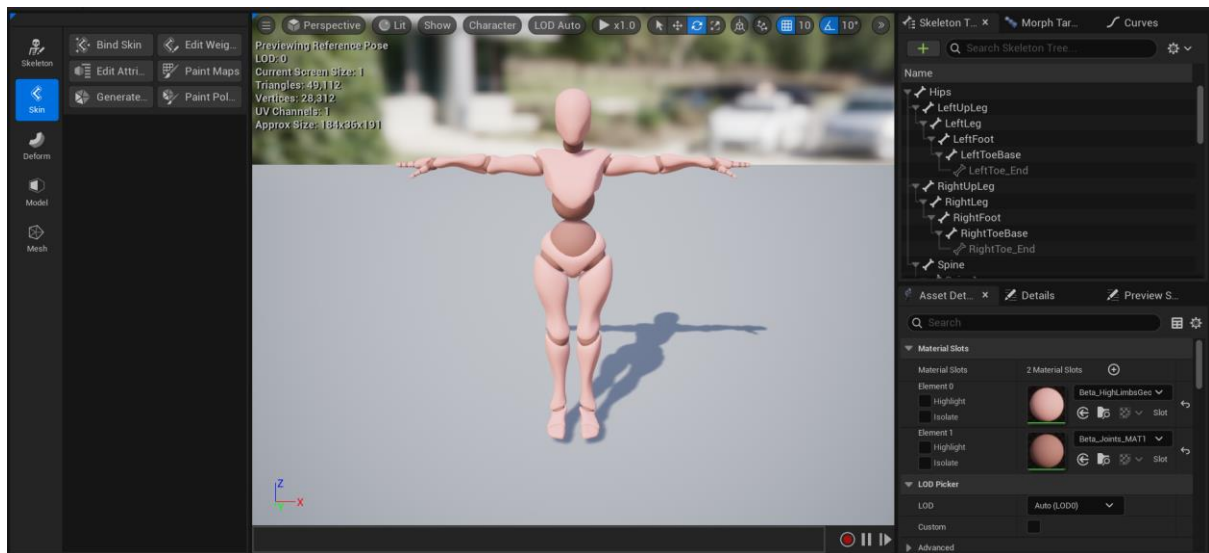


Figure 3. 9 Control Rig Editor

3.4 Mixamo

Mixamo is a web-based service which started off in Stanford University which was then later acquired by Adobe System in 2015. The website has free 3D characters and 3D animations.

Mixamo 3D characters are a collection of 3D models each with unique designs that can be downloaded and used in any game engine or 3D software tool.

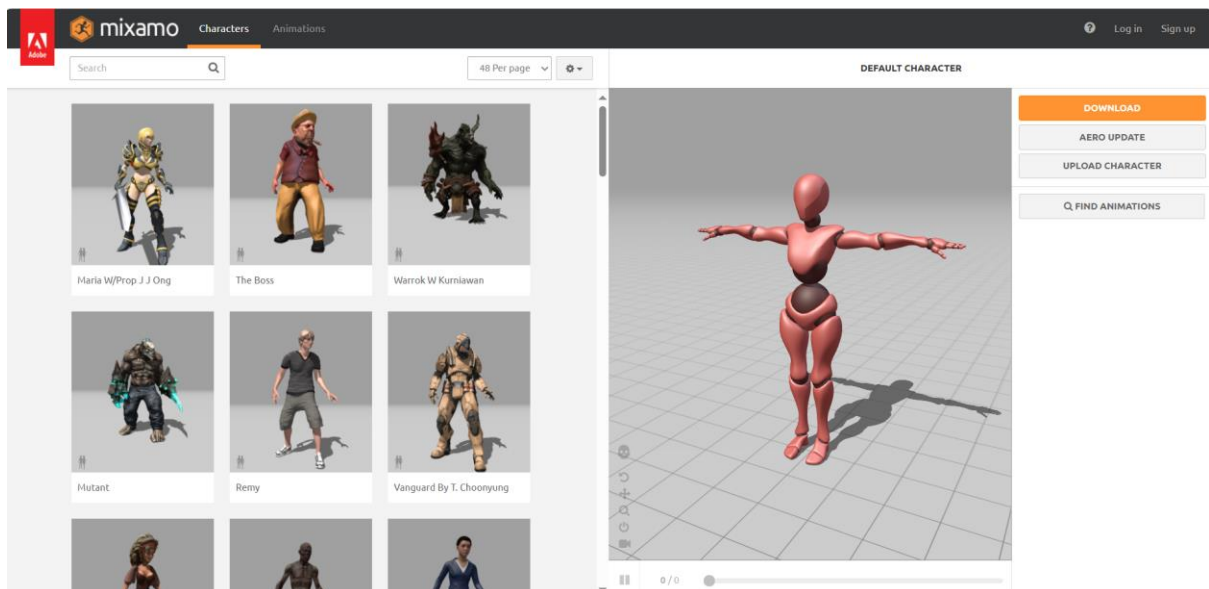


Figure 3. 10 Mixamo Characters Page

It also has a huge suite of 3D animations for all genres of games such as Combat, Adventure, Sport, Dance, Fantasy and Superhero. These animations are very useful to indie or mid-tier studio most used animations include idle, walking, running, or jumping etc.

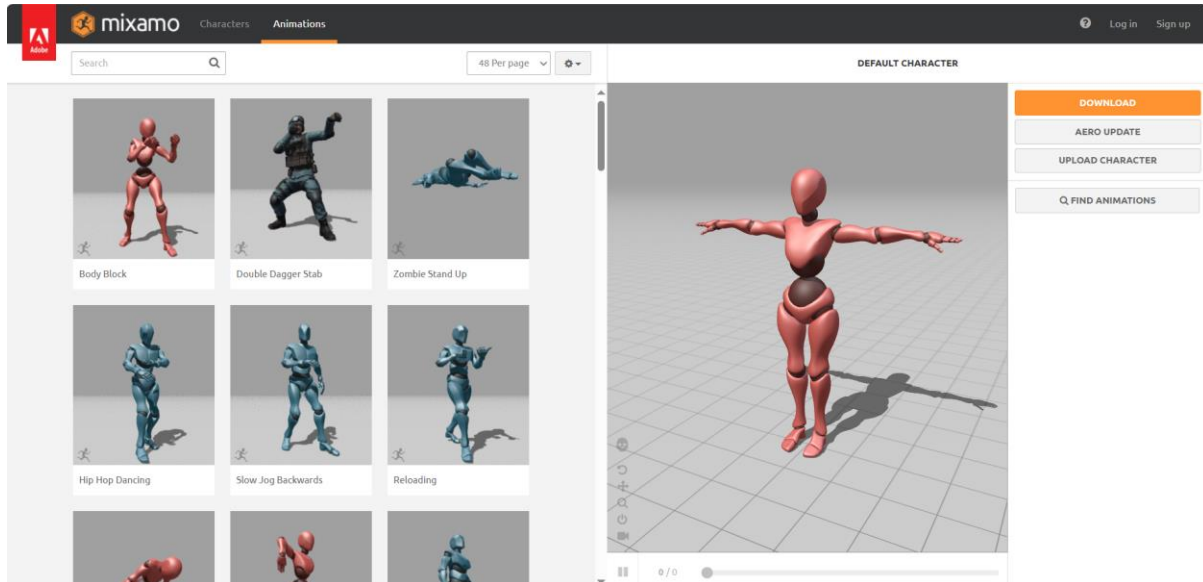


Figure 3. 11 Mixamo Animation Page

3.5 Prototype Implementation

The prototype works on a complex and multi layered logic hence Blueprints was chosen over C++ for easy prototyping and testing. Allowing easy compiling and error fixes through console. Blueprints are useful for prototyping but not main implementations as blueprints are slower than C++ “Since Blueprints add a layer of abstraction on top of Unreal Engine’s core functionality, they tend to run slower compared to code written directly in C++” (codeinfinity.com – Andrii Chornyl, 2024). Since this project is not the final implementation that will be shipped to conserve time and debug things quicker Blueprints were chosen. They can be seamlessly tested and implemented without a problem with this project.

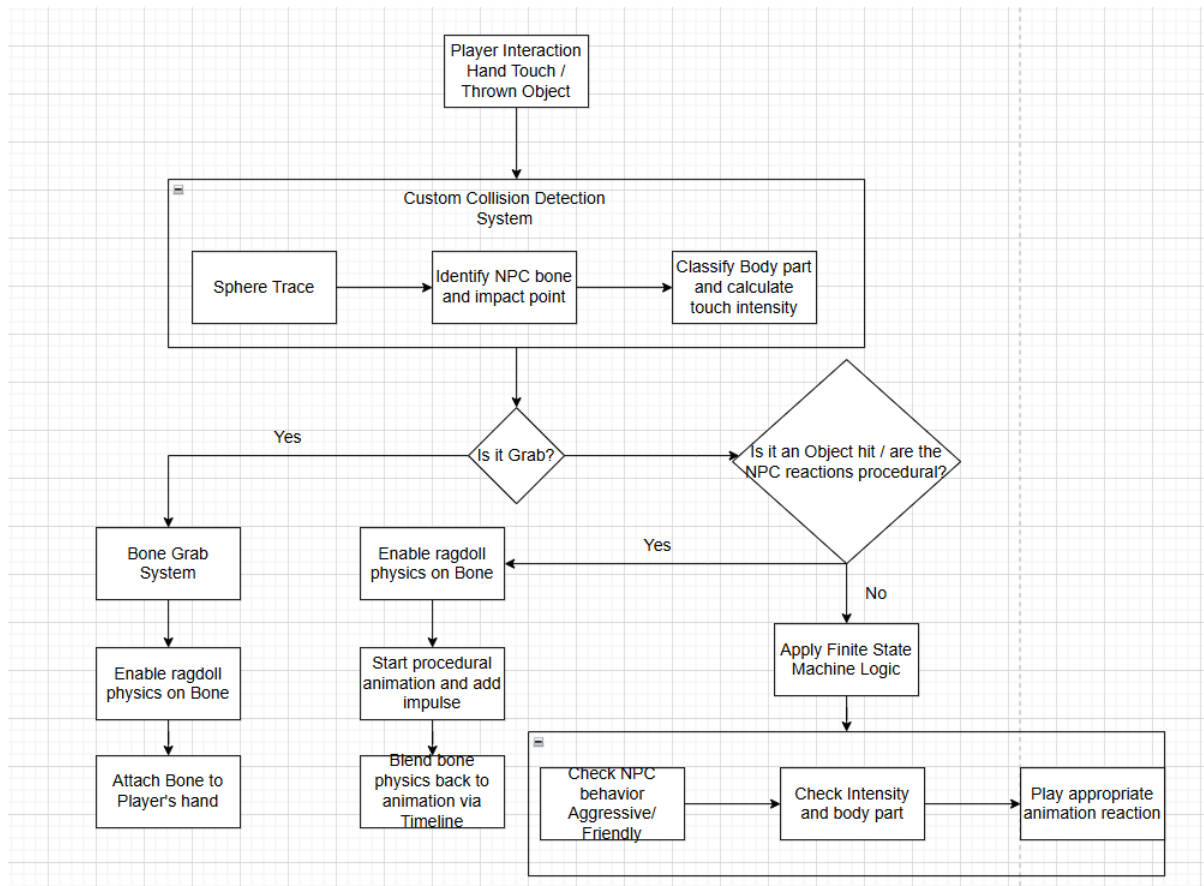


Figure 3. 12 Flowchart Explaining Implementation Logic

This flowchart briefly demonstrates how the NPC system works, the actual implementation was implemented with many levels connecting different Blueprints effectively to run this complex NPC system. This flowchart is for reference and easy understanding of how the overall logic and flow of actions.

3.5.1 Input - Hand-Held Motion Controller

These are the controls explaining what buttons or triggers were used as input for controlling motion.

Quest Touch Controllers

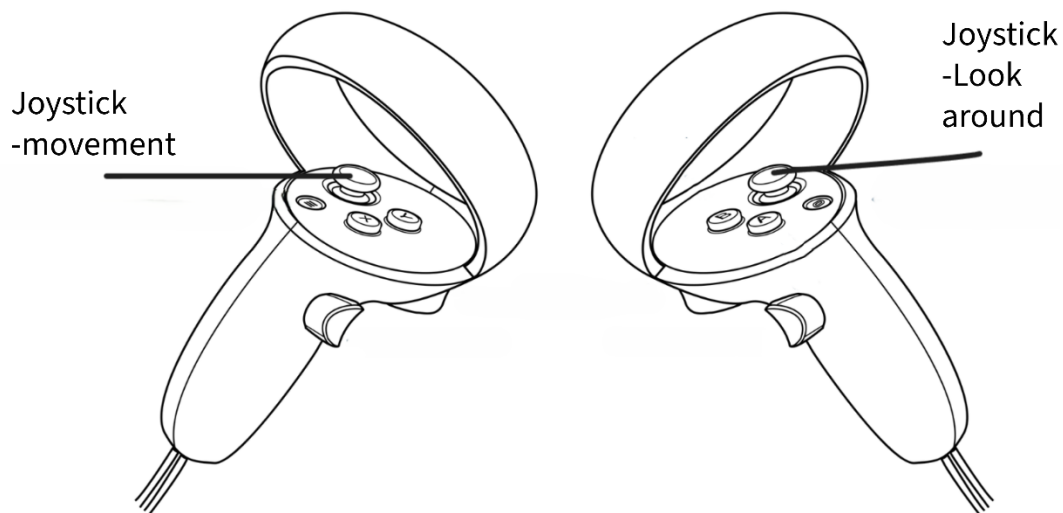


Figure 3. 13 Quest Touch Controller - Joystick

Quest Touch Controllers

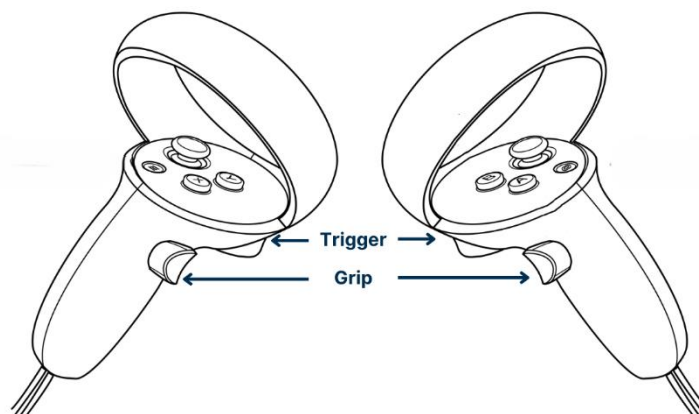


Figure 3. 14 Quest Touch Controller - Triggers

The Grip trigger was used to grab objects in the virtual world. The trigger button was used to shoot through the gun (used in prototype).

3.5.2 Basic VR Character

Unreal Engine has a basic VR Template that is available for free and can be used by developers to fast-track development, giving a good starting point. Hand animation is built in with template making the prototyping faster.

The VR player blueprint is called VR Pawn.

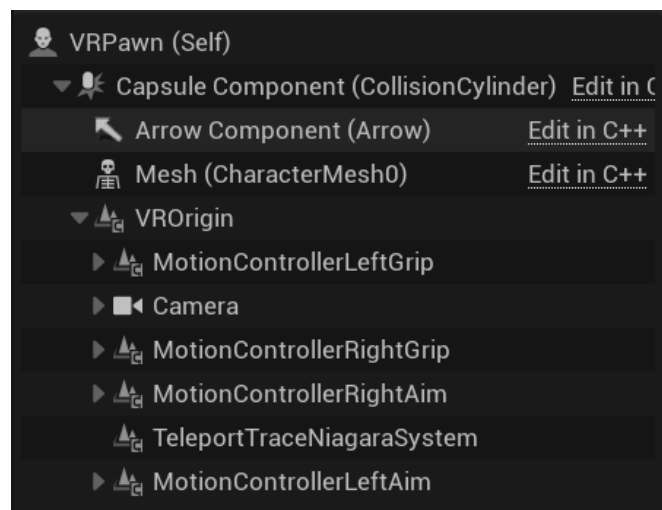


Figure 3. 15 VR Pawn Blueprint Components

These are the important components of VR Pawn Blueprint. The main components are the parent component – Capsule Component, Mesh which includes hand 3D meshes, Camera which is controlled by the VR headset and what player sees and Motion Controller Grips for Left and Right.

There are two types of player movement in VR:

Teleport and Snap Turn: The player can teleport by pointing with handheld motion controller and rotate 90 degrees with right joystick.

Continuous movement: This was used in the prototype because it is more immersive and similar to real world. Moving the joystick forward and backward will let you move the same way in VR. You can strafe left and right by moving the joystick left and right respectively. Right joystick can move left and right to rotate the same way in a smooth motion.

Continuous movement was built with Blueprints logic. 'Enhanced Input Action Locomotion' blueprint node gives coordinates of left controller's joystick, when triggered the player's

blueprint is moved forward or right with ‘Add Movement Input’ function, scaling the movement speed depending on the joystick rotation. For Smooth turning when the right controller joystick is triggered the player’s blueprint is rotated in the world on Z axis.

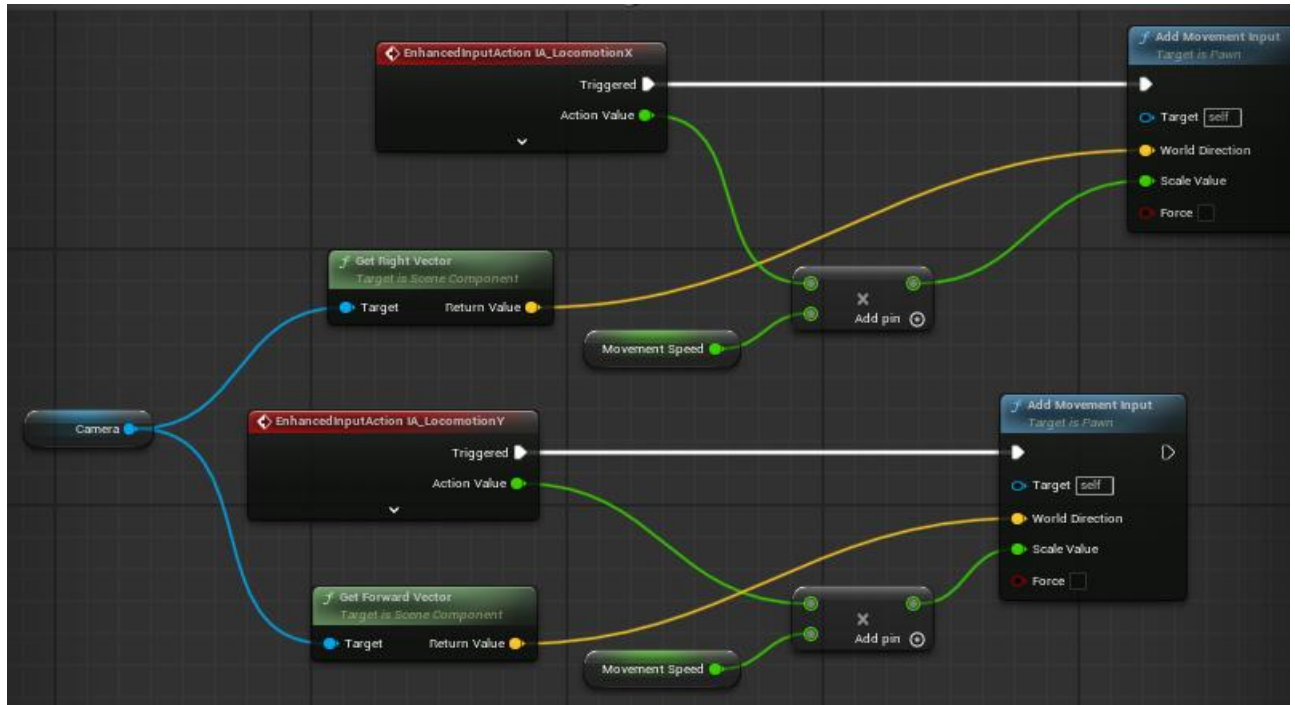


Figure 3. 16 Continuous Movement Blueprint Nodes

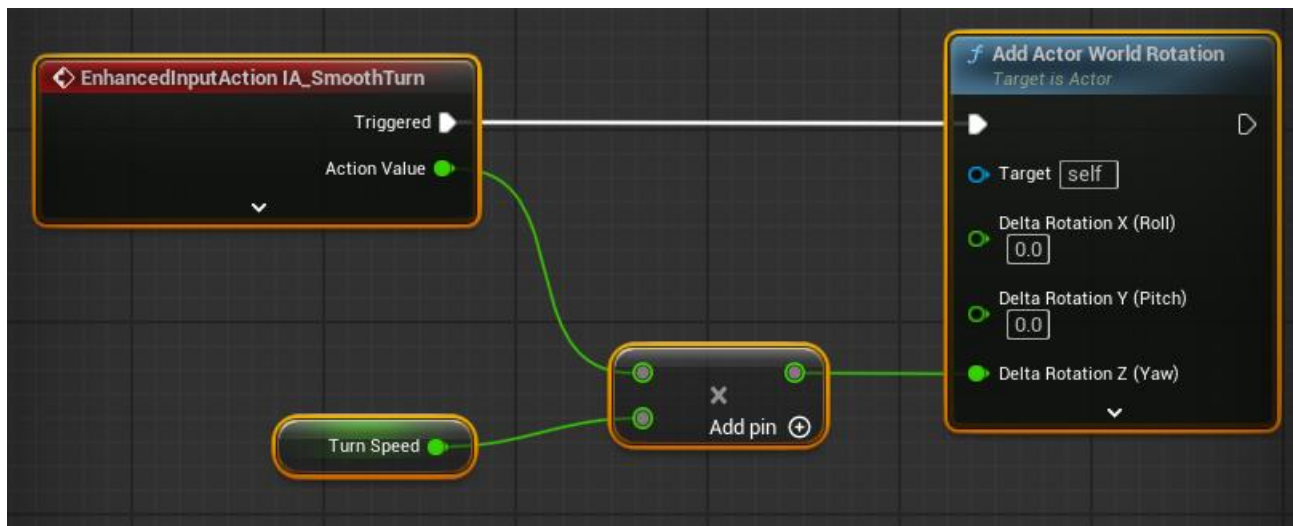


Figure 3. 17 Smooth Turn Blueprint Nodes

3.5.3 Basic Non-Playable Character AI

For designing the base NPC a third person template provided by UE5 which includes a mannequin blueprint with in-built basic animations was used. This mannequin blueprint is for on screen 3rd person games with camera overhead the player's shoulder. The blueprint was modified to fit the needs of this project by removing settings that make the blueprint a player like being able to control it with the keyboard of the laptop and modifying animation, movement setting to convert it into a non-playable character.

An empty Behavior tree was created and paired with a 'Blackboard', to access values. An 'AI controller' Blueprint class was created which was assigned to the NPC to run the behavior tree.

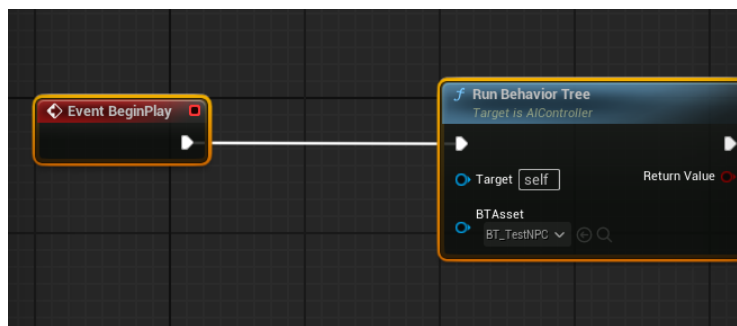


Figure 3. 18 AI Controller – Run Behavior Tree Nodes

3.5.4 Implementing new collision system

This project required a custom collision system to identify the collision between the player's hands and the NPC body. The collision system should be able to identify what part of the body the player is touching and how much force is being used for the touch. This system has been divided into 3 main functions. (For the prototype just, the right hand was used since this is a tech demo)

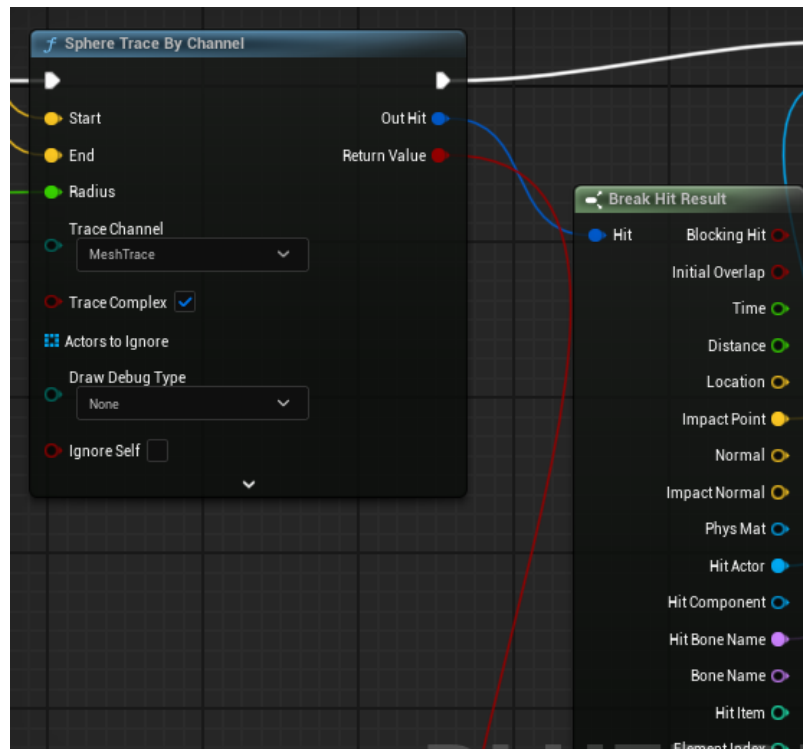


Figure 3. 19 Custom Collision System – Sphere Trace Blueprint Nodes

First *TraceBoneName* function that takes the player’s hand game actor’s location was taken and a 3D ‘Sphere Trace’ (The game engine checks within a specified radius around the given point for actors that are in the sphere) was implemented. The hit result was used to get NPC; it’s bone name and impact point assigned to global variables of the blueprints. If the NPC is valid then a custom event called *Hitdetected* is implemented to the specific NPC.

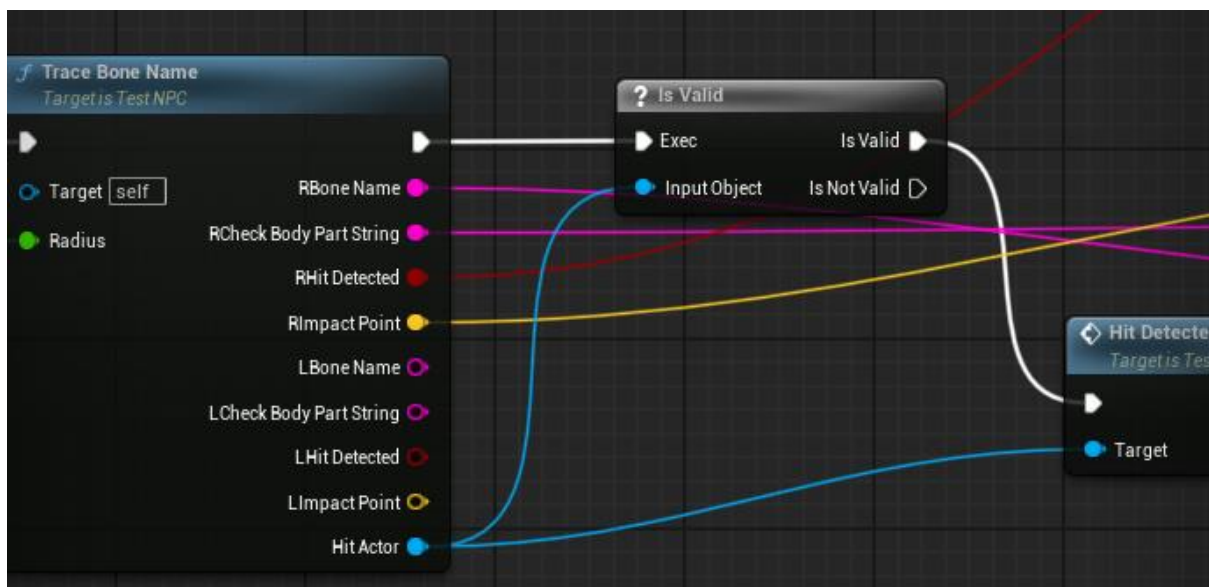


Figure 3. 20 Custom Collision System – Trace Bone System, Hit Detected Nodes

The second main function called *CheckBodyPart* takes the bone name from *TraceBoneName* function and classifies it into one of the 5 categories: Spline, Arm, Leg, Head, None. This is assigned to a global variable called 'BodyPart'.

Finally, the third main function is *TouchIntensity* which further classifies the touch to either high or low assigning it to a 'TouchClass' global variable (These global variables play a big role later on). To calculate the intensity of the player's hand, we get the player's blueprint and cast it into 'VRPawn' blueprint then the Right motion controller's linear velocity vector length is obtained then multiplied to 5 to get a normalized value. Then it is classified into a high intensity touch or a low intensity touch with a specific range assigned by the developer (in this case it is 0-1000=Low and 1001-10000=High).

These are the base functions that are executed before performing any other functions. After this a Finite State Machine is constructed using these new collision values.

3.5.5 Finite State Machine with custom collision system

A Finite State Machine is a mathematical computation model. It functions as a machine that can stay in one of the finite states, transitioning from state to state with specific rules called transition rules. This is used for animations in Unreal engine with each animation being constituted as one state. Custom animation transition rules are assigned to move between states.

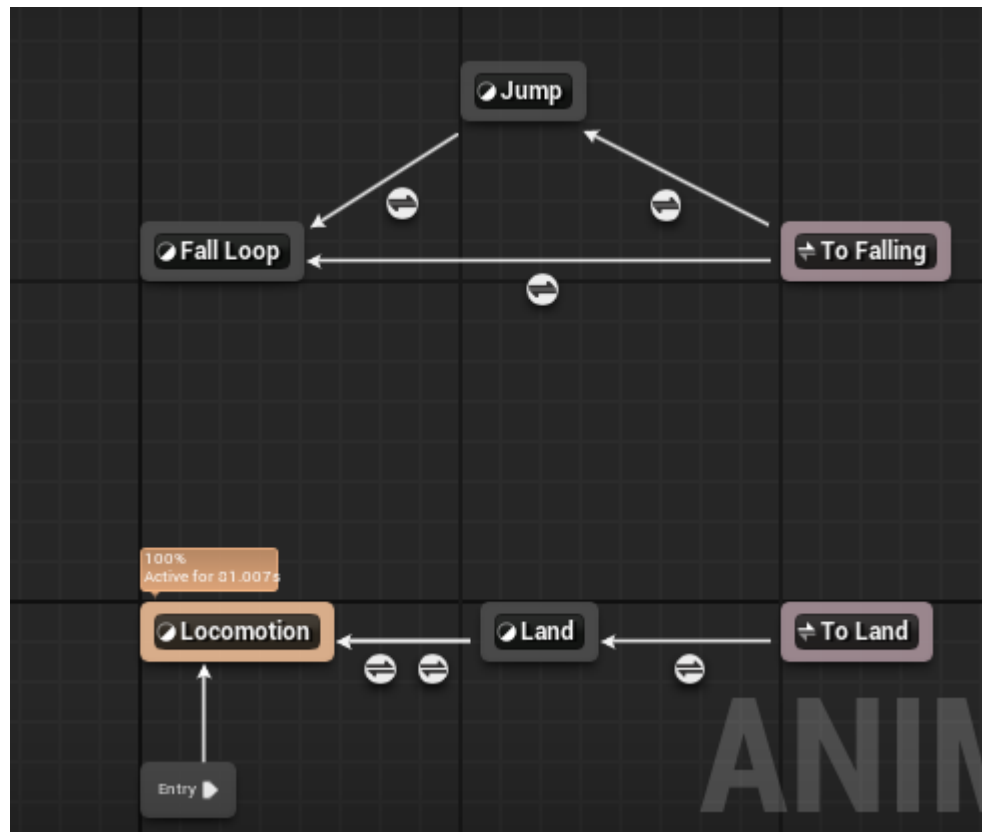


Figure 3. 21 Finite State Machine Editor

The template NPC comes with a basic animation finite state machine. Locomotion state machine handles normal transitions like idle, walk/run. Jump, Fall Loop are states that handle jump and falling condition. Land state is activated just before reaching the ground. ‘To Land’ and ‘To falling’ are called state aliases. They can be called and redirected to the state that has been assigned.

Locomotion is modified for this project to create a finite state machine capable of triggering a different animation for every type of interaction between the player and the NPC for example: the reaction when player touches the arm with high intensity is different from the reaction from touching chest with low intensity. There are a total of 2 sets of 8 reactions: two for each of the four body parts. One set for aggressive NPC and another for friendly NPC (This behavior can be set by the developer using a Boolean variable: ‘bAggressive’ which is accessible as a global variable). Here is a flow chart with all different reactions and their result (The flowchart has been divided into two parts for easy readability)

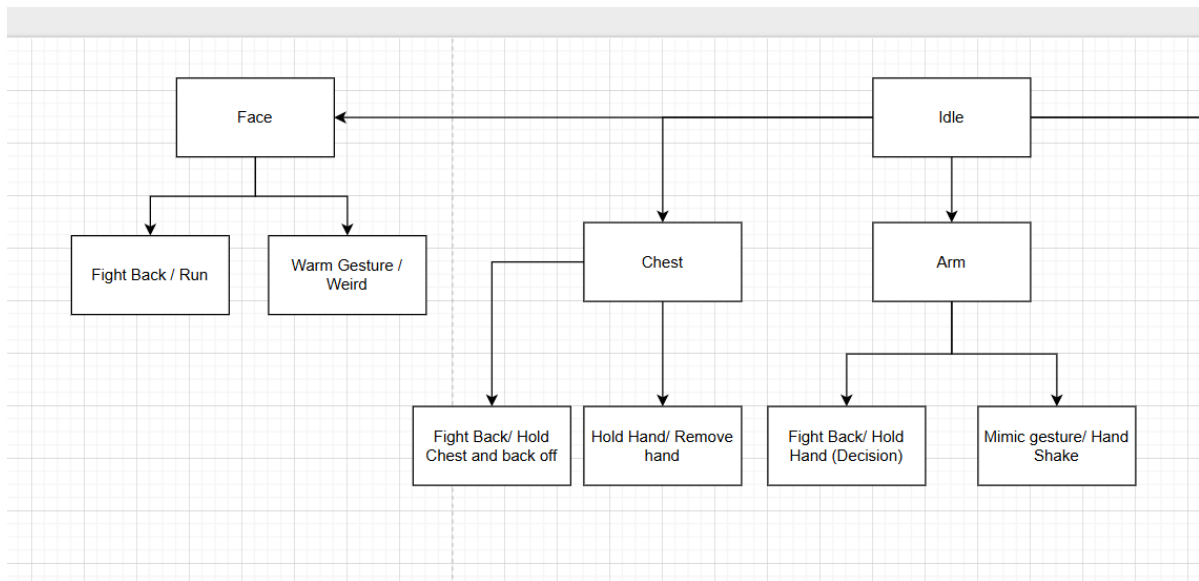


Figure 3. 22 Finite State Machine Interactions Flowchart – Part 1

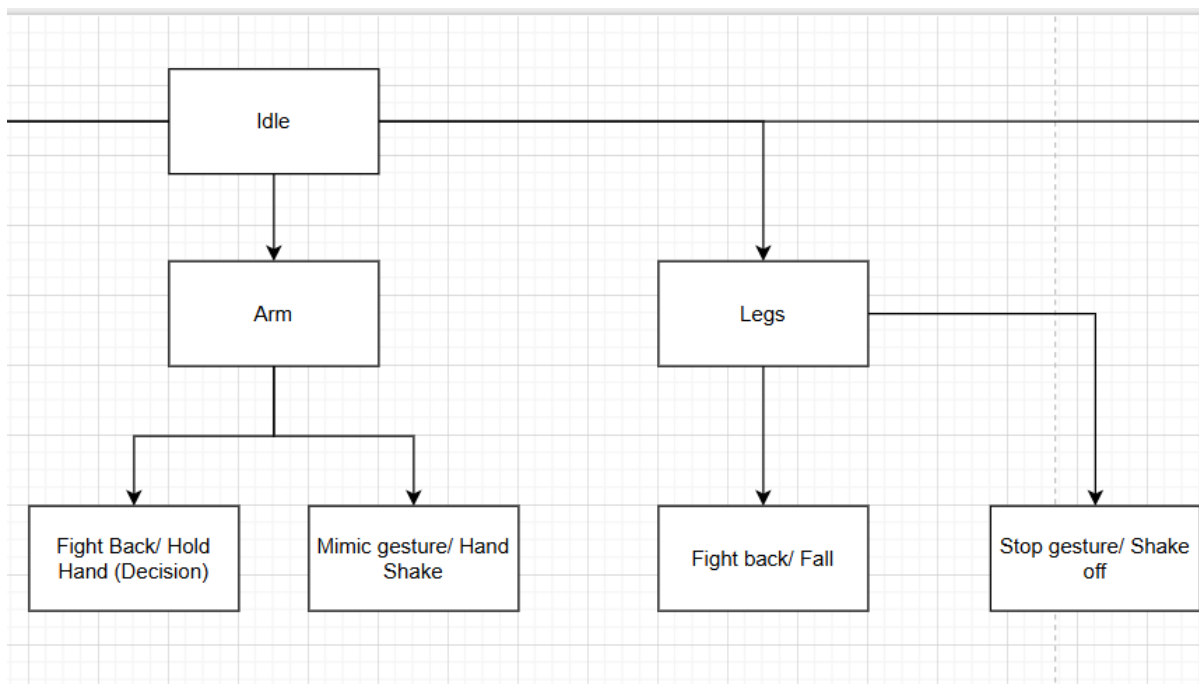


Figure 3. 23 Finite State Machine Interactions Flowchart – Part 2

The in-game behavior finite state machine is activated when ‘bInteract’ Boolean variable (Player having the ability to grab the NPC explained more later) is false. This behavior system is intended for crowd-based NPCs. When the NPC is hit with player’s hand it triggers conduit node (A specialized state that acts as an intermediate point between states) then moving to aggressive or friendly nodes after which it is categorized to one set of touch intensity (high or low) animations then categorized according to the body part. All the states reroute to idle state alias when the animation ends.

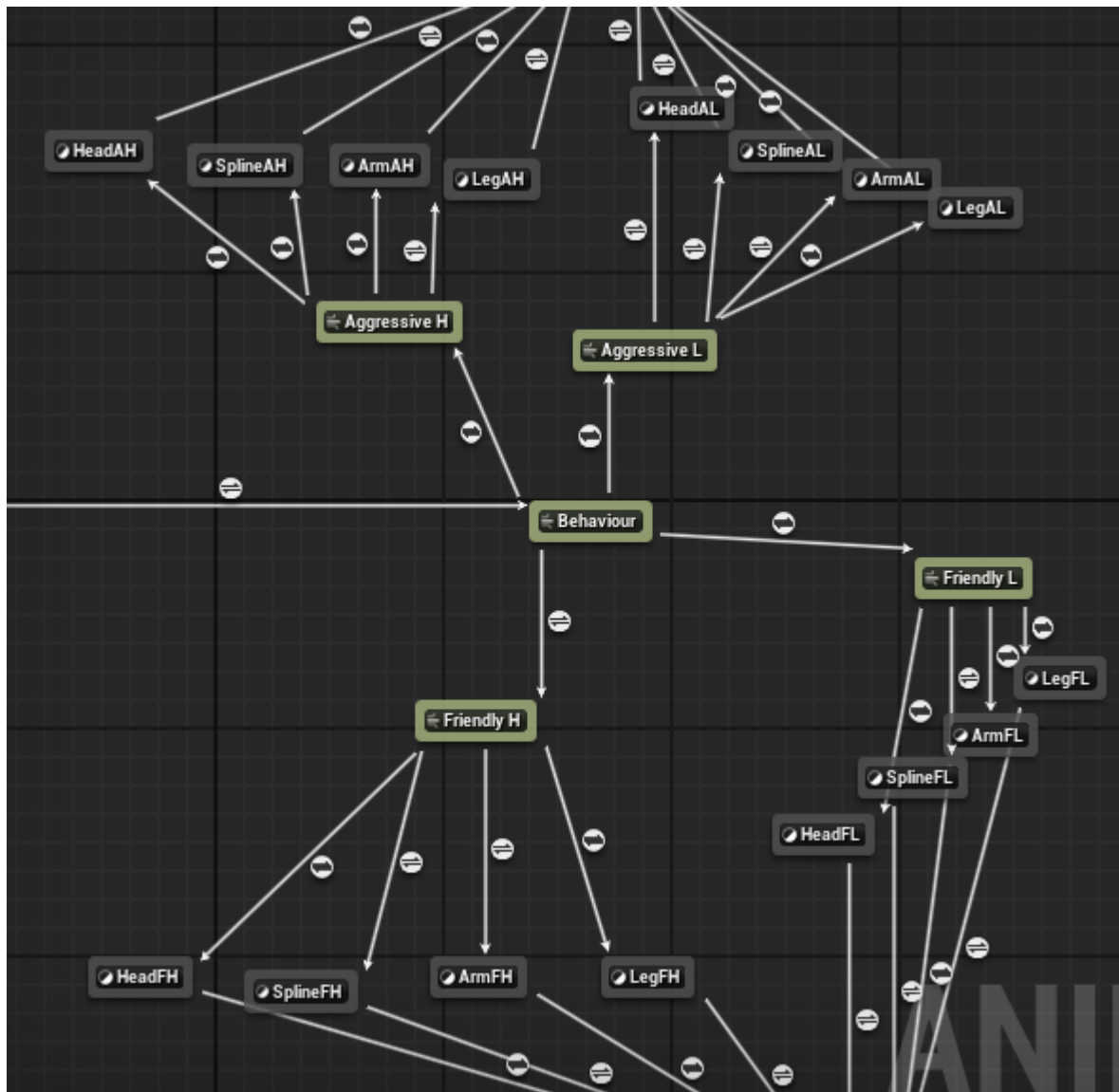


Figure 3.24 Animation Blueprint Editor – FSM for all NPC body parts and intensities

These are set in Animation Blueprint specific to the NPC created. This blueprint also has an event graph with which the values ('bAggressive' Boolean, 'Hit' Boolean, 'body part' and 'touch intensity') are transferred from the NPC to the animation blueprint.

3.5.6 Implementing Object Detection

To take the interactions between the player and the NPC further object detection by the NPC is implemented. When the player throws an object or shoots the NPC, it provides a natural reflex reaction, moving the body limbs. This is done with ragdoll physics.

Ragdoll Physics: "Ragdoll physics is a category of procedural animation that displays human-like figures with more realistic motion." (howstuffworks.com – Nathan Chandler, 2015)

Ragdoll physics are built into game engines as part of game physics.

Specific game objects collision is checked with the NPC to get a controlled reaction. In this project a cube and bullet were chosen to test the system. All game actors of cube and bullet in the level are collected into an array. A For each loop goes through the array of objects and a throwable event is executed.

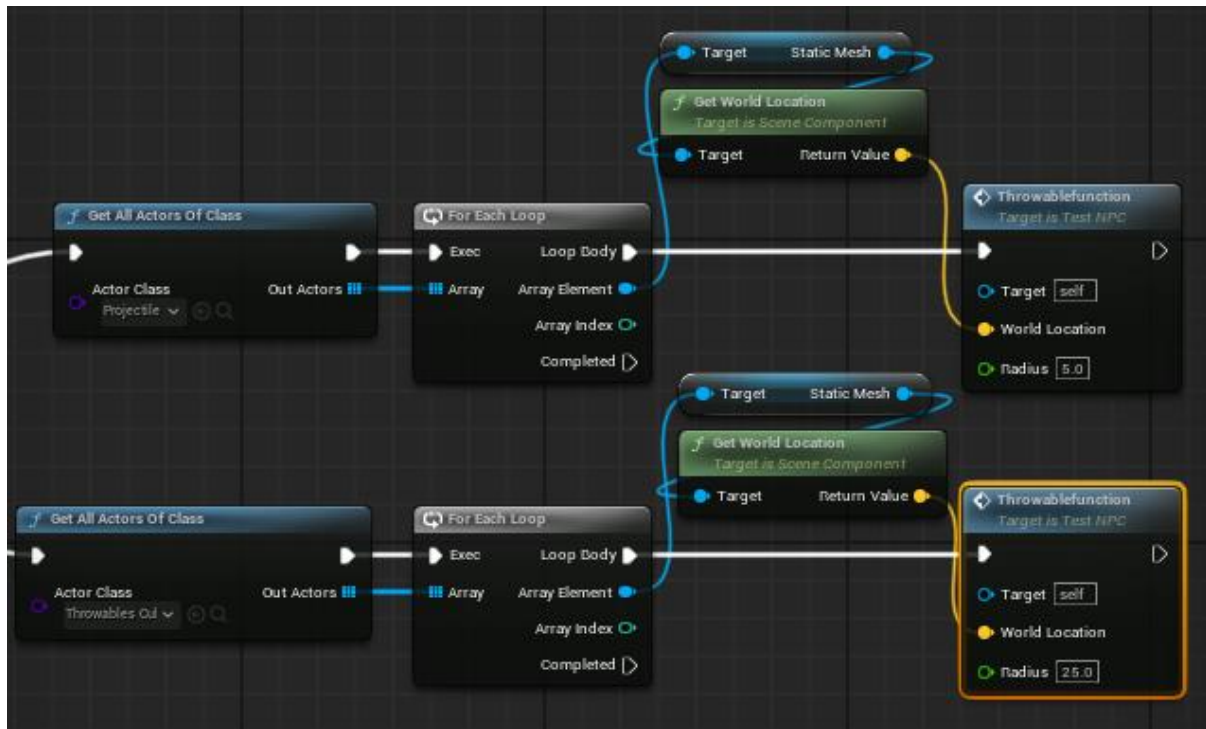


Figure 3.25 Object Hit Detection Blueprint Nodes

Throwable events continuously executes a Sphere trace similar to bone detection following the thrown object, when the object hits the NPC the impact point, bone name variables are set for further implementation of ragdoll physics.

3.5.7 Advanced interaction: Realistic ragdoll physics

To perfectly implement and animate ragdoll body movement the project uses an in-built blueprint component called physical animation. Physical animation is setup using functions available in the engine first it is attached to the mesh component which has the 3D model and the bones. Unreal has an in-built ragdoll physics system that can be simulated using a function call. The functions that are used in this project are 'Set Simulate Physics' (which simulate physics for the whole body) and 'Set Simulate Physics below' (which takes bone name and simulates physics to the bone and the children of the bone). After setting the mesh component's physical animation and executing sphere trace of both hand and the object, assigning both bone name and impact point variables. 'Set Simulate Physics below' is executed inputting the bone name. For example: if the bone name that was hit is right

shoulder, the function enables physics for the right elbow, fingers and other joints following human anatomy and disabling the default animation playing (allowing the bone to move with game physics). Then after a condition check to see if the intensity of touch is high, a ‘Timeline’ is implemented.

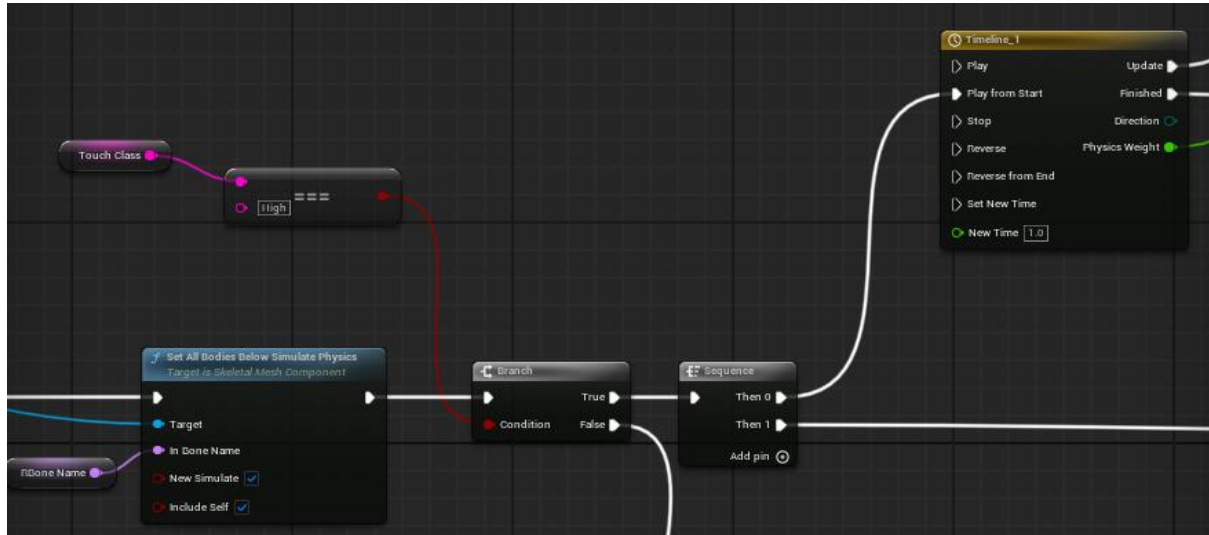


Figure 3. 26 Procedural Animation - Simulating NPC Physics connection to Timeline Nodes

‘Timeline’ takes the physical weights of the bones and smoothens the reaction by following a curve making physics more natural. The graph curve in Timeline guides the bones providing a realistic hit reaction.

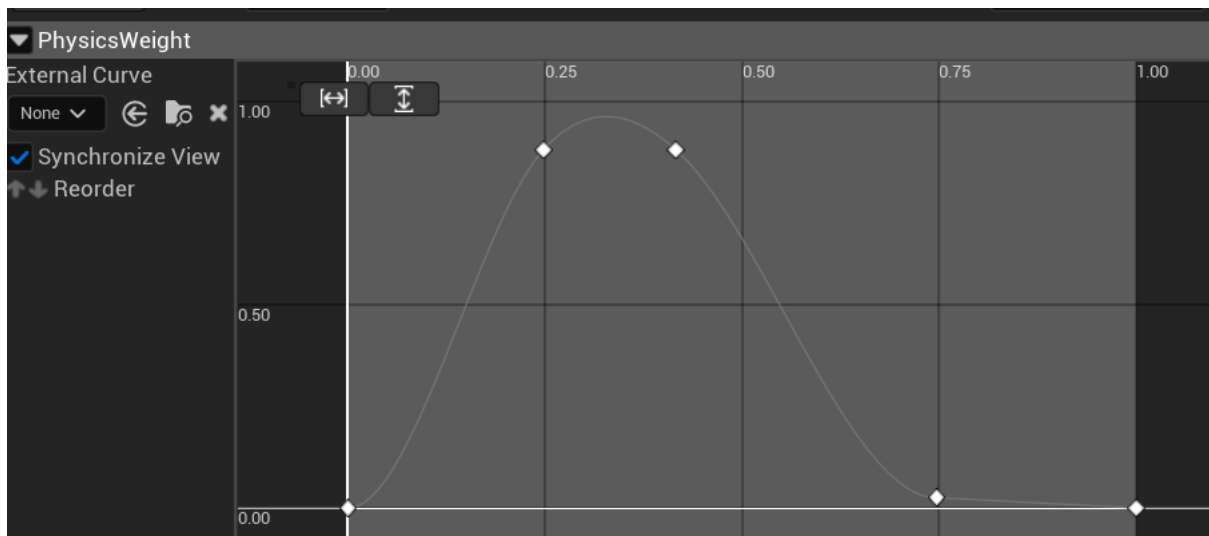


Figure 3. 27 Timeline Editor – Physics Weight Graph

An inbuilt function is used which continuously updates the physics weight of the bone along the graph at the end of the graph the reaction ends and the ragdoll physics are disabled with ‘Set Simulate Physics Below’ function and the bones return to the previous animation. This

controlled approach for physics reaction gives many possibilities and enables different animation reactions that can be tedious if set in a finite state machine.

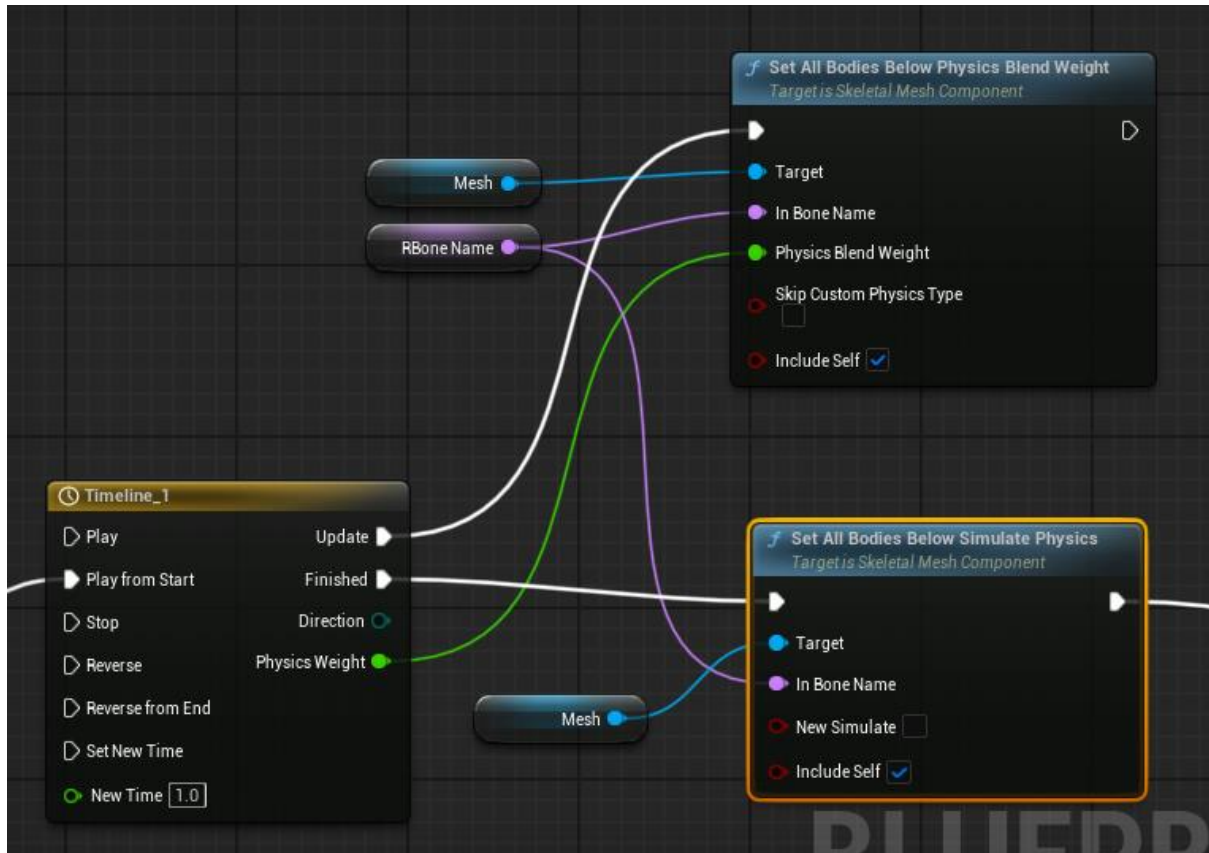


Figure 3. 28 Procedural Animation - Smooth Physics Bones Blend

Immediately after ragdoll physics is enabled an impulse is added to the bone to generate a realistic hit effect. By using the impact point which is normalized and added to a function called ‘Add Impulse’ which takes the impulse amount with direction and bone name as inputs. This gives the bones a stronger reaction guides the touch interaction in the hit direction with a specified force making the motion feel more natural.

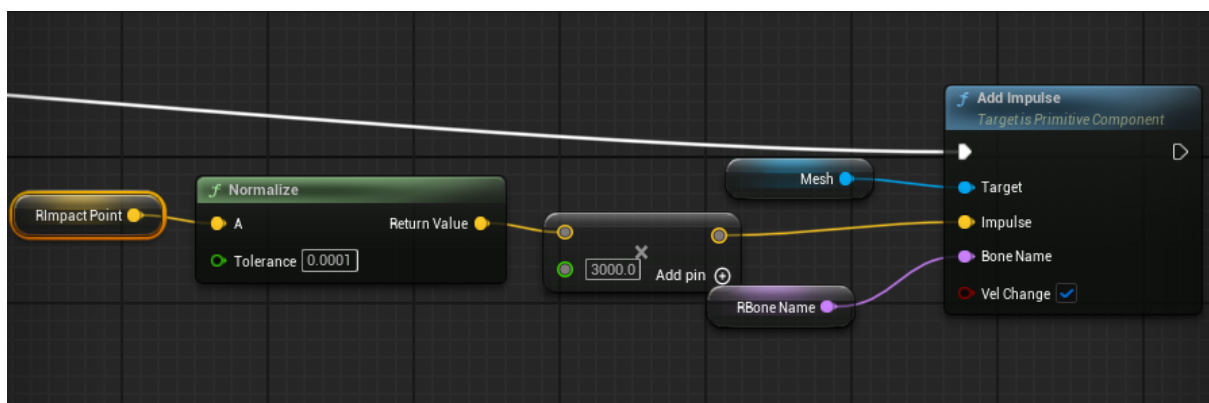


Figure 3. 29 Procedural Animation – Add Impulse Nodes

This impulse is used by both the player's hand and the throwable object. After triggering the impulse from the hit the timeline is executed moving the hit bones in a natural way.

3.5.8 Advanced Interaction: NPC bones grab system

The VR Template's player blueprint VR Pawn comes with its own grab system. This grab system can be used to grab game objects which are marked grabbable by adding a 'Grab Actor Component' to the game objects blueprint. By adding this Grab actor component, the player will be able to pick up the object with their hands and control it.

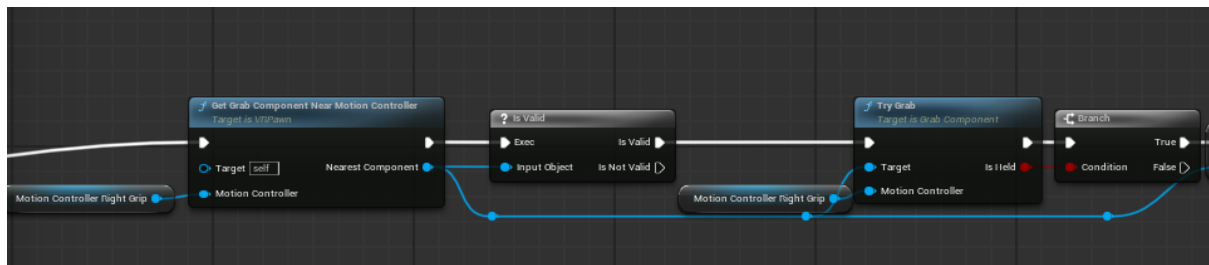


Figure 3. 30 Default Grab System Nodes

Grab component near the player's hand is identified then the grab is implemented using 'TryGrab' an inbuilt function. But this cannot be used with an NPC, if a grab component is added to NPC, we will be able to control and lift the whole NPC body. Since that is not the aim of this project a new grab system was implemented which was added to the current grab system.

To implement this new grab system first we get all the NPC actor blueprint, then use a for each loop and get the sphere trace values performed before. To grab the bone of the NPC a special function is used called Grab Component at location with rotation (inbuilt function) which takes the location of the component, the bone name if any and its rotation, finally the NPC mesh component and the physics component of player's right hand (this is important to assign the grabbed bone to that specific hand).

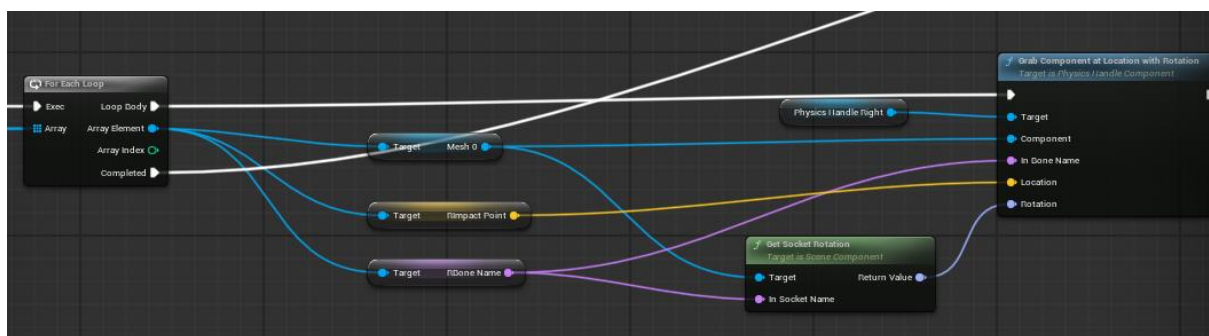


Figure 3. 31 Custom Grab Bone System – Set Bone Location Nodes

As explained before when a sphere trace is performed by the player's hand, the detected bone and its children bones ragdoll is activated releasing the bone of any animation. This allows the Grab Component at location with rotation function work letting the player to move the bone around. Once the player releases the grab trigger the release component function releases the grabbed bone and the bone return to its designated place according to the animation.

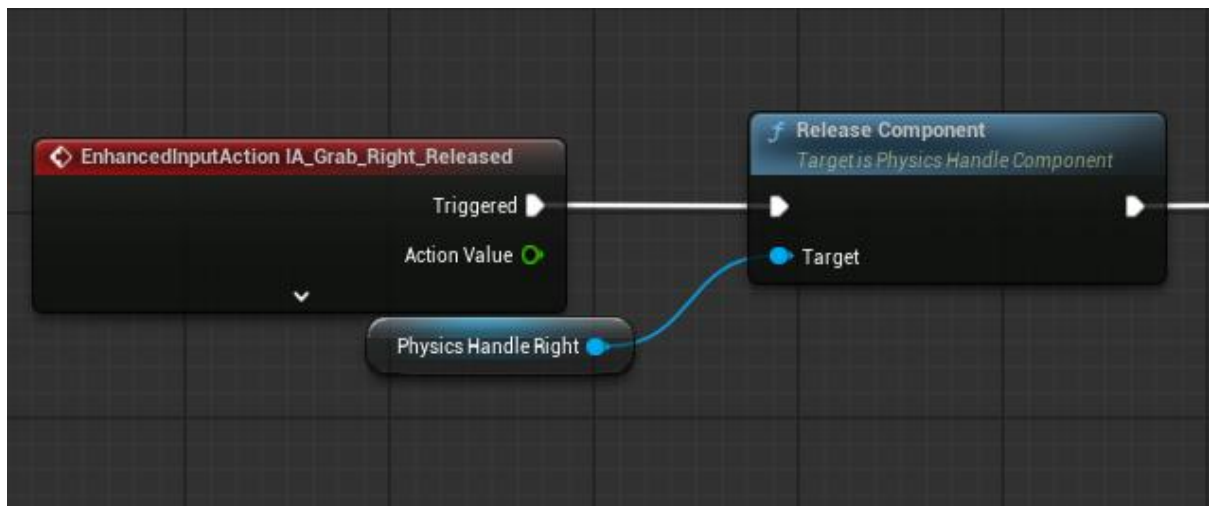


Figure 3. 32 Custom Grab Bone System – Release Grab Nodes

To make sure the bone returns back to animation Set Simulate is disabled if no hand is near the bone, no bone is grabbed, no throwable object hit the body.

3.5.9 Implementing AI behavior

Now that the NPC has been created with all the main functionalities stated in the proposal, A complex FSM integrating new collision detection system, NPC object detection and flinch reflex, reaction when punched and grab bone system. The next part includes using Behavior Tree, AI controller and Blackboard to control the NPC for the main study test level.

To make the test level playable and engaging a boxing level was created with two NPCs that can move towards the player and box. To create the logic of this behavior a Behavior Tree was used which allows the developers to integrate AI logic into NPCs.

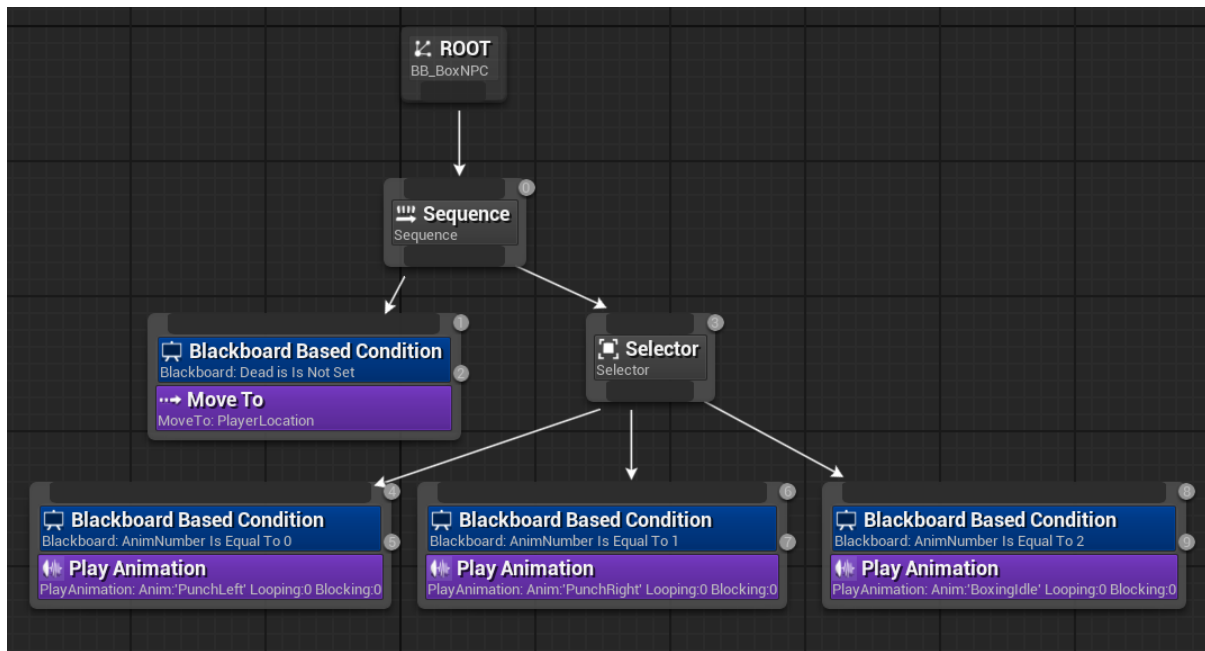


Figure 3.33 AI Behavior Tree – Boxing NPC Nodes

- Sequence: This is an inbuilt AI functionality which runs connected nodes from left to right.
- Selector: Inbuilt functionality that checks and runs each node from left to right and stops when at least one of the nodes is executed.
- Move To: Inbuilt function that moves the parent of this behavior tree i.e. the NPC to the given location.
- Play Animation: This function overrides the Finite State Machine animation playing the animation specified in the function.
- Blackboard Based Condition: This is an If state check. If the condition is true, it executes its paired function.

We use AI controller Blueprint to set the ‘Blackboard’ values that are used in the behavior tree. PlayerLocation variable gets the player location in the level constantly and updates it to the AI controller. After assigning the NPC a location to move to Selector is executed.

Selector checks the Blackboard conditions from left to right and executes the first one that is true. Here AnimNumber variable is an integer that randomly assigns a number from 0-2. This is to avoid a noticeable pattern, playing different animations: Punch Left, Punch Right and Boxing idle each time the NPC reaches the player.

3.5.10 Designing and Implementing level for playtesting

This project includes testing the system with participants to check how well Dynamic NPC system works when it comes to immersion and realistic movement. To get the most out of the participants results a 3-stage level is designed for the participants to get accustomed to VR and to static NPCs and complex NPCs, finally they are taken to the boxing fight level to test the immersion in an actual game level.

Stage 1- Introduction

The first stage includes introducing the participants to VR controls and the game objects they will be using later in the testing, this includes a basic cube, and a gun. This was necessary to get consistent results from the participants and not let inexperience with VR affect the study results. The participants can get accommodated with the movement and interactions then proceed to the next section.

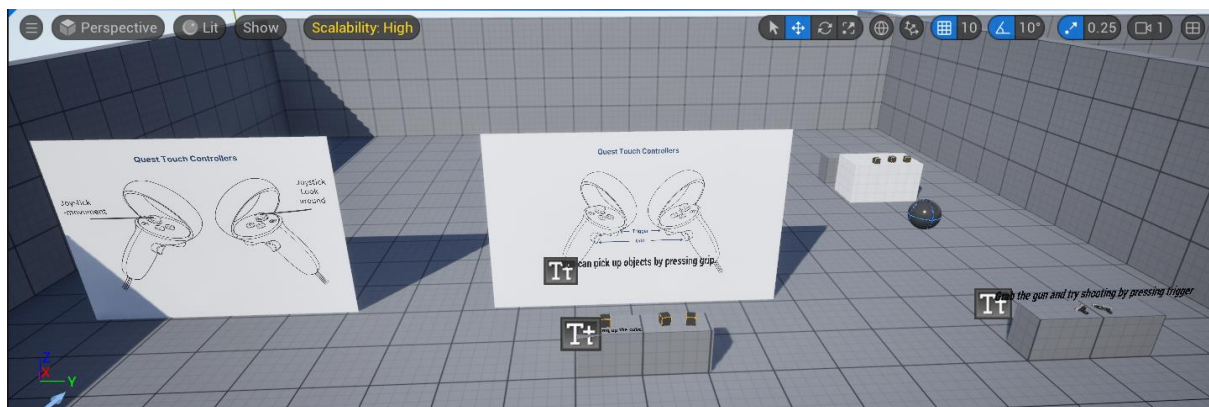


Figure 3. 34 Tutorial Level 1



Figure 3. 35 Tutorial Level 2

There are also two basic NPC in this scene. One stands still and the other runs on a fixed triangle path in a loop. This is to show the difference between a static NPC and the complex NPC which they are going to test.

Stage 2 – Introduction to Dynamic NPC

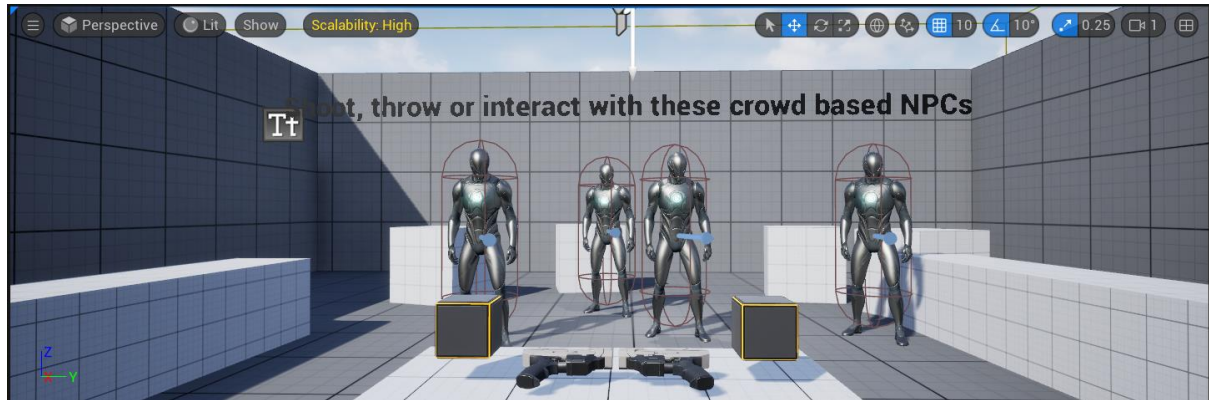


Figure 3. 36 Dynamic NPC Introduction Level

This level consists of 4 NPCs of different behaviors: NPC without FSM with Grab bone system, NPC punch without Grab bone system, NPC with FSM aggressive behavior, NPC with FSM friendly behavior. This was done to showcase the versatility of the NPC interactions. All the NPCs have the same system underneath, only a few Booleans were changed to create all these different variations. The participant can shoot, throw, punch and interact with all the NPCs to test out the system and understand how the NPCs work before moving to the main test scene.

Stage 3 – Boxing level

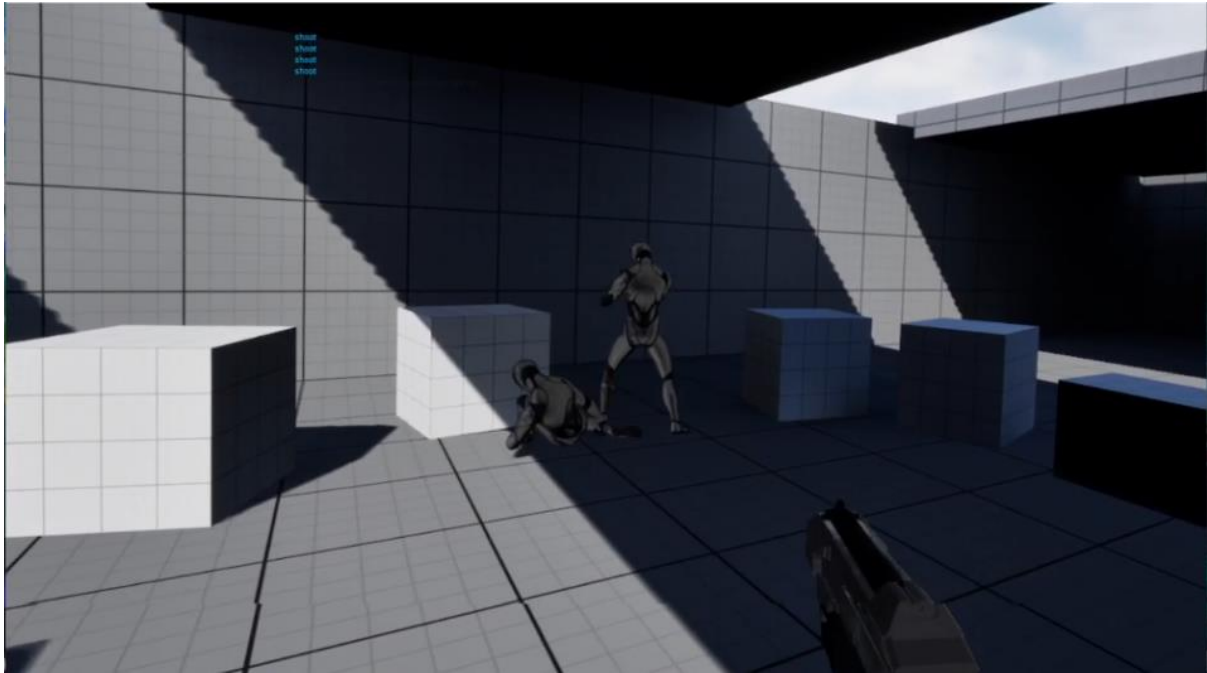


Figure 3. 37 Main Study Gameplay – Boxing Level

This is the final test level where the participant can fight the boxing NPCs with a gun or punch them or throw cubes. All these deal damage to the NPC. The goal of this level is to check for the player's immersion and the realistic movement from the NPC. The player roams around the level shooting and attacking the NPC until it dies. The participant is forced to constantly move and attack to create an engaging gameplay.

3.6 User Evaluation

3.6.1 Playtesting Planning

The evaluation was designed to understand the effectiveness of the NPC system created thereby addressing the research questions of this study. The questions created for the participants specifically focused on immersion, realism, and believability of NPC interactions. To get the most out of the participants experience both quantitative and qualitative questions for richer feedback from the user experience.

The sessions lasted for around 20-40 minutes, giving the required participants enough time to get comfortable in VR and test effectively. The participants were first briefed on Participant Information Sheet, and their consent was taken through consent forms capturing both their experience level and research focused questions for better results analysis.

Then Participants entered the level, each participant spending different amount of time in each stage. They were then given the Immersive Experience Questionnaire, and follow up question for more insights.

3.6.2 Participant Recruitment and Setup

Recruitment Strategy:

Participants were recruited through University groups relevant to the course and personal networks to ensure a mix of VR experiences levels. The target was 4-6 participants, 4 participants were found with different experiences for an in-depth analysis.

Lab Environment:

Testing was conducted in City, University of London's interaction lab with adequate space for the participants safe movement. The setup included:

- Meta Quest 3 headset connected to the Laptop via Meta Cable for faster and smooth transfer of visuals from Unreal Engine to the Headset.
- The Laptop HDMI was connected to the University system. A Webcam was set up to the University system to record the participant movements and responses for later analysis.
- A screen recording software called OBS Studio was installed to capture the VR gameplay and the session recording for cross analysis.

3.6.3 Participant Tasks and Expected Behavior

The study demo level was designed to elicit different types of interactions and observe participant responses, also allowing them to immerse themselves.

Stage 1:

The Participant start by going through a small tutorial, image showing VR controls. This gives the participants a clear understanding of their movement in Virtual world.

The Participants are then shown virtual 3D cubes with controls to pick up the cube. This allows the participant to understand the grab game mechanics in VR, which they will use in the main study. The participants were also given a gun that they can shoot since this will be

an important part of the study. The participants will shoot cubes in front of them and get used to aiming the gun. The author observes their aim and suggests tips or guides the participant, to make sure all participants are above a certain skill level.

In the Next section there two NPC that were designed specifically to show a non-reactive NPC, this gives the participants some insight into the dynamic NPCs to recognize the difference in immersion.

Stage 2:

At this stage, participants are presented with four variations of NPCs from the implemented system, additionally the same cubes and gun are given. The participants are allowed to freely interact with these NPCs. The author observes and suggests interactions depending on their initiative. These NPCs helps the participants understand the rules of the interactions and what can be done, promoting immersion and believability, helping the author with analysis of participants reactions.

Stage 3:

Once the participants are ready, they will move on to the final stage. Here the participants will fight two boxing NPCs, they are given a gun and two throwable cubes to defend themselves. Participants are suggested to observe the movements of NPCs body and responses to player's reaction. This is the main study level, letting the participants fight and interact with NPCs to understand immersion. The level was designed to demonstrate the use of NPCs in a game scenario, and to understand the player immersion, allowing the participants to assess and give feedback on believability of the motion.

3.6.4 Data Collection for Analysis

Three different types of data was captured.

1. Immersive Experience Questionnaire (IEQ): Ten questions with 5 points Likert scale to understand player's immersion, the believability of the NPC, their engagement and presence in VR, the questions help the participants to give their feedback effectively and help the author assess the results and correlate with other data.
2. Qualitative Interview: Questions were asked regarding the experience gaining more insight and having a conversation with the participant, to understand their rating in the previous section, also giving them the opportunity to express their feeling of immersion.

3. Video recording: This helps the author better understand their reactions and the system's feasibility in real testing, also to get more of an insight into the bugs that might have been missed, and the actual immersion of participants or anything they might have missed to mention.

The participants were constantly observed and checked for discomfort or motion sickness common for VR, they were informed that the data is anonymous and can be withdrawn at any time.

To conclude, this evaluation method was chosen capture all forms of results, helping with answering the research questions of this project.

4 Chapter 4 – Results:

4.1 Introduction

This chapter showcases the results of the participant testing after creation of the prototype as described in methodology above. The demo level created was tested with participants and anonymous data was collected through Immersive Experience Questionnaire (IEQ) and a short interview to understand how much of a change the Dynamic NPC system had made on their gameplay; how realistic they felt while playing the level and other questions to answer the research questions. These results helped in understanding the user immersion and the believability of the NPC. Detailed analysis of each question and how participants felt during can be found below.

As for the participant testing, four participants participated in the testing. They were given the participant information sheet and consent form then they were videotaped and recorded after which they participated in IEQ and interview.

4.2 Participant Demographic

The four participants who participated were a diverse group from people new to VR to experienced. When asked about their experience in VR.

- 2 participants self-identified as Beginners
- 1 participant self-identified as Intermediate
- 1 participant self-identified as Experienced

This allowed for analysis of results on whether the prior VR experience influenced their opinion on the interaction system.

4.3 Questionnaire results

The participant size is small; this might impact the strength of the results. The participants were given 10 questions to rate how much they agree with each statement on a 5 – point Likert scale (1- Strongly Disagree, 2- Disagree, 3- Neutral, 4- Agree, 5- Strongly Agree).

Question 1: I lost track of time while I was playing

When asked about their immersion in the game while testing the VR prototype. Three out of four participants agreed with that statement. The sample mean was 4.25 which shows the participants felt the prototype was immersive. The standard deviation was 0.5

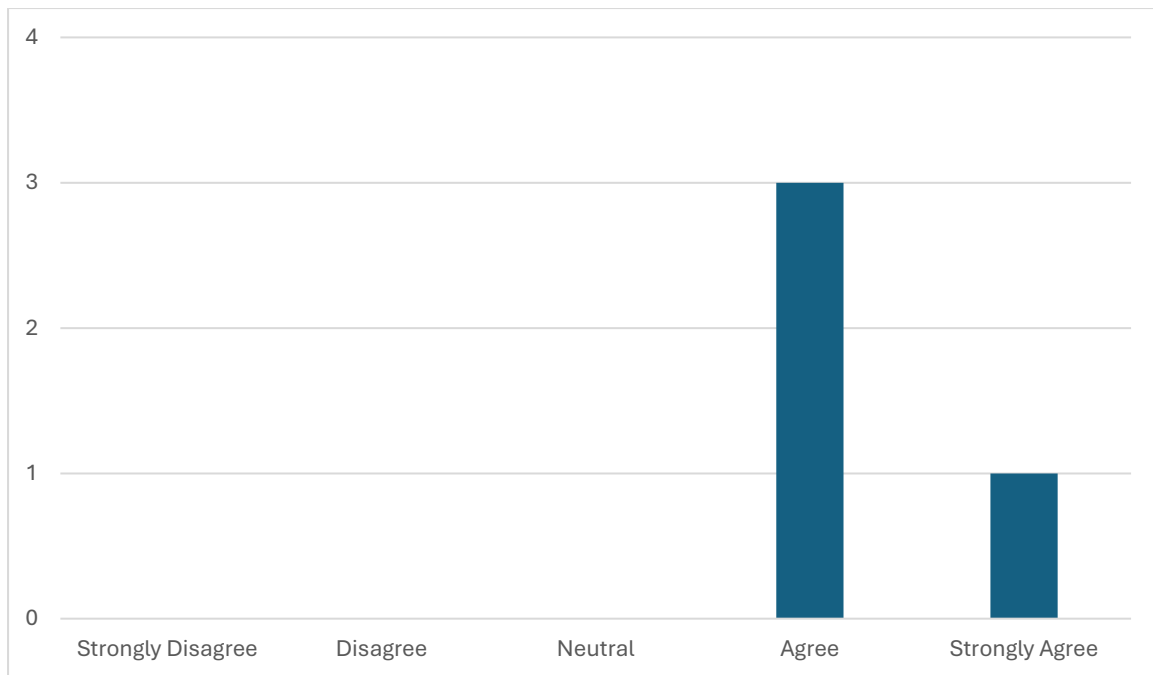


Figure 4. 1 Q1 Frequency Bar Chart

Question 2: I felt like I was really inside the game world

When asked another question about the Virtual reality immersion. Two of the four participants agreed that the world felt real. The sample mean was 4.00 with a standard deviation of 0.82, which shows that participants overall agreed. This answer is helpful to determine player's state of immersion when it comes to the VR world which can be used as a base to understand the NPC system immersion.

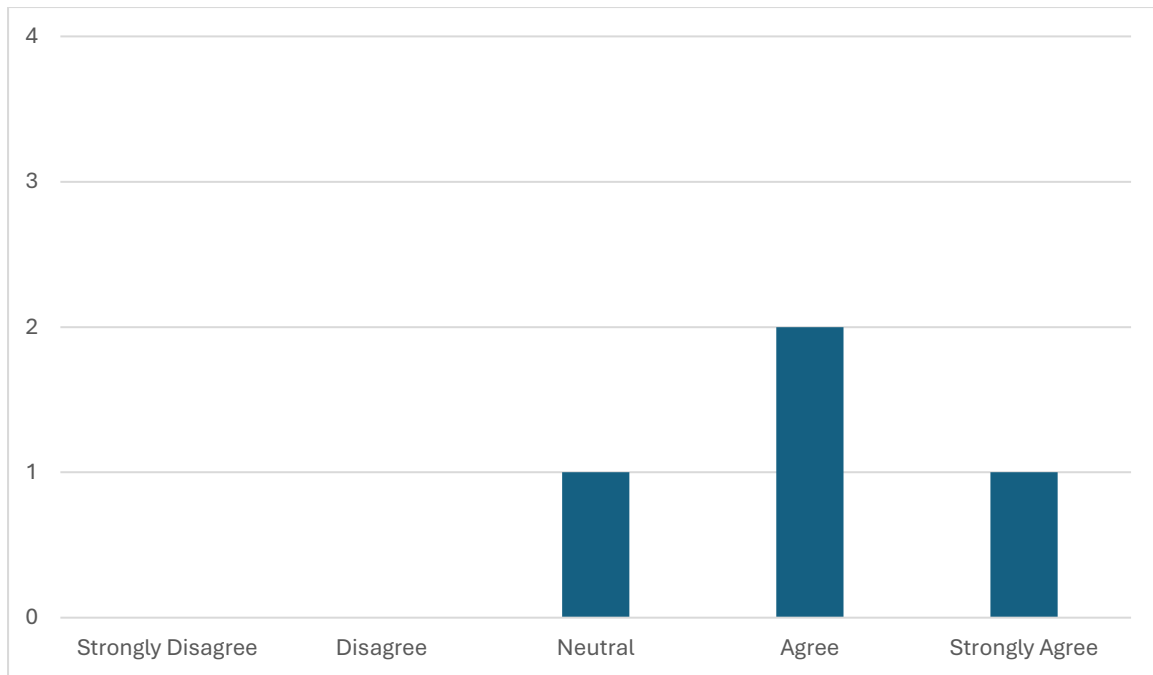


Figure 4. 2 Q2 Frequency Bar Chart

Question 3: I was focused on the game and not easily distracted

This question was asked to gauge how engaged the participant was while testing the prototype. Three out of four participants Strongly agreed that the prototype was engaging. The sample mean was 4.75, one of the highest in this questionnaire with a variation of 0.5 clearly showing that the prototype was engaging.

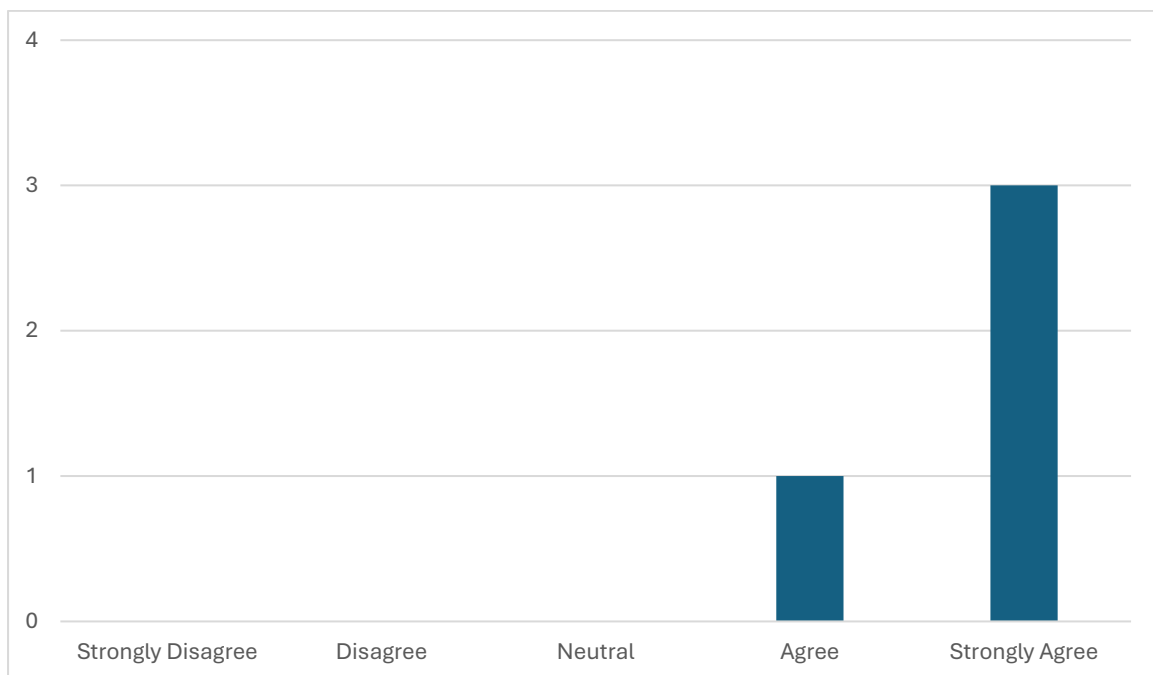


Figure 4. 3 Q3 Frequency Bar Chart

Question 4: I felt the characters move like real people when interacting

When asked about the NPC realism and believability to check for the prototypes immersion. All participants unanimously agreed that the prototype was immersive. Mean was 4 with no deviation. This shows that the prototype blend between different complex systems worked well under testing and was able to convince the participants that they were real people.

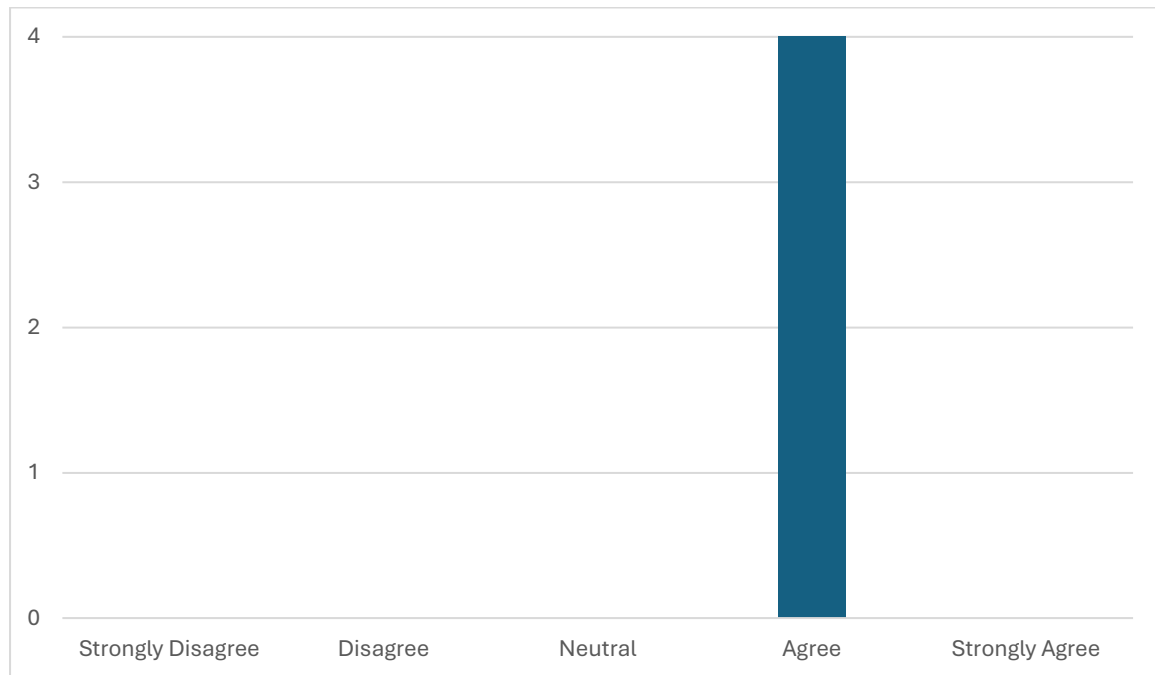


Figure 4. 4 Q4 Frequency Bar Chart

Question 5: I felt the characters respond to my movements realistic

This question was asked to understand if they felt NPC was unrealistic or if there were any unnatural movements. Three of the four participants agreed, which shows that the dynamic NPC system constructed in this project has worked successfully providing reactions to the player's actions. The other participant strongly agreed with the realistic movement of the NPC. This was likely to happen because of the flexibility of the prototype's creation considering all possibilities of player's interaction. The Mean of this question was 4.25 with 0.5 deviation.

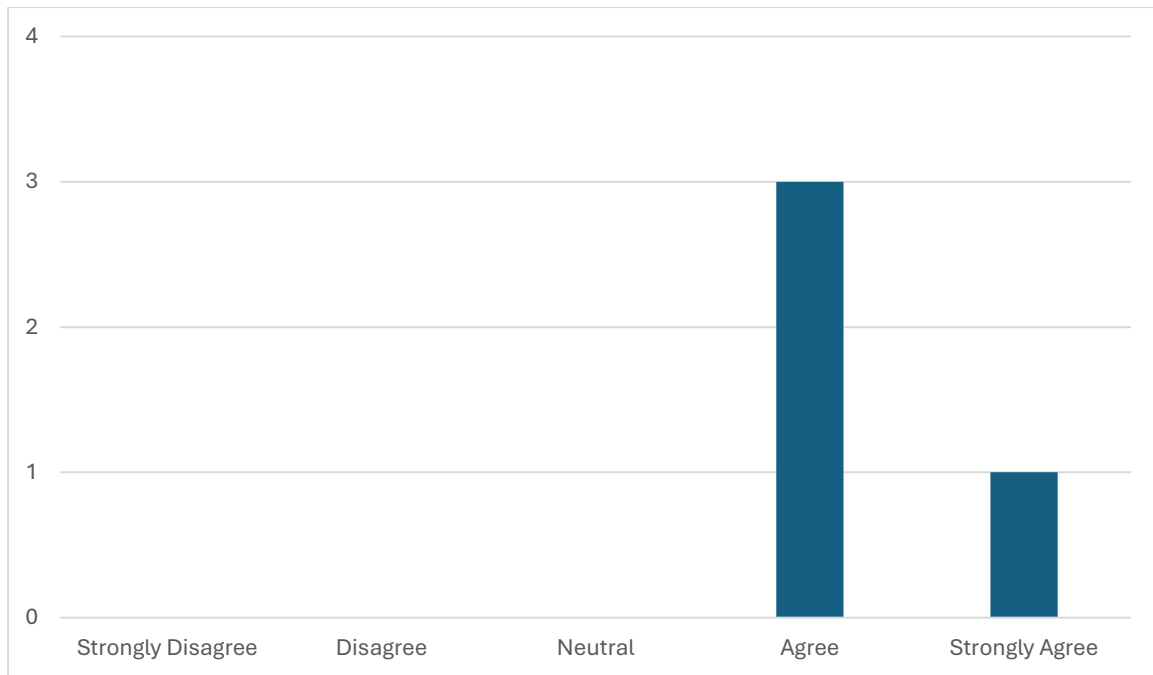


Figure 4. 5 Q5 Frequency Bar Chart

Question 6: I would like this in more VR games

Three out of the four participants strongly agreed with the statement that they want to see these types of interactions in more VR games. This shows that there is a lack of interaction between players and NPCs in current VR market proving that these interactions will hugely benefit the current VR world. The Mean of this question is 4.75 with 0.5 variation.

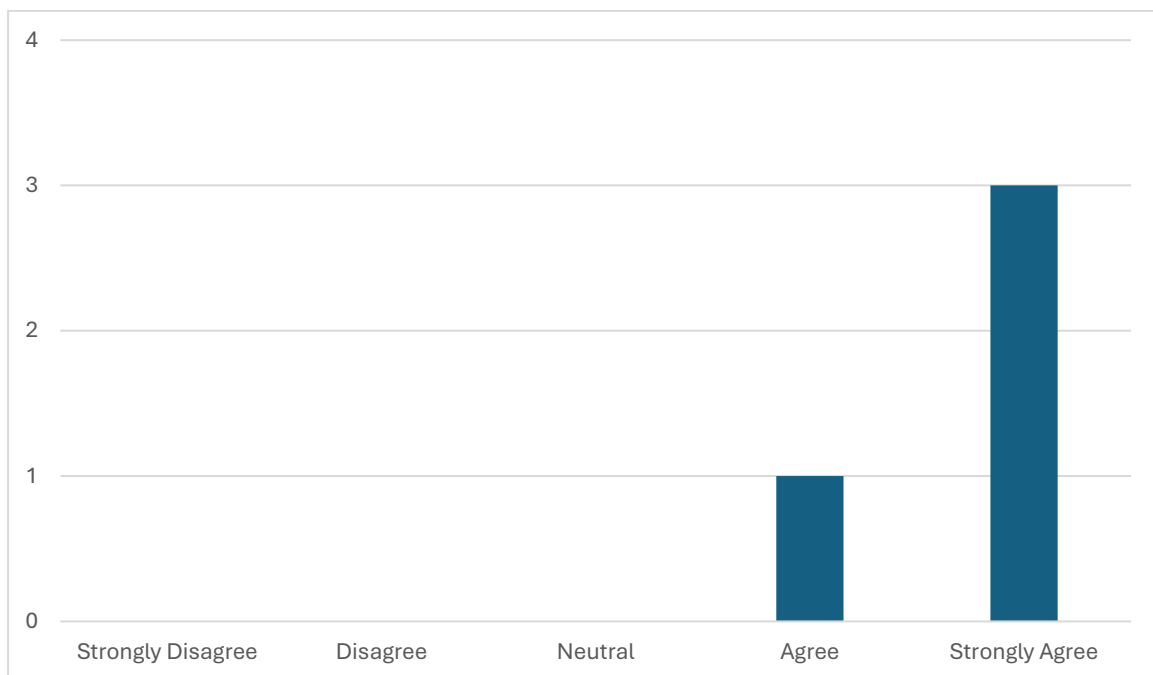


Figure 4. 6 Q6 Frequency Bar Chart

Question 7: I felt this was better than other VR games interaction

Two out of the 4 participants agreed that these interactions were better than other VR games interaction with one participant strongly agreeing with this statement. By this it can be determined that VR games interactions need improvement and the current game titles need to push toward immersion for a more engaging player base. Note: one participant was neutral about this, and the participant is experienced in VR, this shows there is still more potential to this prototype.

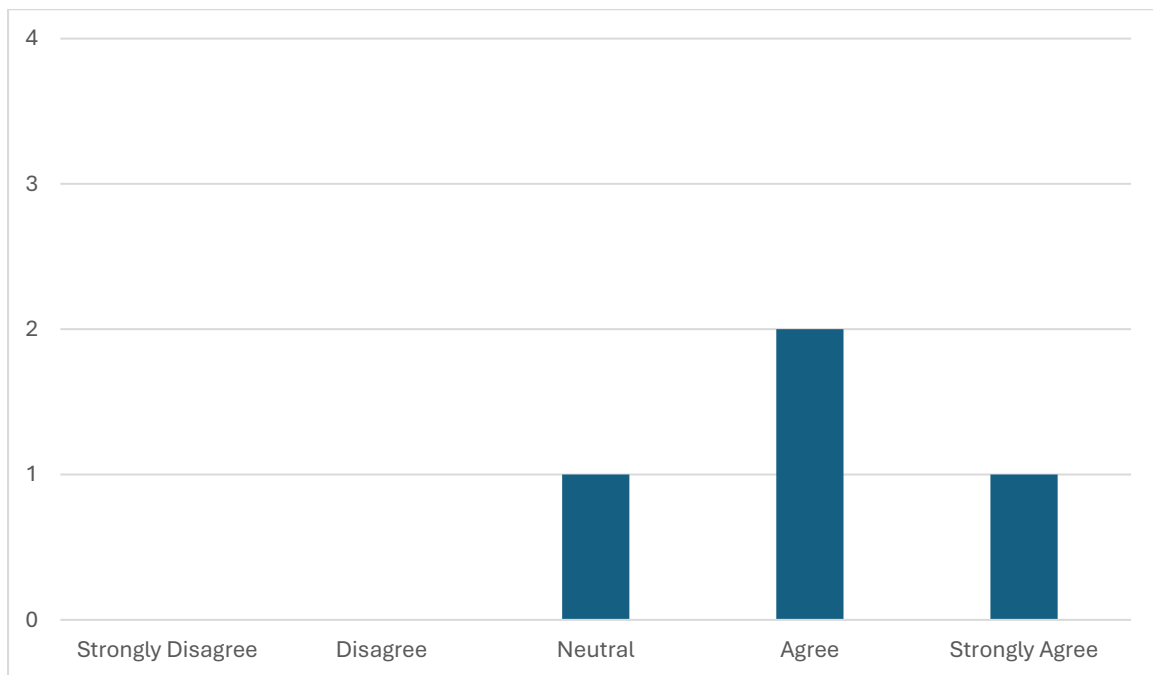


Figure 4. 7 Q7 Frequency Bar Chart

Question 8: The NPC behavior made them seem not static objects

The Mean of this statement was 4.25 with 0.95 variance but two out of the four participants strongly agreed with this statement and one of them is the experienced VR participant, which adds weight to the statement, clearly showing the NPC system was effective and responsive to player's action.

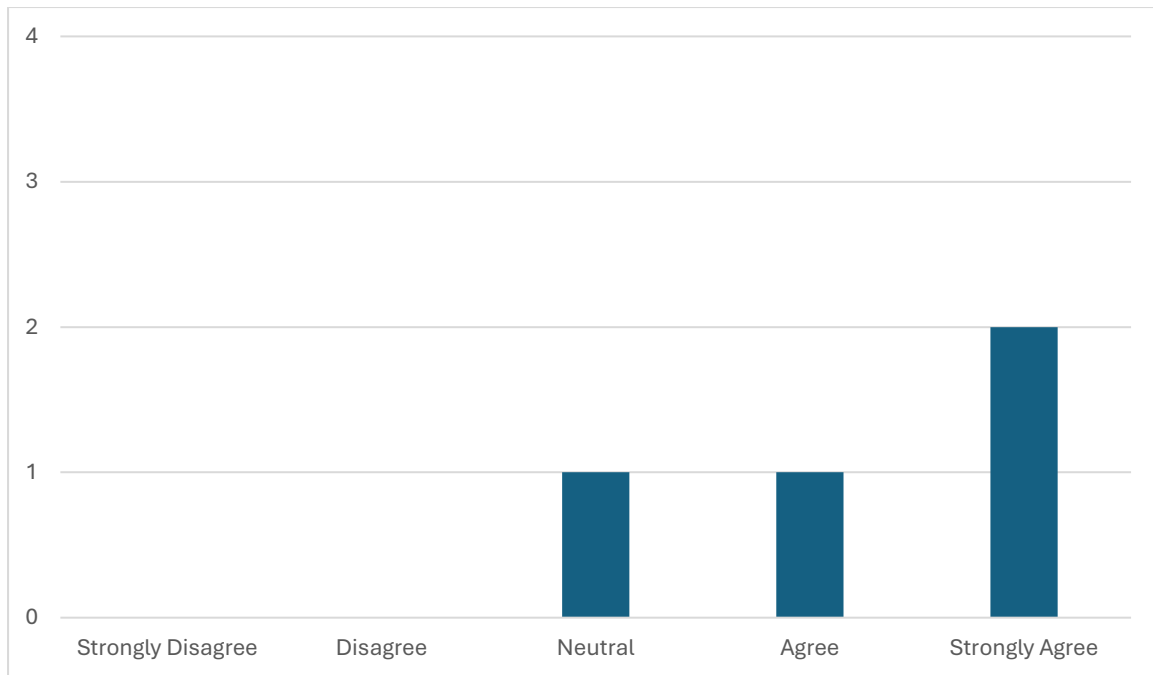


Figure 4. 8 Q8 Frequency Bar Chart

Question 9: I would prefer future VR games to include this type of NPC responsiveness

Three out of the four participants strongly agreed to this question. This shows a clear pattern in the participants that the dynamic NPC system worked as intended providing players with the satisfaction of interacting with a fun and engaging NPC. This strongly supports the need of this system in VR games, and this system will serve as a strong stepping-stone for the interactions to come in future.

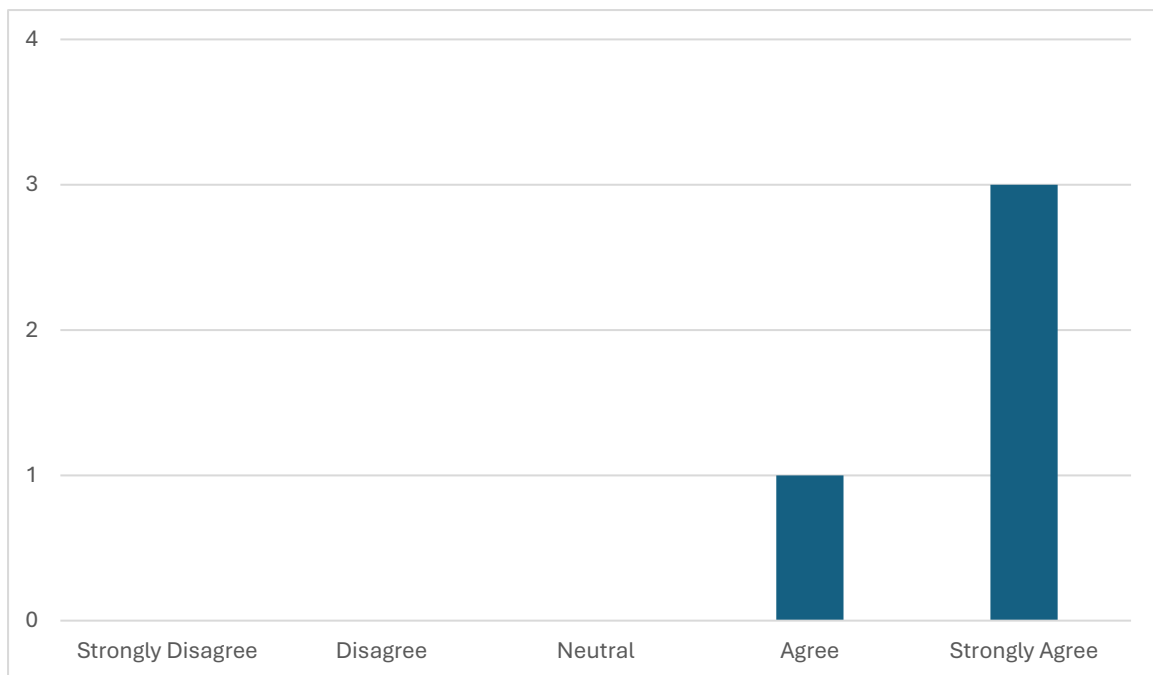


Figure 4. 9 Q9 Frequency Bar Chart

Question 10: The prototype ran smoothly without causing me significant discomfort or confusion

This question was asked to improve on any possible problems with the system. Since this is a prototype and a tech demonstration there were a few minor bugs that hindered the testing process causing this question to get a mean of 3.75. This was bound to happen because of the time limit of the project execution and the prototype phase of the project.

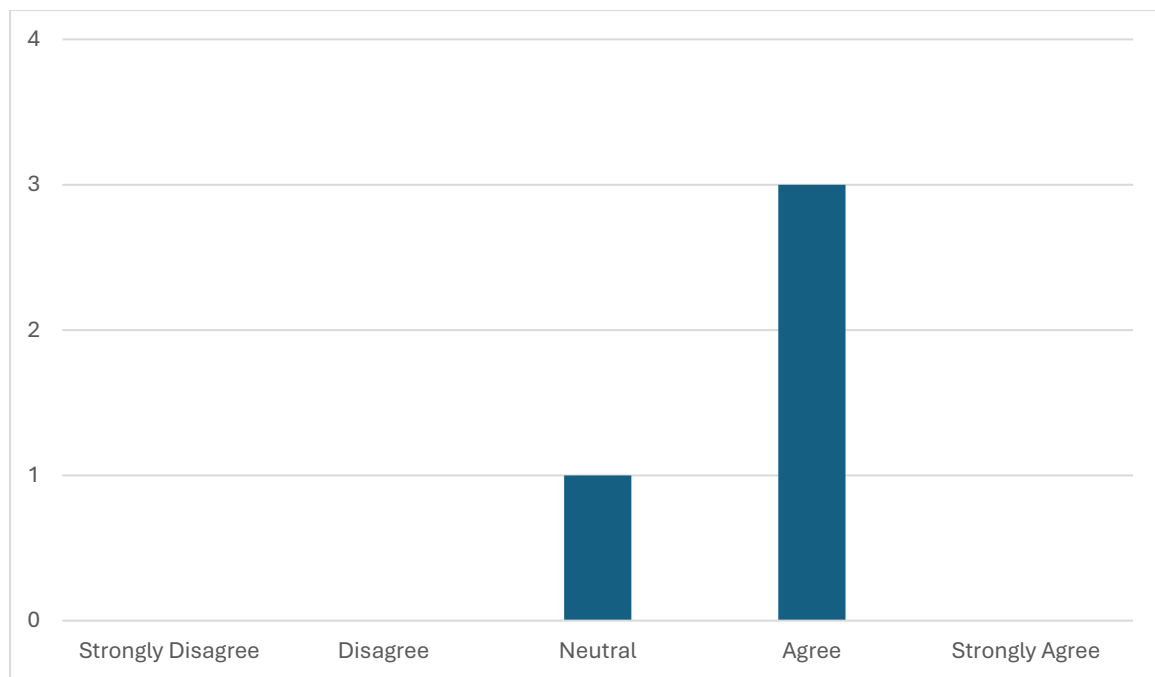


Figure 4. 10 Q10 Frequency Bar Chart

Overall, the quantitative data indicates a very positive response for the dynamic NPC system. Questions 3, 6 and 9, which were questions surrounding the engaging gameplay, and the need of these responsive interactions, showed a highly positive reaction with a mean of 4.75. This suggests the project has successfully met its main objectives of creating a realistic and believable NPC and pushing immersion in VR.

4.4 Qualitative Results

These are the analysis from the qualitative question asked after the participants completed the quantitative questions. The answers received from the participants for the questions are summarized and analyzed below.

Question 1: In your own words, how did the NPC's interactions affect your feeling of immersion in VR?

Participants consistently noted that the reactive behavior of the NPC made them feel they were interacting with real people, and they felt immersed, a participant noted that the interaction was satisfying. Overall, the feedback was positive, one participant suggested an improvement to the project by including haptic feedback for players' responses. This was initially part of the project but due to time constraints it couldn't be fulfilled, but it can be included in a future version of this system to make the interactions feel more natural.

Question 2: Were there any moments when the NPC's reactions felt unrealistic or distracting? Please describe it.

For this question participant noted the minor bugs that appeared during the playtesting. Since this is a prototype made under time constraint the author predicted to encounter some bugs while testing this is especially true when dealing with complex NPC body physics. Some of the bugs noted by participants and author while testing was fixed between playtests. But it is to be noted that none of the bugs heavily effected participants' immersion. Some participants were confused about the variation NPCs responses due being beginners to VR and struggling a little with the complex interaction even when guided. Overall, the bugs were noted and some of them were fixed, no complaints regarding the NPC body movements.

Question 3: How believable was this NPC compared to the other NPC in the scene or other NPCs in games?

The feedback on this question was very positive. Many participants noted that the NPC were realistic and not static. One participant had difficulty with the final stage of the testing because of their expertise with VR they had a hard time multitasking, so they couldn't observe the reactions of the NPC. But the recorded gameplay footage showed reaction. All the other participants felt the NPC reacted to players' movements. Overall, the prototype worked as intended reacting to player's response dynamically and making it immersive.

Question 4: How would you improve the way NPCs respond to your intentions?

This question was asked for future developments of the prototype and there were a few useful suggestions from the participants. A participant noted haptic feedback which was suggested earlier. The other participants wanted it to be even more accurate and realistic to human interactions suggesting ideas like "interlocking hands with the NPC" and recording

movements for all different moves and angles. Since this is a prototype, it is hard to notice and trigger reactions from different angles, but these can be refined with further development.

Question 5: What movements felt most natural or most unnatural?

Most participants noted the natural movement of body parts when shot or grabbed showing clear reaction. The participants enjoyed the grab interaction and the satisfying reactions when shooting. For unnatural movements the participants noted some unexpected bugs that came up while testing in particular one participant had an issue with orientation in VR this bug was immediately fixed after the testing and no other participant encountered that bug. Finally, a participant suggested showing body decay and damage when the NPC is weak. This is a feature that the author thought to include in the future beforehand to make it more natural and will be part of the future development of this project.

Question 6: Can you name a specific game you have played where these interactions would have improved immersion?

Two of the participants talked about Batman: Arkham Shadow VR (Rocksteady Studios, 2024) about the NPCs being not reactive and feel static, this was also explored in literature review before. The game can benefit from NPC interactions like these, and it was made in UE5, the same game engine used to make this project. A sandbox VR game which they couldn't recall was suggested and participants mentioned in some of the previous questions about many NPCs just looking around or following a path without a goal. The opinions of the participants clearly support the prototype being a superior version.

Question 7: What kind of games or genres do you think would benefit most from this type of NPC responsiveness?

All the participants suggested including these types of NPC responsiveness in combat and role-playing games. Some participants also suggested including these interactions in flat screen games noting that it would help to clearly see where the NPC has been shot. This supports the intention of this prototype, including these advanced interactions in games that increase immersion by showing the clear movements of the body.

Question 8: Beyond games, can you think of other areas (e.g., training, simulation, education) where dynamic NPC touch interaction might be useful?

The participants had unique and diverse suggestions that can be explored with these interactions. The suggestions included a Virtual pet with similar interactions in VR to help with mental health by including haptic feedback, for surgery or emergency situations training with few modifications it can be used to train emergency rescue workers in VR, it can be used to educate future game developers for how physics work in game engines and help them understand how different interactions can be used. Overall, the suggestions from participants were great ideas that can be explored.

5 Chapter 5 – Discussion:

5.1 Project Objectives and Outcomes

The aim of this project was to create a realistic and believable NPC system that can react to player's touch interactions dynamically adapting animations to give a reflex reaction, to address the identified gap between advanced object physics and simplistic NPC interactions in current VR games.

All core objectives were successfully completed. The implementation delivered a successful VR prototype combining inspired mechanics of games such as Blade and Sorcery, Half-Life: Alyx, and Red Dead Redemption 2 that features a custom collision detection system to identify the player's touch location and intensity which was then integrated into a Finite State Machine with multiple animations reactions depending on the collision and exhibit different reactions (Friendly, Aggressive) then procedural animation was added into the system blending pre-made animations with physics driven reactions for players high intensity touch or object impacts. Finally, a bone-based grab system that allows the player to grab NPC body part and manipulate their position physically, making the character more present and responsive.

Other external objectives were carried out to make the gameplay more engaging like including damage, changing animations, and AI controllers to change NPC position, further the dynamic NPC system was designed in way that it can be easily modifiable and can act as a plugin for developers or designers, giving them the ability to easily change animations, AI behavior and other settings (This hasn't been covered in methods completely because of the word limit).

5.2 Results and Observations

The test results confirm the project's core objectives regarding believability and immersion. It is safe to assume that from the high mean scores from the previous chapter clearly indicate that the participants found the NPC interactions realistic, engaging and immersive. The low standard deviation in all questions gives a firm idea that the players' opinions are close to each other.

The qualitative questions gave more insight as to what interactions worked and were engaging. Most of the participants found the NPC movements natural and realistic. Claiming the NPC body parts moving accurately and satisfyingly when shot or grabbed. The new collision system worked smoothly most of the time but the author observed things that can be improved while playtesting with others, mainly the collision system uses a sphere as the trace to detect collisions but since spheres are circular the impact point and angle when the hand touches the bone is sometimes calculated wrong, but the only other option is using a line trace which was found to be even more inaccurate and harder to detect, to overcome this the hand mesh as a whole can be used for touch detection for a more precise interaction.

Further animation triggered by the FSM are dynamic and change according to touch but are still simple, taking one of the suggestions provided by the participant to connect hands or high five a NPC and make it more emotional and interactable can help take the immersion further. In one instance the NPC fell on the floor this was an unexpected bug that occurred due to the NPC leg ragdoll physics being enabled when the NPC is hit because of which the NPC collided with the environment around it falling down or rotating. Object hit detection and advanced procedural worked smoothly but when participant punched with a greater force the NPC response isn't strong enough this needs to be scaled and adjusted to be more dynamically. The AI behavior tree worked well and as intended kept the participant engaged and constantly moving.

Other than few bugs that were expected as this is a prototype of a complex system, participants could see how these interactions could benefit the current VR gaming world. Participants' suggestions for future development of this game mechanic showed the clear trajectory of the use of this prototype.

While testing it was noted that players were ignorant about the NPC interactions initially until prompted, this was to be expected as many NPC in games are static, the participants were nudged to do specific actions to generate reactions from the NPCs. This should have been expected; a better approach would be to make the NPC to try and interact with the player first so that players understand they can interact with the NPC in different ways. Since this is a prototype, this can be overlooked because in a real VR game there is a narrative or goal behind interactions. Another important factor was the limited time even though the participants were taken to the main game scene after slowly introducing them to the different

interactions they found the game scene difficult because of its fast gameplay instead the NPC could have been slowed down giving the player more time to react.

One major problem that wasn't noticed while author was testing was orientation of the player; this was fixed immediately but this was caused mainly due to the participant turning in real life physically instead of using the controls to turn. Otherwise, all the participants found the interactions easy and natural; some were curious about interactions implementations.

5.3 Comparisons

In comparison to the VR titles mentioned in the literature review, the interaction mechanics implemented represent advance in NPC responses, being more natural and flexible than reactions in Blade and Sorcery and more interactive and less scripted than Half-Life: Alyx and almost replicating flat screen industry standard NPC movement in Red Dead Redemption 2. Of course, the work of professional development teams with years of development time and resources is hard to compare to a student project under tight constraints.

Yet this system succeeded in being novel and engaging to its testers. With no critique suggesting that it is poorly executed or did not improve on other VR games interactions. In this project effort was focused on combining the core mechanics that provide a unique approach to NPC interactions developing on strengths of other games.

The system performance proved adequate for a prototype, but some optimizations will be needed for a shipped product. Blueprints is a good tool for prototyping, but this system needs to be redeveloped in pure C++ for a more efficient system.

5.4 Results Validity and Scope

The project satisfied its core hypothesis to be correct. Given the small number of participants the results cannot be completely conclusive on a broader scope. The opinions of the participants from the questionnaire strongly support the conclusion that having NPCs respond to all possible variations of interactions significantly helps with realism and immersion in VR. Since the sample population mostly included casual or beginner VR users having trouble with interactions was expected for a prototype, the results are conclusive.

Testing the system's immersion and believability with diverse participants with beginner to experienced users ensures that the system is accessible. More opinions from experienced VR developers and VR enthusiasts would help in future for refining the system for professional applications but keeping things simple and more accessible will benefit the gaming community.

5.5 Project Contribution

The project provides valuable insights into “how NPC interactions in VR can be made more realistic and immersive using a custom changeable system that reacts to player's actions” by reading the design and implementation of the prototype where a customizable NPC interaction system was created, a system that lets designers program different reactions by receiving inputs from the system increasing the number of permutations of reactions and the other research question “Is it possible to design and implement a non-playable character which reacts to player's actions based on force and area that triggers a more realistic and believable reaction in VR” can be answered by understanding the implementation and the results from the testing of the prototype showing the natural way that NPC reacted. Further answering “To what extent can we increase the realism of virtual reality environment by including more reactive NPCs” this can be determined from the answers and suggestions provided by the participants qualitative questions. This shows that the project answers the questions not just through technical implementation but also through user testing validating the approach.

Furthermore, it proves that even individual developers can design and implement creative interaction systems that push the VR NPC interactions. New VR developers could use this project as a plugin and expand or build on it to create more interactive games. Even experienced developers could take advantage of the system and implement physical interactions in VR.

This project also suggests that by taking inspiration from real-world interactions, reacting differently based on touch and area, games can be made more engaging and immersive. Moving beyond static interactions, to a more responsive and physical virtual characters.

6 Chapter 6 – Evaluation, Reflections, and Conclusions:

6.1 Lessons Learned

This project provided valuable opportunities to learn important lessons about VR and games development and project management. Time management and careful planning were crucial for completing this complex project. Though previous experience with Unreal Engine 5 helped many things had to be learned and examined for VR and physics interactions, requiring an effective schedule. An important lesson was learned when some interactions were interrupting other interactions in the level, but version control helped with overcoming these problems by reverting back and forth between versions to identify the problem. This reaffirmed the importance of frequent backups after achieving a specific task.

Researching and implementing complex interaction mechanics took significant time, especially when balancing physics-based reactions with finite state machines. These were overcome by looking for innovative ways to implement and structure the interactions. The iterative approach, while time-consuming, proved vital for a polished interaction system. Testing and debugging consumed as much time as the implementation, adjusting impulse values, animation timeline graphs, and a easy to follow finite state machine.

Finally, this project demonstrates that creating believable character interactions in VR includes a complex multi layered system which must work in sync without being intrusive to each other to create a realistic, immersive experience.

6.2 Future Work

Many exciting extensions could upgrade the believability of this dynamic NPC interaction system, evolving into a complete framework for VR character interactions. Based on both the participant feedback and insights from the project, the following improvements are proposed.

This system could be expanded with more reaction animation also including facial expressions and character voice to make the NPC more emotional. The finite state machine can be expanded and structured better to include other behaviors beyond friendly/ aggressive.

Different interactions could be implemented, for example: interlocking NPC and player hands and other more complex social gestures suggested by participants. Haptic feedback system can be a nice addition to making the experience more immersive. The AI could be improved for advanced games creating NPC-NPC interactions to make the NPC more believable in social scenarios.

The framework could be adapted beyond games such as VR training simulations for healthcare or rescue workers, where realistic human interaction is crucial. By improving the bone-grab system with more complex inverse kinematics for more natural limb movement.

Performance could be improved by converting blueprints into functional C++ code making it more suitable for shipping games or other applications. Also giving the option to sell it as a plugin in game marketplaces like Unreal Marketplace making it available for a broader audience.

6.3 Bugs to Be Fixed

During testing, some technical issues were identified that need to be addressed for smoother system.

The most significant issue occurs when NPC body physics is overloaded with quick consecutive interactions, causing unnatural body movement. While adjustments to physics constraints reduced this during testing, a better system structure is needed. In some rare cases, the collision detection system failed to detect bones when too close to the NPC body. This required participants to repeat the interactions by moving away slightly breaking the immersion.

The transition between grabbed bone to previous animated position was little quick making it unnatural this can be fixed by creating smoother transitions. When player grabs a bone sometimes the hand is detached and not following the grabbed bone, this can be fixed by welding the bone and the player's hand temporarily. In some cases, the force applied by the player is registered as lower than the force actually applied due to the force being recorded even after the collision, this can be fixed by taking the first value after the collision.

6.4 Conclusion

The hypothesis of this project proposed that NPC interactions in Virtual Reality could be significantly enhanced by using a system that responds dynamically and procedurally to player's touch considering different possibilities such as location and intensity of the touch, hence improving immersion and believability. The results discussed in previous chapters, supported by both quantitative and qualitative questions and answers validate this hypothesis.

The successful implementation of all core objectives and the positive responses from the participant regarding immersion, engagement, and the desire to see similar systems in future VR games shows the value of this approach. While there are enhancements to be made this prototype serves as strong steppingstone for further development in dynamic physical interactions in VR.

The prototype testing also shows that advanced innovative interactions can benefit with experience in virtual reality, expanding the user base and encouraging developers to explore interactions beyond one-dimensional reactions. The positive response from all the participants suggests the attraction to such interactions. Finally, after completing this project the author is confident that the interaction system developed can significantly help future work, contributing towards a more immersive, responsive and realistic virtual world in gaming.

Glossary

Term	Definition
AI Controller	A component in Unreal Engine that possesses a Pawn and is responsible for running a Behavior Tree.
Animation Blueprint	A visual scripting system in Unreal Engine used to control the animations of skeletal mesh. It contains an Event Graph, Anim Graph for logic and finite state machines.
Blackboard	A data storage component used in conjunction with Behavior Tree to hold variables.
Blueprint	Unreal Engine's visual scripting system. It allows developers and designers to create game logic and mechanics by connecting nodes without code.
Bone	A component of a skeleton used to animate a 3D character model.
Collision System	A system in a game engine that detects when two objects come into contact.
Control Rig	An editor in Unreal Engine used to create rigs for skeletal meshes.
Game Components	Game elements such as models, scripts, audio etc
Game Mechanics	Gameplay related rules.
Grab System	The mechanics that allow a player to pick up, and manipulate objects.
Impulse	A physics term in game engines to a single, instantaneous force applied to an object.
Inverse Kinematics (IK)	A technique in animation and robotics used to calculate the required positions of a chain of joints.
Level	“ A level is all or part of your game's “world”. Levels contain everything a player can see and interact with, like environments, usable objects, other character and so on” (“Levels in Unreal Engine Unreal Engine 5.6 Documentation Epic Developer Community” n.d.)
Non-Playable Character (NPC)	Any character in a game that is not controlled by the player. Their behavior is controlled by the game's AI
Physics Asset	An asset that defines the collision properties and physical representation for a skeletal mesh

Procedural Animation	Animation that is generated in real-time by an algorithm, rather than being pre-animated.
Prototype	A pre-production stage of a game
Ragdoll Physics	A type of procedural animation that uses the game's physics engine to simulate the body of a character.
Skeletal Mesh	A 3D model that is rigged with an underlying skeleton that can be animated
Unreal Script	A Unreal Engine 3 programming language
Virtual Reality	A three-dimensional world that is computer generated and can be interacted with by a person wearing special equipment.
VR Pawn	The base class for the player's character in VR within Unreal Engine.

References

- [1] Carroll, D. (2022) Attention and Communication in Virtual Worlds. MAI. Trinity College Dublin. Available at: <https://publications.scss.tcd.ie/theses/diss/2022/TCD-SCSS DISSERTATION-2022-044.pdf> (Accessed: 24 March 2025).
- [2] Nadine Aburumman, Marco Gillies, Jamie A. Ward, Antonia F.de C. Hamilton. (2022) Nonverbal communication in virtual reality ', International Journal of Human Computer Studies, Volume 164, 102819, Available at: <https://www.sciencedirect.com/science/article/pii/S1071581922000489> (Accessed: 24 March 2025).
- [3] Witmer, B.G. and Singer, M.J. (1998) Measuring Presence in Virtual Environments Available at: <https://papers.cumincad.org/data/works/att/c7a3.content.pdf> (Accessed: 20 June 2025)
- [4] Slater, M. (2009) Place illusion and Plausibility Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC2781884/> (Accessed: 21 June 2025)
- [5] Fathoni, K., R Y Hakkun, H A T Nurhadi (2020) Finite State Machines for Building Believable Non-Playable Character Available at: https://www.researchgate.net/publication/342980694_Finite_State_Machines_for_Building_Believable_Non-Playable_Character_in_the_Game_of_Khalid_ibn_Al-Walid (Accessed: 21 June 2025)
- [6] Mori, M. (1970) The Uncanny Valley Available at: https://www.researchgate.net/publication/254060168_The_Uncanny_Valley_From_the_Field (Accessed: 21 June 2025)
- [7] Kunze et al., (2014) The OPPORTUNITY Framework Available at: <https://www.ingentaconnect.com/contentone/ben/swcc/2011/00000001/00000002/art00004> (Accessed: 25 June 2025)
- [8] Brown, E., Cairns, P. (2004) A grounded investigation of game immersion Available at: https://www.researchgate.net/publication/221515445_A_grounded_investigation_of_game_immersion (Accessed: 10 September 2025)
- [9] Jennett, C., Cox, A. L., Cairns, P., Dhoparee, S., Epps, A., Tijs, T., Walton, A. (2008). Measuring and defining the experience of immersion in games. Available at:

<https://www.sciencedirect.com/science/article/abs/pii/S1071581908000499> (Accessed: 29 August 2025)

[10] Lankoski. P, Helio. S (2002) Approaches to Computer Game Design Available at: https://www.researchgate.net/publication/221217312_Approaches_to_Computer_Game_Design (Accessed: 20 September 2025)

[11] SUPERHOT Team (2017) SUPERHOT VR [Video game]. SUPERHOT Team Available at: https://store.steampowered.com/app/617830/SUPERHOT_VR/ (Accessed: 10 May 2025).

[12] WarpFrog (2024) Blade and Sorcery[Video game]. WarpFrog Available at: https://store.steampowered.com/app/629730/Blade_and_Sorcery/ (Accessed: 27 March 2025).

[13] Stress Level Zero (2019) BONEWORKS [Video game]. Stress Level Zero Available at: <https://store.steampowered.com/app/823500/BONEWORKS/> (Accessed: 27 March 2025).

[14] Valve (1998) Half-Life [Video game]. Valve Available at: <https://store.steampowered.com/app/70/HalfLife/> (Accessed: 20 June 2025).

[15] Valve (2004) Half-Life 2 [Video game]. Valve Available at: https://store.steampowered.com/app/220/HalfLife_2/ (Accessed: 20 June 2025).

[16] Valve (2020) Half-Life: Alyx [Video game]. Valve Available at: https://store.steampowered.com/app/546560/HalfLife_Alyx/ (Accessed: 27 March 2025).

[17] Rockstar Games (2019) Red Dead Redemption 2 [Video game]. Rockstar Games Available at: https://store.steampowered.com/app/1174180/Red_Dead_Redemption_2/ (Accessed: 10 May 2025).

[18] Rocksteady Studios (2024) Batman: Arkham Shadow VR [Video game]. Oculus Studios Available at: https://www.meta.com/en-gb/experiences/batman-arkham-shadow/3551691271620960/?srsltid=AfmBOor2J6tZ_hCndj7GeuzDI6HTb4SrGismx-XxXAlGw_qo4qdelZE (Accessed: 17 September 2025)

[19] Naughty Dog LLC (2020) The Last of Us Part II [Video game]. PlayStation Publishing LLC Available at: https://store.steampowered.com/app/2531310/The_Last_of_Us_Part_II_Remastered/ (Accessed: 15 June 2025)

[20] Ninja Theory (2017) Hellblade: Senua's Sacrifice [Video game]. Ninja Theory Available at: https://store.steampowered.com/app/414340/Hellblade_Senuas_Sacrifice/ (Accessed: 11 May 2025).

[21] Atari, Inc. (1978) Space Invaders [Video game]. Atari, Inc Available at: <https://apps.apple.com/gb/app/space-invaders/id305608072> (Accessed: 20 July 2025)

[22] VR terms A-Z Glossary | Coursera Available at: <https://www.coursera.org/articles/vr-terms> (Accessed: 30 September 2025).

[23] Levels in Unreal Engine | Unreal Engine 5.6 Documentation | Epic Developer Community Available at: <https://dev.epicgames.com/documentation/en-us/unreal-engine/levels-in-unreal-engine> (Accessed: 18 September 2025)

[24] Wikipedia.org n.d. | Procedural Animation Available at: https://en.wikipedia.org/wiki/Procedural_animation (Accessed: 25 August 2025).

[25] Chandler, N. (2015) How Ragdoll Physics Works | howstuffworks.com Available at: <https://electronics.howstuffworks.com/ragdoll-physics.htm> (Accessed: 25 August 2025).

[26] Chornly, A. (2024) Blueprint vs C++ | Codeinfinity.com Available at: <https://codeinfinity.com/blueprint-vs-c-which-one-to-choose-for-your-next-unreal-engine-project/> (Accessed: 25 August 2025)

[27] Velardo, V. (2019) The Reality of Red Dead Redemption 2's AI Available at: <https://medium.com/the-sound-of-ai/the-reality-of-red-dead-redemption-2s-ai-part-1-c276e9da2763> (Accessed: 20 September 2025)

[28] Adobe (2011) Adobe Mixamo [3D animations]. Available at: <https://www.mixamo.com/#/> (Accessed: 24 June 2025)

APPENDIX A – Project Proposal

Name: Sai Srikar Chirumarri

Student Number: 240009189

Title: Dynamic NPC Touch interaction in Virtual Reality

Supervisor: Stuart Scott

1. Introduction

Virtual Reality games prioritize immersion and the player's interactions with the world around them (Hand Physics, immersive object interaction, etc.) But they are behind in the field of Non player characters (NPCs) and interactions between the player and NPCs. The NPCs are often unresponsive 3D models.

The lack of these mechanisms breaks the player's immersion especially when there are different game mechanics in the rest of the game that solely focus on increasing player immersion. This project – Dynamic NPC Touch interaction in VR addresses this gap by developing NPCs that are reactive and responsive to player physical touch interactions e.g., pushing, grabbing etc. or any other different player reactions that might involve throwing objects. These NPCs are also aware of the player's intensity of the reaction and the type of reaction exerted by the player towards the NPC. This prototype will use Unreal Engine taking advantage of procedural animations, Finite State Machines and hit detection function for the force exerted to enhance realism.

Aim of the project:

The project aim is to improve immersion in VR games by designing NPCs that react realistically to physical touch interactions.

This leads to these research questions:

1. How non-playable character interaction in virtual reality can be more realistic and immersive using a custom changeable system that reacts to player's actions?

2. Is it possible to design and implement a non-playable character which reacts to player's actions based on force and area that triggers a more realistic and believable reaction in virtual reality?

3. To what extent can we increase the realism of virtual reality environment by including more reactive non-playable characters?

These interactions take into consideration the player's place of touch, the intensity of touch to understand of it's aggressive or friendly and provide the required response. The purpose of this project is to enhance that aspect of game immersion and achieve realism. This mechanic will have a significant impact to the games.

The product of this project is a VR prototype demonstrating NPC reactions to touch intensity and location of the touch and a realistic NPC that reacts to objects thrown at them. Provide a template to make custom NPCs and use them in the game.

Beneficiaries:

The main beneficiary of this project is the author who is benefited by gaining experience in Unreal Engine 5 which is an industry standard game engine. Also understand the complexities of procedural animations, finite state machines and VR prototyping.

Other benefactors include VR developers that can use this template to develop NPCs in their own game with these features. This also contributes the rapidly developing VR research field in understanding how this effects the immersion. The gaming community being able to experience enhanced immersion to many future game titles.

2. Critical Context

Literature Review:

1. A 2022 study by Trinity College Dublin (Attention and Communication in Virtual Worlds) found that unresponsive NPCs disrupt the "plausibility illusion," a key driver of immersion.
2. ScienceDirect (2022) identified nonverbal cues (e.g., gestures, eye contact) as critical to social presence in VR, yet most NPCs lack these traits.
3. Superhot VR: Lose track of what's real. Commit yourself, body and mind. Confront the evocative, elegantly brutal world of SUPERHOT VR. Enemies pouring into the room from all sides, dozens of bullets coursing through the air... Wait. Something seems different in here... (Game steam page)
4. Blade and Sorcery: Blade & Sorcery is a built-for-VR medieval fantasy with full physics driven melee, ranged and magic combat. Become a powerful warrior, ranger or sorcerer and devastate your enemies. (Game steam page)
5. Boneworks: BONEWORKS is an Experimental Physics VR Adventure. Use found physics weapons, tools, and objects to fight across dangerous playscapes and mysterious architecture. (Game steam page)

6. Half-life: Alyx: Half-Life: Alyx is Valve's VR return to the Half-Life series. It's the story of an impossible fight against a vicious alien race known as the Combine, set between the events of Half-Life and Half-Life 2. Playing as Alyx Vance, you are humanity's only chance for survival. (Game steam page)

Context and Gaps:

Some games like Superhot VR are the industry standard which feature some excellent NPC player interactions with NPCs reacting to player's touch dynamically depending on the player's touch location with ragdoll physics reacting accordingly. Half-life: Alyx also features NPC that react to the objects that player throws. Hard bullet is another game that features good NPC player interaction where you can hold NPCs fight them, shoot them, etc. Many other VR games like Blade and Sorcery and Bone works are good with physics-based object interactions but lack the player NPC depth. This type of interaction has become a normal mechanism in non-VR games, and this is the next step for VR games. Some studies such as Attention and communication in virtual reality (Trinity College Dublin, 2022) and nonverbal communication in virtual reality (ScienceDirect, 2022) have clearly stated a lack in NPC believability in virtual reality and how it can be more immersive with a reactive non-playable character.

NPCs in VR are designed as targets or obstacles, not as a person with actions. When it comes to non-VR titles like Red Dead Redemption 2 and some of the old assassin's creed games where NPCs dynamically respond to player behaviour (e.g., fleeing if threatened, interacting or moving if the player is passing through or commenting on the player's appearance).

Recent advancements in procedural animation and AI behaviour trees offer promising tools for NPC design:

1. Hellblade: Senua's Sacrifice blends motion capture with procedural systems to generate fluid combat animations.
2. UE5's Control Rig and Animation Blueprints enable developers to create dynamic, physics-driven animations without manual keyframing.

Despite these tools, no VR project has fully utilised them to create touch-reactive NPCs. This project fills that gap by combining UE5's physics engine, finite state machines (FSMs), and procedural animation systems where required.

This project integrates force detection with Unreal Engine's collision system and uses FSMs, procedural animations to create NPCs that adapt to player reactions.

3. Approaches: Methods & Tools for Design, Analysis & Evaluation

This project will use Unreal Engine 5 with Visual Studio 2022 using an Agile development process for VR prototyping and development. This iterative process will allow continuous testing, prototyping and incremental changes to the NPC system.

Phase 1: Force Detection System

1. **Collision Mapping:** Use UE5's collision system to detect collisions between player hands and NPC hitboxes or trigger areas (e.g., shoulder, torso).
2. **Force Calculation:** Measure impact force using UE5's Physics Handle Component. To identify a gentle tap or a shove by the player.
3. **Intent Recognition:** Classify interactions as "friendly" or "hostile" based on force thresholds.

Phase 2: Finite State Machine (FSM) Design

1. **States:** Define NPC behavioural states like Idle, friendly, Defensive, Startled, etc.
2. **Transitions:** Trigger state changes based on force and touch location. For example: A shove on the torso: Transition from Idle to Defensive, A gentle touch on the shoulder: Transition from Idle to friendly.

Phase 3: Procedural Animation Integration

1. **Stagger Reactions:** Use Inverse Kinematics (IK) to simulate natural limb movements when NPCs are pushed or when object is thrown at them.

Use Unreal Engine 5 to set up VR environment and use the physics engine in the game using the collision system to get the force, location of the interaction. Design of finite state machines for NPC animations to express the nature of the NPC for example: a friendly interaction like waving or a fear response like defending themselves by raising their hands. And integrate procedural animations like stagger, fall back or defend using Unreal Engine Animation blueprints. Then playtesting and improve NPC interactions.

Analysis:

To validate the system, a mixed-methods evaluation will be conducted:

1. **Quantitative analysis:**

- Force Accuracy: Measure if the system correctly detects force ranges (e.g., 95% precision for forces between 2–10N).
- Animation Latency: Ensure reactions occur within 0.2 seconds of touch to avoid lag-induced disorientation.

2. Qualitative analysis:

- Immersive Experience Questionnaire (IEQ): Assess immersion using a 5-point Likert scale (e.g., "The NPCs felt realistic").
- Semi-Structured Interviews: Ask participants to compare reactive vs. static NPCs (e.g., "Did the NPCs enhance your sense of presence?").

Participant Recruitment:

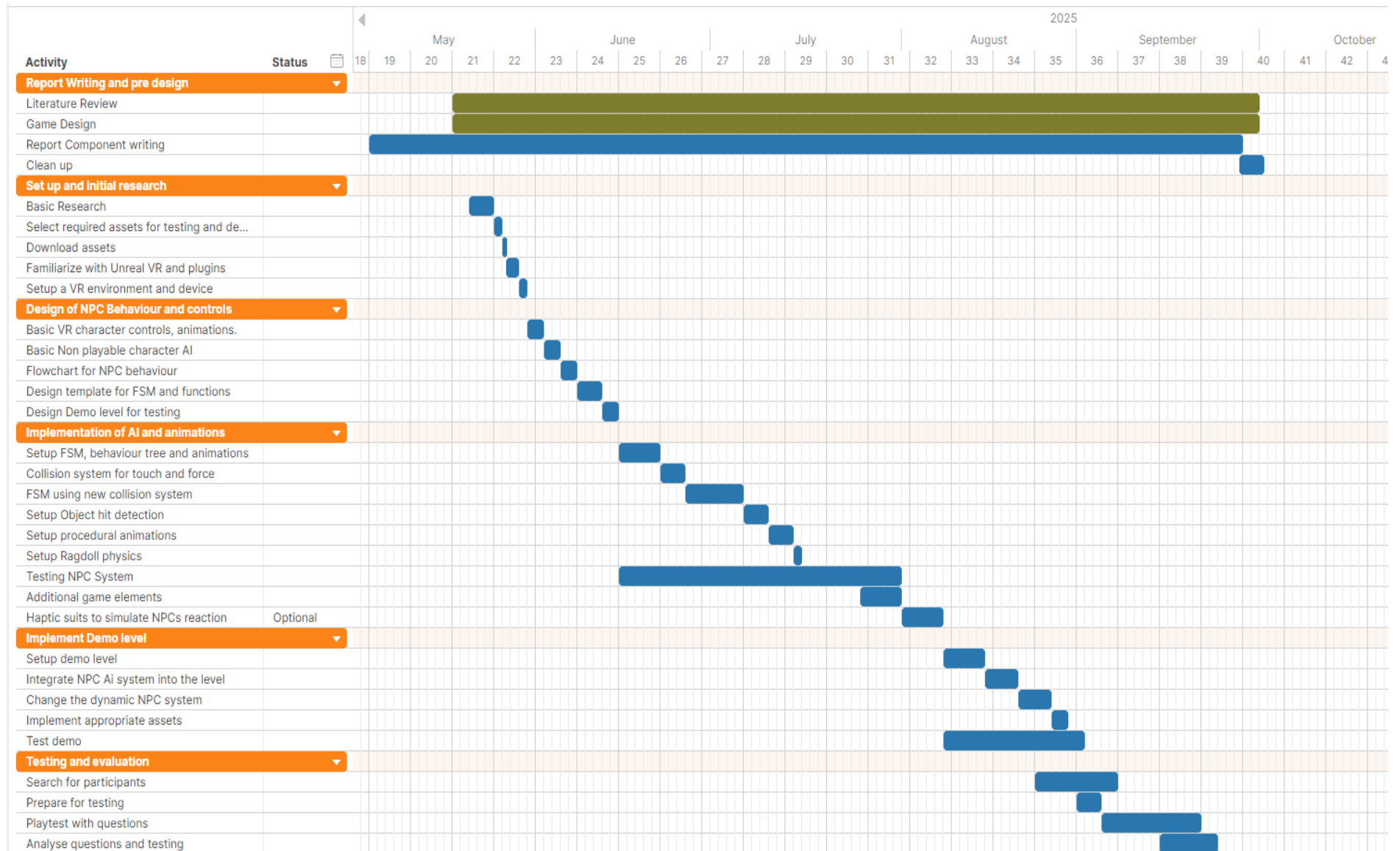
- Target: 12–15 VR users (balanced between novices and experts).

4. Work Plan

The work plan includes all the tasks that need to be completed in a proper order to achieve the intended results. The timeline of the project is rough estimation of how long each task is going to take in reality some tasks may take shorter time and some tasks may take longer as it is possible to encounter many hurdles due to the complexity of the finite state machines and non-playable character AI.

No.	Task Name
1	Setup a VR environment
2	Basic VR character controls, animations and hand physics
3	Basic non-playable character AI
4	Extended collision system to detect touch region and intensity
5	Advanced finite state machines that respond to the interactions
6	Ragdoll physics
7	Object hit detection and Procedural Animation
8	Additional game elements
9	(Optional) Haptic suits to simulate NPCs reaction
10	Prototype game featuring the interactions
11	Testing and Evaluation
12	Project report and documentation
13	Project submission preparation

Dynamic NPC Touch Interaction in Virtual Reality



5. Risks

Ethical issues: The participant's safety is considered reducing the play time to reduce dizziness and other VR effects that might occur to players with no experience in VR. Get anonymous response from the participants and obtain consent

Risk for the key tasks. (1-5)

Task	Likelihood of failure	Severity of failure	Impact (Likelihood *Severity)	Mitigation
Basic Research	1	3	3	Important aspects have been selected
Setup an Unreal VR environment	2	4	10	I have prior knowledge of Unreal and required assets. I will use assets or plugins that are working.
Basic VR character controls, animations	3	4	10	Previous experience with Unity VR and Unreal blueprint and C++ system.
Basic non-playable character AI	3	4	10	Prior knowledge with Unreal AI and control rigs.
Design template for FSM and functions	2	4	8	I will ensure the system is dynamic and easily changeable with a proper structure
Design Demo level	2	3	10	I will ensure the level fits the NPC system and showcases all aspects and is immersive.
Setup FSM, behaviour trees and animations	3	4	10	I have used FSM, behaviour trees and have worked with unreal animations and control rigs

Collision system for touch and force	4	5	15	I will ensure I follow unreal documentation and any relevant resources to implement it properly
FSM using collision system	5	5	15	I have prior knowledge with blueprints and combing FSM
Setup object hit detection	3	4	10	I have prior knowledge with Unreal physics and collision system.
Setup Procedural animations, Ragdoll physics	3	4	10	I have identified the required resources and methods to implement this.
Haptics suits to simulate NPCs reaction	5	3	8	I will take guidance and use documentation of the suits to understand and integrate it.
Implement demo level	4	5	15	I have prior experience with Unreal levels and gameplay setup. I will make sure I follow the documentation and any relevant tutorials to implement this level properly. While making sure to showcase the aspects of the NPC system.
Run play testing, with questionnaires.	3	5	15	I will make sure the testing goes smoothly by properly testing everything well in advance and ask the participants the right questions and take the proper feedback.

References:

Carroll, D. (2022) Attention and Communication in Virtual Worlds. MAI. Trinity College Dublin. Available at: <https://publications.scss.tcd.ie/theses/diss/2022/TCD-SCSS-DISSERTATION-2022-044.pdf> (Accessed: 24 March 2025).

Nadine Aburumman, Marco Gillies, Jamie A. Ward, Antonia F.de C. Hamilton. (2022) Nonverbal communication in virtual reality ', *International Journal of Human Computer Studies*, Volume 164, 102819, Available at: <https://www.sciencedirect.com/science/article/pii/S1071581922000489> (Accessed: 24 March 2025).

SUPERHOT Team (2017) SUPERHOT VR [Video game]. SUPERHOT Team Available at: https://store.steampowered.com/app/617830/SUPERHOT_VR/ (Accessed: 10 May 2025).

WarpFrog (2024) Blade and Sorcery[Video game]. WarpFrog Available at: https://store.steampowered.com/app/629730/Blade_and_Sorcery/ (Accessed: 27 March 2025).

Stress Level Zero (2019) Boneworks [Video game]. Stress Level Zero Available at: <https://store.steampowered.com/app/823500/BONEWORKS/> (Accessed: 27 March 2025).

Valve (2020) Half-Life: Alyx [Video game]. Valve Available at: https://store.steampowered.com/app/546560/HalfLife_Alyx/ (Accessed: 27 March 2025).

Rockstar Games (2019) Red Dead Redemption 2 [Video game]. Rockstar Games Available at: https://store.steampowered.com/app/1174180/Red_Dead_Redemption_2/ (Accessed: 10 May 2025).

Ninja Theory (2017) Hellblade: Senua's Sacrifice [Video game]. Ninja Theory Available at: https://store.steampowered.com/app/414340/Hellblade_Senuas_Sacrifice/ (Accessed: 11 May 2025).

Research Ethics Review Form for MSc Projects**Computer Science Research Ethics Committee (CSREC)**

<https://www.city.ac.uk/about/governance/committees/cs-research-ethics>

A.1		
1.1	Does your research require approval from the National Research Ethics Service (NRES)?	NO
1.2	Will you recruit participants who are covered by the Mental Capacity Act 2005?	NO
1.3	Will you recruit any participants who are covered by the Criminal Justice System, for example, people on remand, prisoners and those on probation?	NO
A.2		
2.1	Does your research involve participants who are unable to give informed consent?	NO
2.2	Is there a risk that your research might lead to disclosures from participants concerning their involvement in illegal activities?	NO
2.3	Is there a risk that obscene and or illegal material may need to be accessed for your research study (including online content and other material)?	NO
2.4	Does your project involve participants disclosing information about protected characteristics (as identified by the Equality Act 2010)?	NO
2.5	Does your research involve you travelling to another country outside of the UK, where the Foreign & Commonwealth Office has issued a travel warning that affects the area in which you will study?	NO
2.6	Does your research involve invasive or intrusive procedures?	NO

2.7	Does your research involve animals?	NO
2.8	Does your research involve the administration of drugs, placebos or other substances to study participants?	NO
A.3		
3.1	Does your research involve participants who are under the age of 18?	NO
3.2	Does your research involve adults who are vulnerable because of their social, psychological or medical circumstances (vulnerable adults)?	NO
3.3	Are participants recruited because they are staff or students of City, University of London?	NO
3.4	Does your research involve intentional deception of participants?	NO
3.5	Does your research involve participants taking part without their informed consent?	NO
3.5	Is the risk posed to participants greater than that in normal working life?	NO
3.7	Is the risk posed to you, the researcher(s), greater than that in normal working life?	NO

A.4		
4	Does your project involve human participants or their identifiable personal data?	YES

PART B: Ethics Proportionate Review Form

B.1		
1.1	Will you ensure that participants taking part in your project are fully informed about the purpose of the research?	YES

1.2	Will you ensure that participants taking part in your project are fully informed about the procedures affecting them or affecting any information collected about them, including information about how the data will be used, to whom it will be disclosed, and how long it will be kept?	YES
1.3	When people agree to participate in your project, will it be made clear to them that they may withdraw (i.e. not participate) at any time without any penalty?	YES
1.4	Will consent be obtained from the participants in your project? Consent from participants MUST be obtained if you plan to involve them in your project or if you plan to use identifiable personal data from existing records. “Identifiable personal data” means data relating to a living person who might be identifiable if the record includes their name, username, student id, DNA, fingerprint, address, etc.	YES
1.5	Have you made arrangements to ensure that material and/or private information obtained from or about the participating individuals will remain confidential?	YES

B.2		
2	Will the research be conducted in the participant’s home or other non-University location?	NO

APPENDIX B – Post Testing Questionnaire

Immersion experience questionnaire.

Study: Dynamic NPC touch immersion in Virtual Reality

Researcher: Sai Srikar Chirumarri

Please mention your previous VR experience

1 – beginner

2 – intermediate

3 – experienced

How experienced are you in VR? _____

Please rate how much you agree with each statement about your experience in the VR prototype. Use the scale below.

1 = Strongly Disagree

2 = Disagree

3 = Neutral

4 = Agree

5 = Strongly Agree

1. I lost track of time while I was playing
☐1 ☐2 ☐3 ☐4 ☐5
2. I felt like I was really inside the game world.
☐1 ☐2 ☐3 ☐4 ☐5
3. I was focused on the game and not easily distracted.
☐1 ☐2 ☐3 ☐4 ☐5
4. I felt the characters move like real people when interacting.
☐1 ☐2 ☐3 ☐4 ☐5
5. I felt the characters respond to my movements realistic.
☐1 ☐2 ☐3 ☐4 ☐5
6. I would like this in more VR games.
☐1 ☐2 ☐3 ☐4 ☐5
7. I felt this was better than other VR games interaction.
☐1 ☐2 ☐3 ☐4 ☐5
8. The NPC behavior made them seem not static objects.
☐1 ☐2 ☐3 ☐4 ☐5
9. I would prefer future VR games to include this type of NPC responsiveness.
☐1 ☐2 ☐3 ☐4 ☐5
10. The prototype ran smoothly without causing me significant discomfort or confusion.
☐1 ☐2 ☐3 ☐4 ☐5

11. In your own words, how did the NPC's interaction affect your feelings of immersion in VR?
12. Were there any moments when the NPC's reactions felt unrealistic or distracting? Please describe it
13. How believable was this NPC compared to the other NPC in the scene or other NPCs in games?
14. How would you improve the way NPCs respond to your intentions?
15. What movements felt most natural or most unnatural?
16. Can you name a specific game you have played where these interactions would have improved immersion?
17. What kind of games or genres do you think would benefit most from this type of NPC responsiveness?
18. Beyond games, can you think of other areas (e.g., training, simulation, education) where dynamic NPC touch interaction might be useful?

Thank you for taking part in this study and giving us feedback.

APPENDIX C – Research Ethics Documentation

INFORMED CONSENT - City St George's (Clerkenwell and Moorgate Campuses)

Name of principal researcher

Sai Srikar Chirumarri

Date and version number

v02

Title of study

Dynamic Non-Playable Character immersive touch interaction in Virtual Reality



		Please initial box
1.	I confirm that I have read and understood the participant information dated 31-08-2025 and version v.01 for the above study. I have had the opportunity to consider the information and ask questions which have been answered satisfactorily.	
2.	I understand that my participation is voluntary and that I am free to withdraw at any point without giving a reason without being penalised or disadvantaged.	
3.	I understand that I will be able to withdraw my data before 1 st October 2025.	
4.	I agree to the focus group/interview being audio recorded	
5.	I agree to the focus group/interview being video recorded	
6.	I agree to take part in this study, which involves playing a VR prototype for approximately 10-15 minutes and completing a short questionnaire (approx. 5 minutes).	
7.	I understand that the responses I provide will be treated anonymously, and no names or identifying information will be published. I understand that anonymised data will be stored securely for 2 years after the end of the study and then deleted.	
8.	I understand the possible minor risks of taking part in a short VR session (e.g., mild dizziness or eye strain). I confirm I will inform the researcher immediately if I feel unwell and understand I can stop the session at any time.	
9.	I agree to City St George's recording and processing this information about me. I understand that this information will be used only for the purpose(s) explained in the	

	participant information and my consent is conditional on City complying with its duties and obligations under the General Data Protection Regulation (GDPR).	
10.	I agree to take part in the above study.	

Sai Srikar Chirumarri

C Sai Srikar

31-08-2025

Name of the researcher

Signature

Date

**PARTICIPANT INFORMATION – City St George’s
(Clerkenwell and Moorgate Campuses).**

Title of study

Dynamic Non-Playable Character immersive touch interaction in Virtual Reality

Name of principal researcher

Sai Srikar Chirumarri

Invitation paragraph

We would like to invite you to take part in a Virtual Reality game testing study. Before you decide whether you would like to take part it is important that you understand why the research is being done and what it would involve for you. Please take time to read the following information carefully and discuss it with others if you wish. Ask us if there is anything that is not clear or if you would like more information. You will be given a copy of this information sheet to keep.

What is the purpose of the study?

This study is part of my master’s dissertation at City University of London. The project looks at how non-playable character (NPCs) in virtual reality can respond to physical touch and interactions (such as pushing, touching specific body parts, the nature of touch, etc) to create a more realistic and immersive experience.

The aim is to understand if these interactions make the NPC behavior improve players’ immersion compared to other NPCs.

Why have I been invited to take part?

You have been invited because you have an interest in non-VR games or VR games and are willing to take part in a short playtest. We are aiming for 6 – 8 participants with a mix VR experience levels from beginners to experienced users. If you are a student at City St



George's choosing to take part or not take part in the study will have no effect on assessments, marks or future studies. Or if you are an employee, you will have no effect on your employment or promotion prospects etc.

Do I have to take part?

Participation in the project is completely voluntary, and you can choose not to participate in part or all of the project. You can withdraw at any stage of the project without being penalized or disadvantaged in any way. It is up to you to decide whether or not to take part. If you do decide to take part you will be asked to sign a consent form. If you decide to take part you are still free to withdraw at any time and without giving a reason.

If you decide to withdraw after taking part, you can ask for your data to be removed before 1st October 2025. After this point, the data will be anonymized, and it will not be possible to remove it.

What will happen if I take part?

If you agree to take part, you will:

1. Come to City St George's, University of London Interaction Lab.
2. Play a VR prototype for about 10-15 minutes. You will wear a VR headset and box with a NPC and see how they respond also notice how the crowd responds to your touch.
3. Complete a short questionnaire about your experience (approx. 5minutes).

The entire session will last around 20-30 minutes.

What are the possible disadvantages and risks of taking part?

VR use may sometimes cause mild side effects while using it such as dizziness, motion sickness, or eye strain. To reduce these risks:

- Play sessions will be short.
- You may stop at any time if you feel uncomfortable.

There are no other known risks associated with this study.

What are the possible benefits of taking part?

While there are no direct personal benefits, your feedback will help improve future VR games and research into immersive experiences.

Data privacy statement

City St George's, University of London is the sponsor and the data controller of this study based in the United Kingdom. This means that we are responsible for looking after your information and using it properly. The legal basis under which your data will be processed is City's public task.

Your rights to access, change or move your information are limited, as we need to manage your information in a specific way in order for the research to be reliable and accurate. To safeguard your rights, we will use the minimum personally identifiable information possible (for further information please see: <https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/lawful-basis/a-guide-to-lawful-basis/public-task/>).

City will use your name and contact details to contact you about the research study as necessary. **If you wish to receive the results of the study, your contact details will also be kept for this purpose.** The only people at City St George's who will have access to your identifiable information will be members of the research team, and, if appropriate, individuals with responsibility for monitoring and auditing at City St George's, including of research projects. There may be occasions when regulatory authorities may access research data in accordance with their statutory powers. City St George's will keep identifiable information about you from this study for **2** years after the study has finished.

You can find out more about how City handles personal data by visiting <https://www.city.ac.uk/about/governance/policies/data-protection-policy>. You can also read City's general privacy notice by visiting <https://www.city.ac.uk/about/governance/policies/general-privacy-notice> . If you are concerned about how we have processed your personal data, you can contact the Information Commissioner's Office (ICO) directly <https://ico.org.uk/>.

What will happen to the results?

- Questionnaire responses will be collected anonymously.
- The audio and video data from the session will be stored in City OneDrive which is encrypted and accessible only to the research team and will be deleted after 2 years. No identifying audio or video will be published.
- No personal identifying information will be recorded.
- Data will only be used for this MSc dissertation project and may be included in academic publications in anonymized form.
- All data will be stored securely and only the researcher, and the supervisor will have access.

Who has reviewed the study?

This study has been approved by City St George's, University of London (Clerkenwell and Moorgate Campuses) Computer Science Research Ethics Committee (CSREC).

What if there is a problem?

If you have any problems, concerns, or questions about this study, you should ask to speak to a member of the research team. If you remain unhappy and wish to complain formally, you can do this through City St George's complaints procedure. To complain about the study, you can contact the Research Governance, Ethics and Integrity Manager using the details below and inform them that the name of the project is Dynamic NPC Touch interaction with Virtual Reality.

Research Governance, Ethics, and Integrity Manager
Research & Innovation Directorate
City St George's, University of London
Northampton Square
London, EC1V 0HB
Email: senaterec@city.ac.uk

Insurance

City St George's, University London holds insurance policies which apply to this study, subject to the terms and conditions of the policy. If you feel you have been harmed or injured by taking part in this study you may be eligible to claim compensation. This does not affect your legal rights to seek compensation. If you are harmed due to someone's negligence, then you may have grounds for legal action.

Further information and contact details

- Researcher: Sai Srikar Chirumarri – sai.chirumarri@citystgeorges.ac.uk, +44 7553859988
- Supervisor: Stuart Scott – stuart.scott.2@citystgeorges.ac.uk

Thank you for taking the time to read this information sheet.

Screening form:

1. Are you 18 years of age or older?
(YES/NO)
2. Are you a student or a staff member of City University of London?
(YES/NO)
3. Do you have any medical conditions that may prevent you from safely using a VR headset? (Link to an official VR health and safety page [Health and safety warnings | Meta Quest | Meta Store](#))
(YES/NO)

4. Have you previously taken part in this VR NPC interaction study?
(YES/NO)
5. Do you feel comfortable interacting with a VR headset for 10-15 minutes
(YES/NO)
6. Are you comfortable completing a short questionnaire in English after the VR session?
(YES/NO)

APPENDIX D – Game Assets

Asset	Uploaded by	License	Date Uploaded	Accessed	Modified	URL
Acknowledging	Unknown (Not Mentioned) Mixamo.com Artist	Free for personal and commercial use	Unknown (Not Mentioned)	24-07-2025	Modified and Retargeted in Unreal Engine Control Rig Editor	Mixamo - Acknowledging
Bouncing Fight Idle	Unknown (Not Mentioned) Mixamo.com Artist	Free for personal and commercial use	Unknown (Not Mentioned)	24-07-2025	Modified and Retargeted in Unreal Engine Control Rig Editor	Mixamo - Bouncing Fight Idle
Cheering	Unknown (Not Mentioned) Mixamo.com Artist	Free for personal and commercial use	Unknown (Not Mentioned)	24-07-2025	Modified and Retargeted in Unreal Engine Control Rig Editor	Mixamo - Cheering
Dismissing Gesture	Unknown (Not Mentioned) Mixamo.com Artist	Free for personal and commercial use	Unknown (Not Mentioned)	24-07-2025	Modified and Retargeted in Unreal Engine Control Rig Editor	Mixamo - Dismissing Gesture
Falling Down	Unknown (Not Mentioned) Mixamo.com Artist	Free for personal and commercial use	Unknown (Not Mentioned)	24-07-2025	Modified and Retargeted in Unreal Engine Control Rig Editor	Mixamo - Falling Down
Getting Hit	Unknown (Not Mentioned) Mixamo.com Artist	Free for personal and commercial use	Unknown (Not Mentioned)	24-07-2025	Modified and Retargeted in Unreal Engine	Mixamo - Getting Hit

					Control Rig Editor	
Hands Forward Gesture	Unknown (Not Mentioned) Mixamo.com Artist	Free for personal and commercial use	Unknown (Not Mentioned)	24-07-2025	Modified and Retargeted in Unreal Engine Control Rig Editor	Mixamo - Hands Forward Gesture
Happy Hand Gesture	Unknown (Not Mentioned) Mixamo.com Artist	Free for personal and commercial use	Unknown (Not Mentioned)	24-07-2025	Modified and Retargeted in Unreal Engine Control Rig Editor	Mixamo - Happy Hand Gesture
Injured Hurting Idle	Unknown (Not Mentioned) Mixamo.com Artist	Free for personal and commercial use	Unknown (Not Mentioned)	24-07-2025	Modified and Retargeted in Unreal Engine Control Rig Editor	Mixamo - Injured Hurting Idle
Leg Sweep	Unknown (Not Mentioned) Mixamo.com Artist	Free for personal and commercial use	Unknown (Not Mentioned)	24-07-2025	Modified and Retargeted in Unreal Engine Control Rig Editor	Mixamo - Leg Sweep
Shaking Hands 2	Unknown (Not Mentioned) Mixamo.com Artist	Free for personal and commercial use	Unknown (Not Mentioned)	24-07-2025	Modified and Retargeted in Unreal Engine Control Rig Editor	Mixamo - Shaking Hands 2
Shoulder Hit and Fall	Unknown (Not Mentioned) Mixamo.com Artist	Free for personal and commercial use	Unknown (Not Mentioned)	24-07-2025	Modified and Retargeted in Unreal Engine Control Rig Editor	Mixamo - Shoulder Hit and Fall

BoxingIdle	Unknown (Not Mentioned) Mixamo.com Artist	Free for personal and commercial use	Unknown (Not Mentioned)	24-07-2025	Modified and Retargeted in Unreal Engine Control Rig Editor	Mixamo- BoxingIdle
Center Block	Unknown (Not Mentioned) Mixamo.com Artist	Free for personal and commercial use	Unknown (Not Mentioned)	24-07-2025	Modified and Retargeted in Unreal Engine Control Rig Editor	Mixamo- Center Block
Dodging	Unknown (Not Mentioned) Mixamo.com Artist	Free for personal and commercial use	Unknown (Not Mentioned)	24-07-2025	Modified and Retargeted in Unreal Engine Control Rig Editor	Mixamo- Dodging
Head Hit	Unknown (Not Mentioned) Mixamo.com Artist	Free for personal and commercial use	Unknown (Not Mentioned)	24-07-2025	Modified and Retargeted in Unreal Engine Control Rig Editor	Mixamo- Head Hit
Hit To Body	Unknown (Not Mentioned) Mixamo.com Artist	Free for personal and commercial use	Unknown (Not Mentioned)	24-07-2025	Modified and Retargeted in Unreal Engine Control Rig Editor	Mixamo- Hit To Body
Idle	Unknown (Not Mentioned) Mixamo.com Artist	Free for personal and commercial use	Unknown (Not Mentioned)	24-07-2025	Modified and Retargeted in Unreal Engine Control Rig Editor	Mixamo- Idle
PunchLeft	Unknown (Not Mentioned)	Free for personal and	Unknown (Not	24-07-2025	Modified and Retargete	Mixamo- PunchLeft

	Mixamo.com Artist	commercial use	Mentioned)		d in Unreal Engine Control Rig Editor	
PunchRight	Unknown (Not Mentioned) Mixamo.com Artist	Free for personal and commercial use	Unknown (Not Mentioned)	24-07-2025	Modified and Retargeted in Unreal Engine Control Rig Editor	Mixamo- PunchRight

APPENDIX E – Additional Content (OneDrive Content, Flowchart, Screenshots)

OneDrive Content:

- Participant Playtesting videos
- Unreal Engine Project (Can be run by downloading the folder, and open with Unreal Engine 5.5)
- A PDF with all the filled consent forms
- Immersive Experience Questionnaire answers provided by the participants
- Informed Consent, Participant Information Sheet, Recruitment Material (Screener Questions, Poster), Testing Immersive Questions, Study guide

OneDrive Link: [https://cityuni-](https://cityuni-my.sharepoint.com/:f/r/personal/sai_chirumarri_city_ac_uk/Documents/Dissertation_Files?csf=1&web=1&e=j5p3mh)

[my.sharepoint.com/:f/r/personal/sai_chirumarri_city_ac_uk/Documents/Dissertation_Files?csf=1&web=1&e=j5p3mh](https://cityuni-my.sharepoint.com/:f/r/personal/sai_chirumarri_city_ac_uk/Documents/Dissertation_Files?csf=1&web=1&e=j5p3mh) The OneDrive can only be accessed by the author and the supervisor. OneDrive link access can be requested through the supervisor.

Here are some additional implementations that are part of the NPC system but couldn't be included due to word limit and a lack of a designated area.

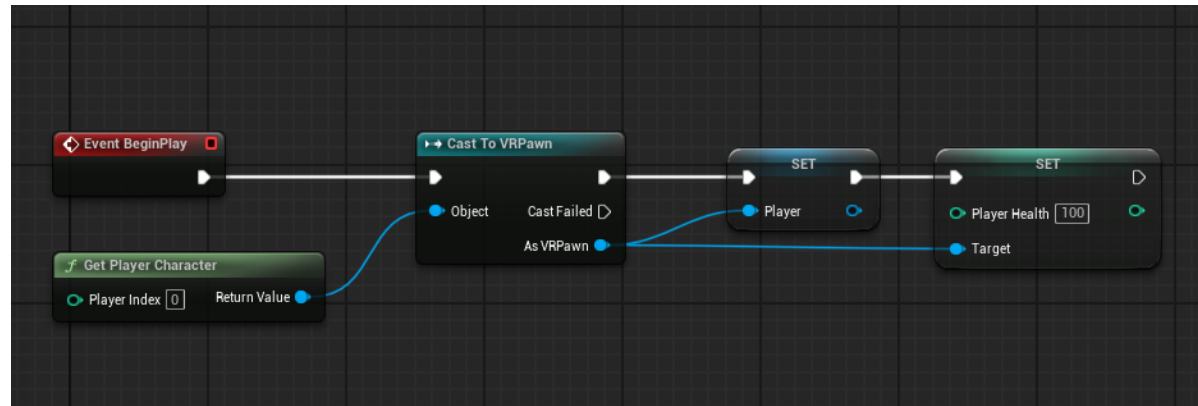


Figure E. 1 Boxing Level – Main Study Area Level Blueprint Beginning Conditions

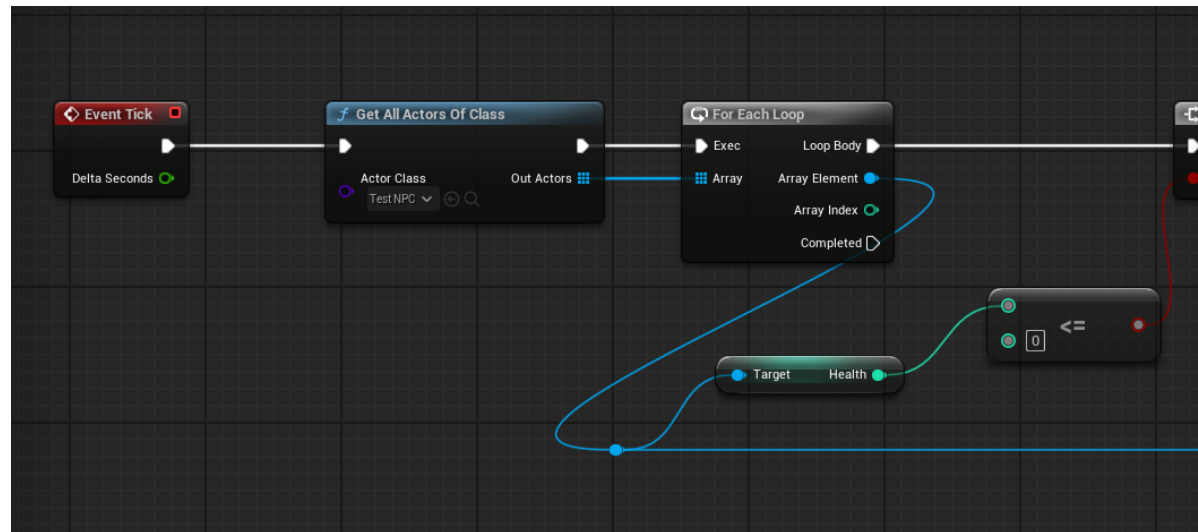


Figure E. 2 Boxing Level – Main Study Area Level Blueprint Death Logic Part 1

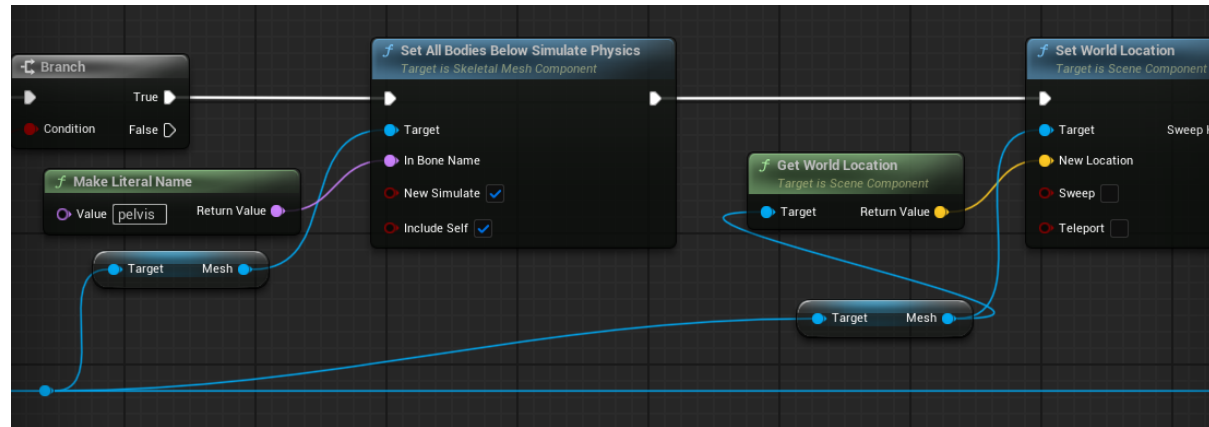


Figure E. 3 Boxing Level – Main Study Area Level Blueprint Death Logic Part 2

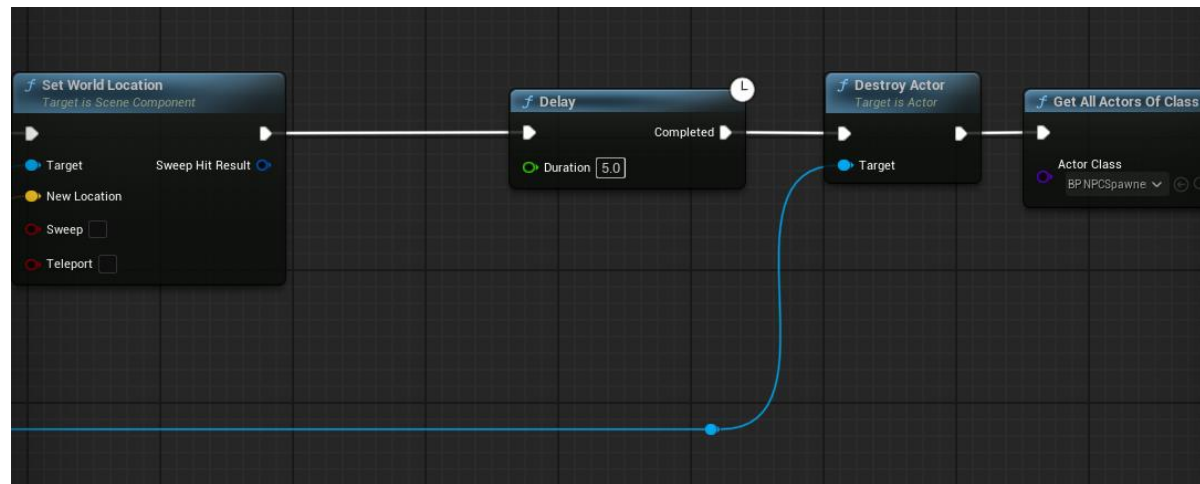


Figure E. 4 Boxing Level – Main Study Area Level Blueprint Death Logic Part 3

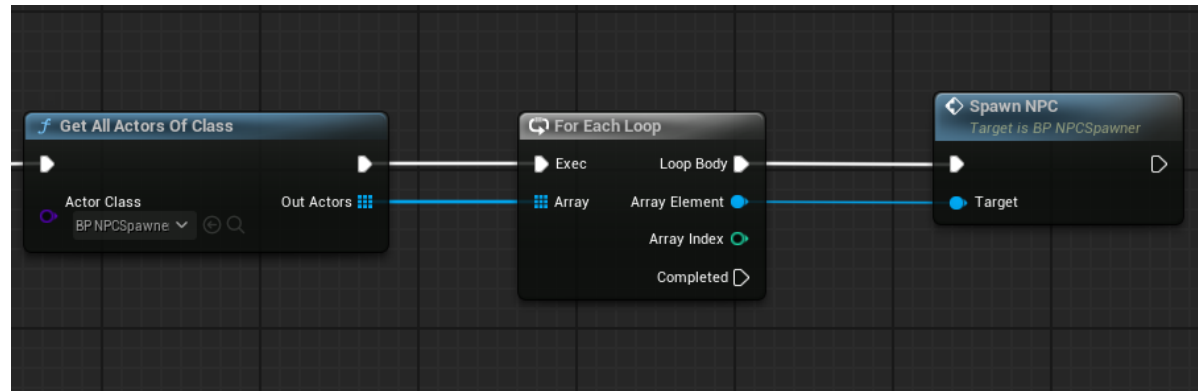


Figure E. 5 Boxing Level – Main Study Area Level Blueprint Death Logic Part 4

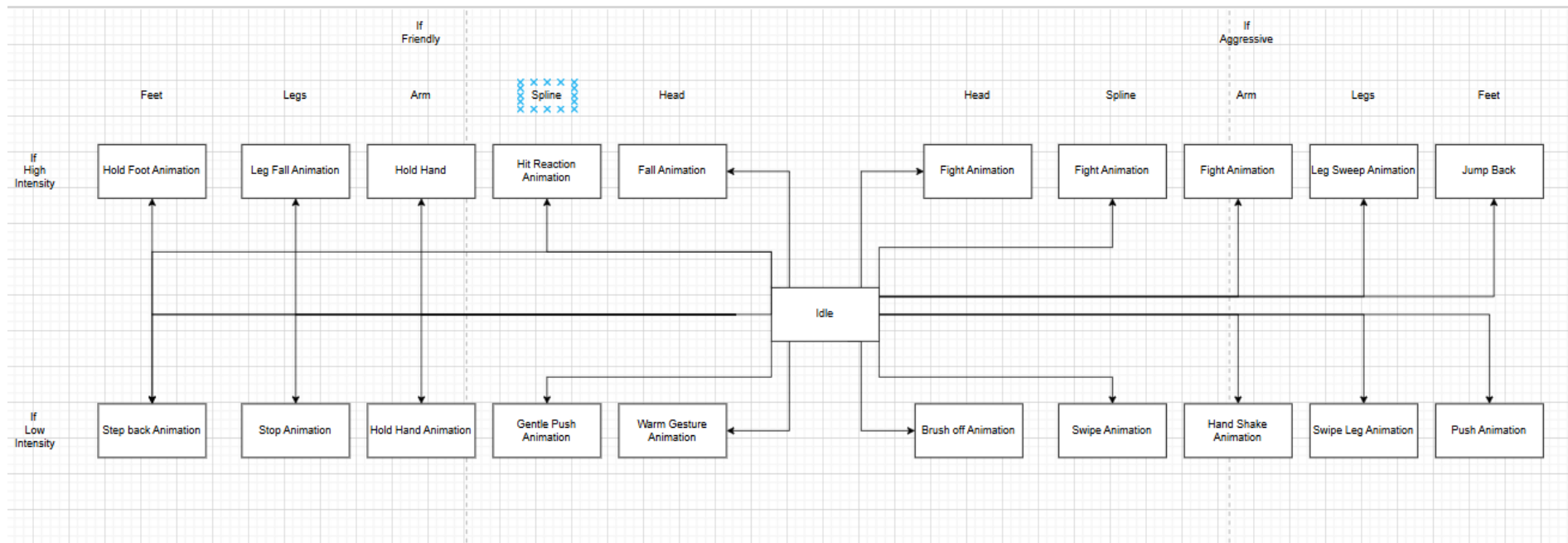


Figure E. 6 Detailed NPC Behaviour Flowchart